Autonomous Observation Dynamics, Minimal All-Future Realizations, and Provenance Information Loss: From Terminating Quotient Refinement and Optimal Windows to Non-Resetting Physical Circuits

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Formal verification. Lean 4 verifies the main constructions and properties of this paper [10]. For Corollary 2.5, the declaration `prefix_has_update_iff_kernel_eq` gives the necessary and sufficient condition for an arbitrary single-valued update. The pinned public selection contains the proofs, their immediate consumers, and the verification materials. The paper proves the general risk identity in Theorem 5.6. Its fixed clock-to-parity instance follows from the formalized counting tables.

Abstract

For a linear system over a commutative ring, we prove that an observation window admits an autonomous update if and only if its kernel equals the all-future invisible submodule. The criterion applies to every single-valued update map. For a fixed observer that reads all columns of a spatial, mass, and clock carrier, $a + 1$ consecutive frames recover every observable spatial coordinate and the independent mass and clock at copy scale $a > 1$. No update map on the full carrier can use fewer frames. A rational encoding and decoding protocol uses two moments. For finite systems with heterogeneous observation outputs, refining the old minimum quotient after adding a readout yields the same equivalence relation as recomputation.

For streaming reception, counted moments make coarsening commute with updates and preserve the initial counted error across all finite continuations. One precision budget thus suffices for exact recovery of the evolving conditional distributions. For a fixed task, population, and weights, the excess coarsening risk is the mean squared distance between the fine and coarse conditional means. In an explicit parity-coupled family, retaining the same pair of channels yields a strict recovery gain at every successor, while replacement improves recovery of the new task at even depths $d \geq 2$ and degrades recovery of the previous task at odd depths $d \geq 1$. In both cases the worse channel has risk at least $(d + 1)/(d + 2)$. A dissipative RLC and gate-level model gives blind recovery of all 1024 independent drive configurations and a 128-bit reception program that completes by its deadline on every raw packet of the specified width. Along runs generated by the hardware, the receiver carries capacitor voltages across every finite sequence of frames without a dedicated reset-to-zero phase. For a decoder with access to one B-bit packet, the mean squared error for elapsed time over n frames is at least $\tau^2[(n/2^B)^2 - 1]/12$, where $\tau > 0$ is a lower bound on frame duration.

Keywords: autonomous observation quotients; shortest observation windows; terminating partition refinement; streaming conditional distributions; excess coarsening risk; non-resetting measurement circuits.

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1 Equal readings can have different futures

A signal takes the value 1 only at spatial position 2. An instrument reads positions 1 and 3 and reports zero at both. A claim that the system is at rest with no signal fails at the next step. The original update shifts the signal to the right, placing it at 3, where the second probe reads 1. Equal local readings at the current step can lead to different future readings.

The state generated by the process carries spatial coefficients, an independent mass coordinate, and an independent clock. This unit signal has mass 1 and clock 3. At copy scale $a = 2$, the complete physical response recorded by the two columns at step $t = 0$ is

$$(0 + 1 + 2 \cdot 3,\ 0 + 1 + 4 \cdot 3) = (7, 13), \tag{1.1}$$

The full response differs from $(0, 0)$ because the two zero spatial samples omit the mass and clock. After one step the spatial samples are $(0, 1)$ and the full response is $(9, 18)$. In the third frame $(t = 2)$ the response is $(11, 21)$. This frame supplies the last independent equation, recovering mass 1, clock 3, and the four spatial coordinates $(0, 0, 1, 0)$ uniquely (Example 3.2, §3.1).

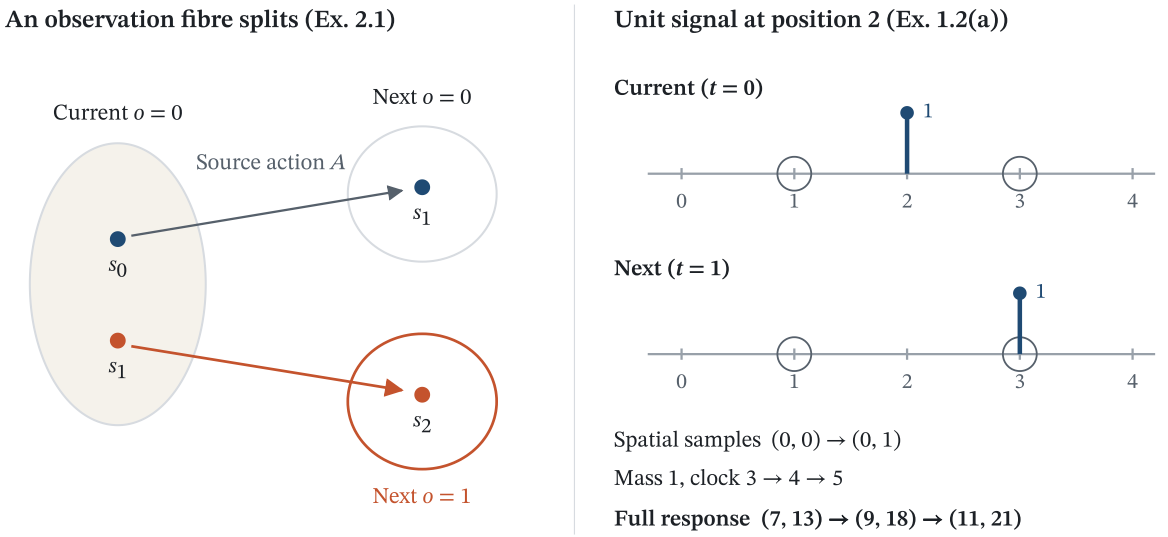


Figure 1. States with equal current readings separate at the next step. Left. Both states initially read $o = 0$. Applying the original action A for one step sends their readings to different cells. Red marks the trajectory that leaves the shared reading cell. Right. At $a = n = 2$, open circles mark the spatial sample positions $1, 3$. Both spatial probes report zero for the unit signal at 2, while the mass and clock contribute to the nonzero full response.

This three-frame example leads to four questions about dynamics, automata, statistical inference, and measurement circuits.

1. **Autonomy and computable minimal realizations.** Under what algebraic condition can a restricted observation evolve on its own? For a finite state space with several operations and heterogeneous readouts, can the number of initially indistinguishable pairs bound the computation of a minimal all-future model? Can a new readout be incorporated by refining the old result?

2. **Shortest observation windows for arbitrary update maps.** For a commutative-ring module and a fixed multicolumn instrument, can a nonlinear or even nonmeasurable update use fewer frames than the first stable linear prefix? At copy scale a, how many frames does the full-column observer need to determine every future reading?
3. **Risk under streaming compression, channel replacement, and multiscale forgetting.** In a receiver with a growing inventory, does coarsening before an update give exactly the same result as updating before coarsening? As the system generates new observation channels, does replacing the old channels monotonically reduce recovery risk?
4. **Non-resetting physical measurement circuits.** Can a circuit with Joule dissipation, capacitor-held voltages, and gate propagation delays keep measuring across arbitrarily many finite frames by directly handing off voltages between frames? What lower bound does the fixed packet width impose on long-history time recovery?

The relevant foundations include the observable decomposition of linear systems, geometric invariant subspaces, and realization over rings [23; 2; 35; 39; 22; 45], smooth nonlinear observability and delay embeddings [18; 40], automaton minimization and coalgebraic behavioural quotients [28; 30; 19; 31; 36], mean ergodic theory and iterative regularization [41; 34; 24; 26; 12], one-pass variance merging and comparison of experiments [42; 8; 5; 32], multiscale memory and completely monotone functions [4; 43; 14; 29], and the thermodynamics of measurement and erasure [25; 3; 37]. We work within one source-native process and whole-ledger framework [16; 17]. The four problems are prefix stability for arbitrary update maps over commutative-ring modules and a sharp $a+1$-frame bound on an infinite-dimensional spatial, mass, and clock carrier, conservation of counted residuals with one initialization budget and parity-dependent risk reversal in deterministically growing channels, incremental refinement for heterogeneous readouts and continuation after a large-state collision, and continuous non-resetting measurement in an acyclic dual-rail RC gate netlist together with a fixed-packet distortion bound in squared seconds.

1.1 Main results and supporting formalizations

Three results organize the paper. Operator approximation, incremental algorithms, and system examples give the supporting results below.

Main Theorem 1.1 (Autonomous observation dynamics). The following conclusions hold in their respective settings of commutative-ring modules, the spatial, mass, and clock Hilbert carrier $G = \ell^2(\mathbb{N};\mathbb{C}) \oplus_2 \mathbb{C} \oplus_2 \mathbb{C}$, and the dissipative RC gate-level circuit model.

1. **Result A. Autonomy for arbitrary update maps and the shortest $a+1$-frame window (Theorems 2.2 and 2.4, Corollary 2.5, Theorems 3.3–3.5).** A linear system over a commutative ring has a unique linear successor on its current observation coimage if and only if $\mathrm{actionDefect}_{A,o} = 0$. The all-future kernel $K_\infty = \bigcap_{t\ge 0} \ker(oA^t)$ is the largest submodule of $\ker o$ invariant under A. In the Hilbert case the orthogonal decomposition is $x = R\pi x + Dx$. The prefix P_ℓ admits a single-valued update on the full source carrier if and only if $\ker P_\ell = K_\infty$. For a full-column observer at fixed copy scale $a > 1$, $a+1$ consecutive frames recover the first na spatial coordinates, the independent mass, and the clock. This

many frames are necessary for an autonomous update. The rational encoding and decoding protocol uses two moments per frame.

2. **Result B. Conservation of counted residuals, continual posterior recovery, and parity-dependent channel risk (Theorems 5.2–5.6, 6.1, and 6.5–6.6).** The receiver state satisfies $\mathcal{M}_f(\mathrm{next}_{y_*}\,\mathsf{F}) = \mathrm{next}_{f(y_*)}(\mathcal{M}_f\mathsf{F})$. The counted residual $c_y\varepsilon_y$ is constant across every finite continuation, so one initialization budget permits exact posterior recovery at every step. For a fixed task, population, and weights, excess coarsening risk equals the mean squared distance between the fine and coarse conditional means. In the copy-channel family coupled modulo 2, a fixed pair of retained channels gives a strict recovery gain at every generated successor. Replacing the channel at each generation improves the new task at even depths $d \geq 2$ and harms the old task at odd depths $d \geq 1$. In both branches the worse code has risk at least $(d+1)/(d+2)$.
3. **Result C. On-time fixed-width packet decoding, capacitor handoff, and time distortion (Theorems 7.2–7.6).** The 128-bit gate arithmetic implements an integer RL energy test using six ADC words per channel. The microprogram decides every raw packet of the specified width by the deadline determined by the gate graph. Accepted results are emitted at the deadline. The 11 continuous voltage decisions in a shared AND-inverter graph (AIG) agree with micro-execution. In hardware-generated frames the registered input and dual-rail gate branches with capacitors hand off their voltages directly and satisfy $\mathcal{V}_{\mathrm{next}}(v,p;0) = \mathcal{V}_{\mathrm{old}}(v,p;t_{\mathrm{end}})$. The frames concatenate with no finite upper bound on elapsed time. Given a single B-bit packet, the mean squared time-recovery error for a history of length n is at least $\frac{\tau^2}{12}[(n/2^B)^2 - 1]$.

Supporting results. (i) **Ergodic truncation and moving-window error (§2.2, §3.5, §4, and §7.1).** Driven nonexpansive Birkhoff averaging and Landweber adjoint recovery (Theorem 2.7) lead to $n2^{-n}\|x\|$ finite-prefix truncation in state-indexed spaces without a uniform operator bound (Theorem 2.8). An infinite-rank free $\mathbb{Z}$-module admits a saturated window at the single prime $\mathrm{cut}(p)+1$ (Theorem 3.6). Native perturbations give cross-step noise amplification $2N+3$, a gain budget κ_{gain}, and three orthogonal error terms in moving empirical spaces (Theorems 4.1–4.4). A signed 60-word ADC output yields a blind quotient isomorphism on 1024 samples and requires at least 10 bits (Theorem 7.1). (ii) **Terminating incremental refinement and system examples (§2.3–§2.4).** Adding a heterogeneous readout gives the same kernel by incremental refinement as by full recomputation (Theorem 2.9). The continuation after a collision uses an added coordinate, with 37.65% lossless compression of a 52-field, 27.54 MB fly state (Theorem 2.10). The three-clock system supports replay from an eight-byte success key and an all-future zero fiber for closed syntax (Theorem 2.11 and Proposition 2.12). (iii) **Singular Gram updates, a forgetting law, and appendices (§6.4–§6.5, Appendices A–C).** These cover preservation of normal-equation kernel fibers and transfer of innovations across cofinal stages (Theorem 6.8), the shifted Gamma–Laplace power law $\Psi_{\alpha,r}(t) = (1+t/r)^{-\alpha}$ (Theorem 6.9), and closed-relation quotients of graph completions, observation accumulation points, and Cayley–Hamilton horizons. Table 1 pairs the positive results with counterexamples and lower bounds.

Table 1. Main results, supporting results, counterexamples, and lower bounds.

Result	Construction or identity	Counterexample or lower bound	Location
A. Arbitrary updates and the shortest $a+1$-frame window	Quotient action iff actionDefect $= 0$, orthogonal recovery K_∞, arbitrary-map criterion over commutative-ring modules, and $a+1$-frame rational inversion from two moments	Same-reading collision (Example 2.1) and a two-frame zero-read collision v_* ruling out arbitrary, including nonlinear or nonmeasurable, $< a+1$-frame maps (Theorem 3.4)	§2.1, §3.1–§3.4, Figures 1–2, Appendix B.4
B. Counted residuals and parity-dependent risk	One initialization gives $\|c_y\varepsilon_y\| < 1/2$, its conservation permits exact posteriors, coarsening commutes with updates, and same-task gain $\mathrm{gGap}_k > 0$ holds at every successor, with a replacement gain at even depths	Opposite and aligned residuals show why the vector cross term is needed (§5.3), five participants have clockGap $= 0$ and full gGap $= 1/5$ (Example 6.2), and odd-depth replacement harms the old task with worse-code risk $\geq (d+1)/(d+2)$ (Theorem 6.6)	§5.1–§5.5, §6.1–§6.3, Figures 3–5 (left)
C. Non-resetting measurement and capacity	128-bit gate-level energy test completes on all raw packets, and capacitors on shared AIG dual rails pass voltages between frames with non-Zeno time advance	An accepted packet cannot be emitted before the deadline (Theorem 7.2), and single-packet time error $\geq \frac{\tau^2}{12}[(n/2^B)^2 - 1]$ eventually exceeds any fixed tolerance (Theorem 7.6)	§7.2–§7.4, Figure 6
Supporting results	$n2^{-n}\|x\|$ truncation in state-indexed spaces, equality of incremental and full refinement, single-prime cut$(p)+1$ window, $2N+3$ noise and three-term error identities, 1024 blind recoveries, singular Gram updates, shifted Gamma power law	An escaping branch precludes factorization through the old view (Theorem 2.10), changing the environment moves an open term out of the zero fiber (Proposition 2.12), short-prefix zeros with later ones (Theorem 3.6), pulse timing residual $\geq 1/4$ (Example 4.5), and impossibility of < 10-bit encoding (Theorem 7.1)	§2.2–§2.4, §3.5, §4, §6.4–§6.5, §7.1, Figure 5 (right), Appendices A–C

1.2 Four examples for hand verification

The following arithmetic examples exhibit the four mechanisms before we turn to modules, functional analysis, and gate netlists.

Example 1.2 (Four hand calculations).

(a) The $(7, 13)$ response behind zero spatial samples and inversion from three frames (§3.1–§3.2). Take copy scale $a = 2$ and $n = 2$ columns. Place the unit signal at spatial coordinate

$k = 2$, with mass $m = 1$ and clock $c = 3$. The columns sample only positions $2(1) - 1 = 1$ and $2(2) - 1 = 3$. The spatial samples at step $t = 0$ are therefore $(h_1, h_3) = (0, 0)$. The full response $r_j(t) = h_{2(j+1)-1-t} + m + 2(j + 1)(c + tm)$ over the first three frames is

$$r(0) = (7, 13), \qquad r(1) = (9, 18), \qquad r(2) = (11, 21).$$

The first two frames give 4 scalars for 4 spatial coordinates and (m, c), a total of 6 independent degrees of freedom on the full carrier. They cannot determine all these coordinates. At the third frame $t = 2$, the first spatial index $1-2 < 0$ is negative and that entry vanishes. The second-column spatial term $h_{3-2} = h_1$ cancels its counterpart h_1 at $r_0(0)$. Subtraction gives

$$m = \frac{r_1(2) - r_0(0) - r_0(2)}{2^2 - 1} = \frac{21 - 7 - 11}{3} = 1, \qquad c = \frac{11 - 1}{2} - 2 = 3,$$

Substitution into the first two frames recovers $(h_0, h_1, h_2, h_3) = (0, 0, 1, 0)$ coordinate by coordinate.

(b) Terminating refinement of four states and incremental addition of a readout (§2.3). Take the four-state set $X = \{0, 1, 2, 3\}$ with transition $0 \to 1 \to 2 \to 2$, $3 \to 0$ and binary readout o_1 whose values are $(0, 0, 1, 0)$. The initial equal-reading unordered pairs of distinct states are $\{0, 1\}, \{0, 3\}, \{1, 3\}$. The first successor test drops $(0, 1) \mapsto (1, 2)$ and $(1, 3) \mapsto (2, 0)$ because their successor readings are $(0, 1)$ and $(1, 0)$, respectively, and hence differ within each pair. Only $(0, 3) \mapsto (1, 0)$ remains. The next test drops $(0, 3)$ because the successor pair $(1, 0)$ was deleted in the first round. The algorithm therefore removes $(0, 3)$ and reaches the discrete minimal kernel within 2 steps. For an incremental example with transition $0 \to 1, 1 \to 1, 2 \to 3, 3 \to 3$ and constant readout $o_1 = (0, 0, 0, 0)$ (old kernel $K_* = X \times X$), add $o_2 = (0, 0, 0, 1)$. Inside K_* filter by o_2 to obtain the current partition $\{0, 1, 2\} \mid \{3\}$, in which $(0, 2)$ still share a cell. One more successor test separates $(0, 2) \mapsto (1, 3)$ because of $o_2(1) = 0 \neq 1 = o_2(3)$, leaving $\{0, 1\} \mid \{2\} \mid \{3\}$, the same relation obtained by recomputing from (o_1, o_2).

(c) Five participants with zero excess clock risk and $1/5$ excess full risk, and parity reversal of channel replacement (§6.1 and §6.3). For five equally weighted participants $i = 0, 1, 2, 3, 4$, the scale-2 copy action gives clock values $(4, 6, 8, 10, 12)$. The fine observation splits by parity into $\{0, 2, 4\}$ and $\{1, 3\}$. Each cell has conditional clock mean $(4+8+12)/3 = (6+10)/2 = 8$, so the merged mean stays at 8. The clock risk is 8 before and after coarsening, with increment clockGap $= 0$. For five orthogonal unit spatial vectors the risk rises from $3/5$ to $4/5$. The full risk rises from $43/5$ to $44/5$ (gGap $= 1/5 > 0$). Now take the family of growing channels coupled modulo 2 at depth d. The old and new copy materials have binary multipliers $(d + 2) \bmod 2$ and $(d + 3) \bmod 2$, respectively. At even depth $d = 2$ the multipliers are $(0, 1)$, so the old channel vanishes, giving new-task decoding risk $\geq 3/4$, and the new channel wins. One step later at odd depth $d = 3$, the multipliers become $(1, 0)$. The newborn channel vanishes, and using it to decode the old task incurs positive risk $\geq 4/5$.

(d) Ten-channel 1024 blind recovery, direct dual-rail capacitor handoff, and the $M = 2^B$ pigeonhole bound (§7.1–§7.4). Ten independent Boolean switches give $2^{10} = 1024$ drive configurations. Since the 60-word ADC readout blindly recovers each configuration, every lossless digital representation requires at least 10 bits ($2^9 = 512 < 1024$). In continuous operation the dual-rail capacitor voltages held at time t_{end} give the initial values for frame $t = 0$. Forced RC

relaxation in the acyclic gate graph converges to the new target rails, so it needs no dedicated interframe discharge phase. A single recording packet has fixed width B and thus at most $M = 2^B$ codes. Once the number of steps reaches $n > M$, different times separated by at least τ seconds share a code fiber. The lower bound $\frac{\tau^2}{12}[(n/M)^2 - 1]$ on squared time-recovery error grows quadratically in n, eventually exceeding any fixed tolerance for a single-packet decoder.

1.3 Reading routes and notation

After Example 1.2, the following sections locate the proofs by subject.

- **Dynamical systems, control, and inverse problems.** See §2.1–§2.2, §3, §4, §8.1–§8.3, and Appendices A and B.4 for the action-defect criterion for arbitrary updates (Theorem 2.2 and Corollary 2.5), Landweber–Birkhoff convergence and $n2^{-n}\|x\|$ prefix truncation in state-indexed spaces (Theorems 2.7–2.8), the sharp $a + 1$-frame observability index for arbitrary maps on the full carrier (Theorem 3.4), and the three-way orthogonal error decomposition in moving empirical spaces (Theorem 4.4).
- **Automata, program semantics, and formal mathematics.** See §2.3–§2.4, §3.5, §5.2, and §8.1 for bounded terminating refinement, its universal minimality and incremental equivalence (Theorem 2.9), automatic coordinate recovery after a collision and eight-byte success-key replay (Theorems 2.10–2.11), the zero fiber of a consecutive interval of closed requests (Proposition 2.12), and the single-prime saturated window (Theorem 3.6).
- **Information theory, statistics, and computational models of memory.** See §2.1 (Lemma 2.6), §5, §6, §8.4–§8.5, and Appendix B for counted residual conservation, the merge-update law for complete receiver states and discrete support recovery (Theorems 5.2–5.6), fixed-task compression risk and parity reversal for new channels (Theorems 6.1 and 6.5–6.6), empirical Hermitian Gram updates allowing singularity (Theorem 6.8), and the shifted Gamma–Laplace forgetting law (Theorem 6.9).
- **Measurement, hybrid control, and thermodynamics of computation.** See §7.1–§7.4 and §8.6 for the blind quotient isomorphism for 1024 commands on a noisy dissipative RLC netlist (Theorem 7.1), on-time micro-execution of the 128-bit gate-level energy test on every raw packet (Theorem 7.2), direct capacitor handoff and non-Zeno time advance of the shared AIG receiver (Theorems 7.3–7.4), and physical rate-distortion lower bounds in squared seconds (Theorems 7.5–7.6).

Notation and terminology. S is the type of the full current state, and $\sigma : S \to S$ is its deterministic actual successor. In an occurrence, the current state generates the event e_s, the whole-ledger writeback L_s, and the successor $\sigma(s)$ [16; 17]. The compressed readout then represents that fixed evolution. We reserve C for the source module or Hilbert carrier (e.g. $C_s, C_{\mathrm{src}}, C_{\mathrm{now}}$). A coarse-cell count carries a subscript as in C_z, a window gain constant is κ_{gain}, and a forgetting curve is $\Psi_{\alpha,r}(t)$. We use D for the orthogonal residual projection onto the all-future invisible kernel (e.g. $Dx, D_N v$), D for the optimal paired distortion of a finite source, and g for a binary decoder. We use R for the isometric orthogonal representative of an observation quotient in the source carrier, $R : M \to C$. The forward readout operator in §2.2 is Θ, the readout index set in §2.3 is $\mathcal{J}_{\mathrm{read}}$, and the full receiver state in §5 is F. We use T for linear reconstruction from a finite

observation window (e.g. $TP_a v, T_N$) and $\Lambda = K^*$ for the empirical time-conditioned transition in §4.4. Inner products $\langle \cdot, \cdot \rangle$ are linear in the second variable and conjugate-linear in the first. The spatial, mass, and clock Hilbert carrier of §3 is G. General finite-source results allow zero weights. Claims of strict samplewise separation explicitly assume positive weights. Full-object recovery means equality of vectors or states in the target carrier. Full-posterior recovery means equality of the conditional probability of each participant. All-future sufficiency means equality under every legal future action word and readout.

Term	Meaning	Location
Participant $i \in I_N = \{0, \dots, N\}$	An indexed sample or event record in a finite history (uniform empirical weight $p_N(i) = 1/(N+1)$)	§1.3, §2.1
Whole-ledger writeback	Deterministic state component recording six conservation and attribution fields during a native transition [17]	§1.3, §2.3
State-indexed space C_s **/ moving empirical space** $(C_{\text{now}}, C_{\text{next}})$	A Hilbert carrier indexed by state $s \mapsto \sigma(s)$. A moving empirical space is the case indexed by consecutive empirical measures L^2.	§2.2, §4.4
Generation depth d **/ material source** s_{d+1} **/ scan index** $j+1$	Respectively the trajectory step s_d, the successor state s_{d+1} to which the copy channel captured at that step is tied, and the positive integer governing the multiplier modulo 2 and the expansion scale	§6.2
Cofinal stage $m \in \mathbb{N}$	A recurring settlement point in tree-structured or multistage evolution, where the empirical Gram matrix $\mathcal{G}_m$ is recomputed from the current request inventory	§2.4, §6.4
Non-resetting handoff (`no_reset`)	Hold the previous frame's dual-rail capacitor voltages as the next frame's initial values. No separate discharge-to-zero phase is inserted. Dissipative RC overwriting still produces Joule heat.	§7.3, §8.6

2 Autonomous quotients for multiple operations, minimal all-future models, and convergent recovery

An observation o partitions the source state space into fibers. The states x, y belong to the same cell precisely when $o(x) = o(y)$. A compressed readout can evolve independently of the source only if the original action maps each whole fiber into one fiber. If two states in a fiber have different readings at the next step, a readout-only updater would require two outputs for the same input.

Example 2.1 (Equal readings, different successors). On the state sequence s_j take successor $\sigma(s_j) = s_{j+1}$ and readout $o(s_j) = \lfloor j/2 \rfloor$, which rounds down and groups states by twos.

Then $o(s_0) = o(s_1) = 0$, but their readings at the next step are $o(\sigma(s_0)) = 0$ and $o(\sigma(s_1)) = 1$ respectively. No map F satisfies $F(o(s_j)) = o(\sigma(s_j))$ for both $j = 0, 1$. The collision holds regardless of whether F is linear, continuous, or computable.

2.1 The action defect, maximal invisible kernel, and the criterion for arbitrary updates

We study observation quotients of modules in the tradition of linear systems over commutative rings [35; 39]. Before taking the intersection over every future reading, we determine when the coimage of one observation can update autonomously and quantify the obstruction when it cannot.

Theorem 2.2 (Action-defect criterion for autonomous update on the current observation coimage). Let R be a commutative ring, C, B an R-module, and $A : C \to C$ and $o : C \to B$ be R-linear maps. Write $Q_0 = C/\ker o$ for the observation coimage and $\pi_0 : C \to Q_0$ for the quotient map. Define the action defect on the invisible kernel by

$$\mathrm{actionDefect}_{A,o} := \pi_0 \circ A\big|_{\ker o} : \ker o \longrightarrow Q_0. \tag{2.1}$$

Then the following hold.

1. **Existence and uniqueness.** There is a unique R-linear successor $\bar{A}_0 : Q_0 \to Q_0$ on Q_0 satisfying $\bar{A}_0 \circ \pi_0 = \pi_0 \circ A$ if and only if $\mathrm{actionDefect}_{A,o} = 0$, or equivalently $A(\ker o) \subseteq \ker o$.
2. **Successor difference within a fiber.** If $x, y \in C$ satisfy $o(x) = o(y)$, then $x - y \in \ker o$ and their next coimage difference is

$$\pi_0(Ax) - \pi_0(Ay) = \mathrm{actionDefect}_{A,o}(x - y); \tag{2.2}$$

   In particular, if there is an $z \in \ker o$ with $o(Az) \neq 0$, no map $\bar{A}_0$ makes the square commute.
3. **One-step joint refinement.** Define the joint observation $o_{(1)} = (o, o \circ A) : C \to B \times B$. Its quotient $Q_1 = C/\ker o_{(1)}$ induces linear maps $q_{\mathrm{cur}}, q_{\mathrm{next}} : Q_1 \to Q_0$ for the present and next readings, satisfying $q_{\mathrm{cur}}(\pi_1 x) = \pi_0 x$, $q_{\mathrm{next}}(\pi_1 x) = \pi_0(Ax)$, and $\pi_1 x = \pi_1 y \iff o(x) = o(y) \wedge o(Ax) = o(Ay)$.

Proof. If $\bar{A}_0$ satisfies $\bar{A}_0 \pi_0 = \pi_0 A$, then for every $z \in \ker o$ we have $\pi_0 z = 0$ and hence $\mathrm{actionDefect}_{A,o}(z) = \pi_0(Az) = \bar{A}_0(\pi_0 z) = 0$. Conversely, if $\mathrm{actionDefect}_{A,o} = 0$, then $\ker o \subseteq \ker(\pi_0 \circ A)$. The universal property of the quotient module gives a unique linear $\bar{A}_0 : Q_0 \to Q_0$ satisfying $\bar{A}_0 \pi_0 = \pi_0 A$. For $o(x) = o(y)$, linearity gives $x - y \in \ker o$ and $\pi_0(Ax) - \pi_0(Ay) = \pi_0(A(x - y)) = \mathrm{actionDefect}_{A,o}(x - y)$. If $o(Az) \neq 0$, then $\pi_0(Az) \neq 0$, contradicting a zero defect. Finally $\ker o_{(1)} = \ker o \cap \ker(o \circ A)$ lies in both $\ker \pi_0$ and $\ker(\pi_0 \circ A)$, so the two readings descend to the joint quotient. ■

The obstruction in Theorem 2.2 is the nonzero visible component that A gives to vectors in $\ker o$. For example, before its saturation cutoff the single-prime free-source readout of §3.5 has state directions with present reading 0 and next reading 1. This witnesses $\mathrm{actionDefect} \neq 0$ and excludes a one-frame quotient action. The joint observation $o_{(1)}$ retains that increment. Intersecting the invisible kernels over all time steps accounts for every later distinction.

Definition 2.3 (All-future evaluations and the autonomous observation quotient). In the setting of Theorem 2.2, for $t \in \mathbb{N}$ and $\ell \in \mathbb{N}$ set

$$e_t = oA^t, \qquad P_\ell x = (e_0 x, \dots, e_\ell x), \qquad K_\infty = \bigcap_{t \geq 0} \ker e_t. \tag{2.3}$$

Elements of K_∞ are called all-future invisible differences. The autonomous all-future model is $M = C/K_\infty$, with quotient projection $\pi : C \to M$.

Theorem 2.4 (Maximal invisible invariant kernel and Hilbert recovery).

1. **Maximality and the all-future quotient.** K_∞ is the largest R-submodule of $\ker o$ invariant under A. There is a unique induced linear action $\bar{A} : M \to M$ and readout $\bar{o} : M \to B$ on $M = C/K_\infty$ with

$$\bar{A}\pi = \pi A, \qquad \bar{o}\pi = o, \qquad \pi x = \pi y \iff \forall t \geq 0,\ oA^t x = oA^t y. \tag{2.4}$$

2. **Orthogonal representatives and residual energy.** Let C be a real or complex Hilbert space, B a normed space over the same scalar field, and A, o continuous linear maps. Then K_∞ is closed. Endow M with the natural quotient norm. There are a linear isometric embedding $R : M \to C$ and an orthogonal projection $D : C \to K_\infty$ such that

$$x = R\pi x + Dx, \qquad \|x\|^2 = \|\pi x\|^2 + \|Dx\|^2, \tag{2.5}$$

$$oA^t R\pi x = oA^t x \quad (\forall t \geq 0), \qquad \|\bar{A}\| \leq \|A\|. \tag{2.6}$$

   Full-object recovery $R\pi x = x$ holds if and only if $Dx = 0$. In general $R\bar{A} = R\pi AR$. If A is symmetric (self-adjoint), the two components commute exactly, with $R\bar{A} = AR$ and $DA = AD$.

Proof. If $x \in K_\infty$, then for all $t \geq 0$ we have $oA^t(Ax) = oA^{t+1}x = 0$, giving $Ax \in K_\infty$. Taking $t = 0$ gives $K_\infty \subseteq \ker o$. Conversely, let a submodule $V \subseteq \ker o$ satisfy $AV \subseteq V$. Induction on t gives $A^t V \subseteq V \subseteq \ker o$, so $V \subseteq K_\infty$. Invariance implies $\mathrm{actionDefect}_{A,\pi} = 0$ in Theorem 2.2 and induces the unique $\bar{A}$. The containment $K_\infty \subseteq \ker o$ induces $\bar{o}$, and writing out membership of the difference in K_∞ yields (2.4).

In the Hilbert case each $e_t = oA^t$ is continuous, so $K_\infty = \bigcap_{t \geq 0} \ker e_t$ is closed. The orthogonal decomposition $C = K_\infty^\perp \oplus K_\infty$ identifies the quotient $M = C/K_\infty$ isometrically with $K_\infty^\perp$ [9]. Compose this isometry with the inclusion $K_\infty^\perp \hookrightarrow C$ to obtain R, and take the orthogonal projection onto K_∞ as D. This gives (2.5). Since $Dx \in K_\infty$, every future evaluation satisfies $oA^t Dx = 0$, so $oA^t R\pi x = oA^t x$. For any $m \in M$, we have $\|Rm\| = \|m\|$. The orthogonal projection $R\pi$ has norm at most 1, giving $\|\bar{A}m\| = \|R\pi ARm\| \leq \|A\|\|Rm\| = \|A\|\|m\|$. Finally, if A is symmetric, for any $y \in K_\infty^\perp, z \in K_\infty$ we have $\langle Ay, z\rangle = \langle y, Az\rangle = 0$ because $Az \in K_\infty$. Thus $K_\infty^\perp$ is also invariant under A, and each orthogonal component commutes with A. ■

Any single-valued update on the full carrier preserves the equal-reading classes of the current window. Theorems 2.2 and 2.4(1) make equality of the window kernel and all-future kernel the criterion. The Lean theorem `prefix_has_update_iff_kernel_eq` proves the equivalence directly for

arbitrary set maps. Theorem 3.4 and Appendix Theorem B.4 determine the shortest windows for the spatial, mass, and clock carrier and for finite-rank free modules, respectively.

Corollary 2.5 (The prefix criterion for arbitrary maps over commutative-ring modules). In the setting of Theorem 2.2 and Definition 2.3, for every prefix index $\ell \in \mathbb{N}$ the following are equivalent.

1. There is an R-linear successor $\bar{A}_\ell : \operatorname{ran} P_\ell \to \operatorname{ran} P_\ell$ on the prefix image $\operatorname{ran} P_\ell \subseteq B^{\ell+1}$ such that $\forall x \in C,\ \bar{A}_\ell(P_\ell x) = P_\ell(Ax)$.
2. There is a set map $F : B^{\ell+1} \to B^{\ell+1}$ with $\forall x \in C,\ F(P_\ell x) = P_\ell(Ax)$, with no linearity, continuity, smoothness, or measurability assumption.
3. The linear prefix kernel is stable under the one-step action, $A(\ker P_\ell) \subseteq \ker P_\ell$, that is, $\mathrm{actionDefect}_{A,P_\ell} = 0$.
4. The prefix kernel equals the all-future kernel, $\ker P_\ell = K_\infty$.

Proof. For (1) $\Rightarrow$ (2) extend $\bar{A}_\ell$ by zero outside $\operatorname{ran} P_\ell$. For (2) $\Rightarrow$ (3) use linearity of P_ℓ and A. Substituting the zero element $x = 0$ into $F(P_\ell x) = P_\ell(Ax)$ gives $F(0) = F(P_\ell 0) = P_\ell(A0) = 0$. For any $z \in \ker P_\ell$ we have $P_\ell z = 0$, hence $P_\ell(Az) = F(P_\ell z) = F(0) = 0$, giving $Az \in \ker P_\ell$. For (3) $\Rightarrow$ (4) the first coordinate of P_ℓ gives $\ker P_\ell \subseteq \ker o$. By $A(\ker P_\ell) \subseteq \ker P_\ell$ and maximal invariance of K_∞ in Theorem 2.4(1), we obtain $\ker P_\ell \subseteq K_\infty$. The reverse containment $K_\infty \subseteq \ker P_\ell$ follows from the definition. The implication (4) $\Rightarrow$ (3) follows from invariance of K_∞ under A. Theorem 2.2(1) applied to the linear map $P_\ell : C \to \operatorname{ran} P_\ell$ proves (3) $\Rightarrow$ (1). ■

Theorem 2.4 and Corollary 2.5 identify the information needed for each objective. Prediction may identify states differing by K_∞, and the shortest stable prefix on the full carrier is determined by stabilization of the linear kernel chain $\ker P_0 \supseteq \ker P_1 \supseteq \cdots$. Exact recovery of x keeps the residual vector Dx alongside the quotient point πx. The residual norm $\|Dx\|$ records its magnitude. For a finite source with an empirical history distribution, the corresponding conditional decomposition gives lower bounds for distortion in terms of entropy.

Lemma 2.6 (Optimal conditional recovery and an entropy–distortion bound for a finite source). Let a finite set I carry probability weights $p_i \geq 0$ with $\sum_{i\in I} p_i = 1$. Equip the source and the observation codomain of $q : I \to Y$ with measurable singletons. The latter may be infinite. For a complex- or Hilbert-valued task $z : I \to \mathcal{H}$, on a positive-weight fiber $p(q = y) > 0$ define the conditional distribution and mean by

$$p(i \mid y) = \frac{p_i \mathbf{1}_{q(i)=y}}{p(q = y)}, \qquad \mu_y = \sum_{i\in I} p(i \mid y) z_i. \tag{2.7}$$

For every decoder $d : Y \to \mathcal{H}$, the squared risk has the orthogonal decomposition

$$\sum_{i\in I} p_i \|z_i - d(q_i)\|^2 = \sum_{i\in I} p_i \|z_i - \mu_{q_i}\|^2 + \sum_{i\in I} p_i \|\mu_{q_i} - d(q_i)\|^2, \tag{2.8}$$

On zero-weight fibers μ_y and $d(y)$ may be chosen arbitrarily, since the corresponding terms have weight zero. In particular, take n equally weighted participants ($p_i = 1/n$) and a real-valued task $t_i \in \mathbb{R}$, and write $m_y = |F_y|$ for the cardinality of a nonempty fiber $F_y = q^{-1}(y)$. The optimal conditional distortion $\mathsf{D}(q, t)$ and the Shannon conditional entropy with natural logarithms [38]

satisfy

$$\mathsf{D}(q,t) = \frac{1}{2n}\sum_y \frac{1}{m_y}\sum_{i,j\in F_y}(t_i - t_j)^2, \qquad H(I \mid q) = \sum_y \frac{m_y}{n}\log m_y, \qquad e^{2H(I|q)} \le \sum_y \frac{m_y}{n} m_y^2. \tag{2.9}$$

Proof. In each fiber satisfying $p(q = y) > 0$, split $z_i - d(y)$ into $(z_i - \mu_y) + (\mu_y - d(y))$. By the definition of μ_y, we have $\sum_{q(i)=y} p_i(z_i - \mu_y) = 0$, so the inner-product cross term vanishes. Sum over the positive-weight fibers to obtain (2.8) [11, Theorem 4.1.15]. The $p_i = 0$ terms have zero weight and can be omitted. For an equally weighted real task, substitute the identity $\sum_{i,j\in F_y}(t_i - t_j)^2 = 2m_y\sum_{i\in F_y}(t_i - \bar{t}_y)^2$ into the first term of (2.8) to obtain $\mathsf{D}(q,t)$. Then apply Jensen's inequality with weights $2H(I \mid q) = \sum_y \frac{m_y}{n}\log(m_y^2)$ to the convex function exp to obtain the last part of (2.9). ■

2.2 Ergodic averages, Landweber adjoint recovery, and finite-prefix truncation in state-indexed spaces

Theorem 2.4 obtains the orthogonal projection $R\pi x \in K_\infty^\perp$ from the closed subspace K_∞. In infinite-dimensional inverse problems and in systems $C_s \to C_{\sigma(s)}$ whose carrier changes at each step, an observer can lack an analytic description of the common kernel, compactness, and uniform bounds on the readouts. The next two theorems use finite Birkhoff averages of driven nonexpansive dynamics and adaptively scaled adjoint residuals. They recover the observable orthogonal representative and its typed successor in the strong topology, with explicit error budgets and without inverse matrices or spectral cutoff parameters.

Theorem 2.7 (Driven nonexpansive recovery and strong Landweber–Birkhoff convergence). Let H, F be Hilbert spaces over the same scalar field $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$.

1. **Driven nonexpansive recovery and the source-condition rate.** Let a continuous linear operator $A : H \to H$ satisfy $\|A\| \le 1$. Given a drive vector $d \in H$, define the affine step $S_d(u) = Au + d$ and the zero-initialized $n+1$-step Birkhoff average by

$$\mathcal{A}_n(A,d) := \frac{1}{n+1}\sum_{j=0}^{n} S_d^j(0). \tag{2.10}$$

For any source element $x \in H$ and residual drive $d = x - Ax$, every $n \in \mathbb{N}$ satisfies the algebraic identity

$$\mathcal{A}_n(A, x - Ax) = x - \frac{1}{n+1}\sum_{j=0}^{n} A^j x; \tag{2.11}$$

If A has no nonzero fixed points (i.e. $Au = u \implies u = 0$), then $\lim_{n\to\infty}\mathcal{A}_n(A, x - Ax) = x$ converges strongly in H. Under the source condition $x = z - Az$ for x ($z \in H$), for every $n \in \mathbb{N}$ the explicit nonasymptotic rate is $\|\mathcal{A}_n(A, x - Ax) - x\| \le \frac{2\|z\|}{n+1}$. A pair of arbitrary drives $d_1, d_2 \in H$ and step count $n \in \mathbb{N}$ satisfy the linear trajectory error budget

$$\|\mathcal{A}_n(A,d_1) - \mathcal{A}_n(A,d_2)\| \le n\|d_1 - d_2\|. \tag{2.12}$$

2. **Strong Landweber–Birkhoff recovery of the observable part.** Let a continuous linear readout $\Theta : H \to F$ satisfy $\|\Theta\| \le 1$ and set $A_\Theta = I - \Theta^*\Theta$. For observation data $y \in F$ the step $S_{\Theta^* y}(u) = u + \Theta^*(y - \Theta u)$ is an adjoint residual correction. For any $x \in H$, the Birkhoff average $\mathcal{L}_n(\Theta, y) := \mathcal{A}_n(A_\Theta, \Theta^* y)$ converges strongly as follows:

$$\lim_{n\to\infty} \mathcal{L}_n(\Theta, \Theta x) = P_{(\ker\Theta)^\perp} x, \tag{2.13}$$

Here $P_{(\ker\Theta)^\perp}$ is the orthogonal projection onto the orthogonal complement of the observation kernel. In particular, when Θ is injective, $\mathcal{L}_n(\Theta, \Theta x) \to x$ recovers the full source strongly. For any data perturbation $y_1, y_2 \in F$ one has $\|\mathcal{L}_n(\Theta, y_1) - \mathcal{L}_n(\Theta, y_2)\| \le n\|y_1 - y_2\|$.

Proof. For part 1, induction on j gives $S^j_{x-Ax}(0) = x - A^j x$. At $j = 0$ both sides are 0. If the formula holds at step j, substitution gives $S^{j+1}_{x-Ax}(0) = A(x - A^j x) + (x - Ax) = x - A^{j+1}x$. Averaging over $j = 0, \dots, n$ yields (2.11). Since $\|A\| \le 1$, the mean ergodic theorem for nonexpansive operators [41; 34] gives strong convergence of $\frac{1}{n+1}\sum_{j=0}^n A^j x$ to the orthogonal projection $P_{\ker(I-A)}x$ onto the closed fixed-point subspace $\ker(I - A)$. If $\ker(I - A) = \{0\}$, the projection is zero, and (2.11) gives $\mathcal{A}_n(A, x - Ax) \to x$. If $x = z - Az$ is a coboundary [24], $\sum_{j=0}^n A^j(z - Az) = z - A^{n+1}z$ telescopes. Combining with $\|A^{n+1}z\| \le \|z\|$ gives $\|\mathcal{A}_n(A, x - Ax) - x\| = \frac{1}{n+1}\|z - A^{n+1}z\| \le \frac{2\|z\|}{n+1}$. For arbitrary d_1, d_2 set $\delta_j = S^j_{d_1}(0) - S^j_{d_2}(0)$. By $\delta_0 = 0$ and $\|\delta_{j+1}\| = \|A\delta_j + (d_1 - d_2)\| \le \|\delta_j\| + \|d_1 - d_2\|$, induction gives $\|\delta_j\| \le j\|d_1 - d_2\| \le n\|d_1 - d_2\|$ $(\forall j \le n)$. The triangle inequality for the average of $n + 1$ terms gives (2.12).

For part 2, since $\|\Theta\| \le 1$, we have $\|\Theta^*\| \le 1$ and for every $u \in H$,

$$\|(I - \Theta^*\Theta)u\|^2 = \|u\|^2 - 2\|\Theta u\|^2 + \|\Theta^*\Theta u\|^2 \le \|u\|^2 - \|\Theta u\|^2 \le \|u\|^2,$$

Thus $\|A_\Theta\| \le 1$. Also $A_\Theta u = u \iff \Theta^*\Theta u = 0 \iff \langle \Theta u, \Theta u\rangle = 0 \iff \Theta u = 0$, so the fixed-point set is $\ker(I - A_\Theta) = \ker\Theta$. As $x - A_\Theta x = \Theta^*\Theta x$, (2.11) and the mean ergodic limit give $\mathcal{L}_n(\Theta, \Theta x) = x - \frac{1}{n+1}\sum_{j=0}^n A^j_\Theta x \to x - P_{\ker\Theta}x = P_{(\ker\Theta)^\perp}x$. If Θ is injective, $\ker\Theta = \{0\}$ and the limit is x. Substituting $d_i = \Theta^* y_i$ into (2.12) and using $\|\Theta^*(y_1 - y_2)\| \le \|y_1 - y_2\|$ gives the data perturbation bound. ■

The operator $A_\Theta = I - \Theta^*\Theta$ is self-adjoint with spectrum in $[0, 1]$, so the Landweber sequence already converges before averaging [26; 12]. The Birkhoff average $\mathcal{A}_n$ in Theorem 2.7 also covers general nonself-adjoint nonexpansive operators A with spectrum meeting other points on the unit circle, such as -1 or complex phases. It gives the operator perturbation estimate used in Theorem 2.8 for finite prefixes in state-indexed spaces.

Now take a sequence of state-indexed spaces. At state $s \in S$ the carrier is a complex Hilbert space C_s, the one-step action is a continuous linear operator $A_s : C_s \to C_{\sigma(s)}$, and the current readout is $o_s : C_s \to B$ where B is a complex Hilbert space. The future evaluation at step j is $E_{s,j} := o_{\sigma^j(s)} \circ A^{(j)}_s : C_s \to B$, with $A^{(0)}_s = I_{C_s}$ and $A^{(j+1)}_s = A_{\sigma^j(s)} \circ A^{(j)}_s$. Since $\|A_s\|$ can exceed 1, $\|E_{s,j}\|$ can grow without bound in j. At finite time n the observer has access to the first n readings $(E_{s,0}x, \dots, E_{s,n-1}x)$.

Theorem 2.8 (The $n2^{-n}\|x\|$ truncation bound for actual prefixes and recovery of the typed coimage successor). In the state-indexed observation system above, with no boundedness assumption on $\|E_{s,j}\|$, define the adaptively scaled observation operator $\widetilde{E}_{s,j} := (1+\|E_{s,j}\|)^{-1}E_{s,j}$ and geometric weights $w_j := 2^{-(j+1)}$. Using the first n operators and the actual readings $y = (y_0, \dots, y_{n-1}) \in B^n$, form the finite-prefix contraction and its drive:

$$F_{s,n} := I - \sum_{j=0}^{n-1} 2^{-(j+1)}\widetilde{E}_{s,j}^*\widetilde{E}_{s,j}, \qquad d_{s,n}(y) := \sum_{j=0}^{n-1} 2^{-(j+1)}(1+\|E_{s,j}\|)^{-1}\widetilde{E}_{s,j}^* y_j, \tag{2.14}$$

Define the finite-prefix recovery operator $\widehat{\mathcal{P}}_{s,n}(y) := \mathcal{A}_n(F_{s,n}, d_{s,n}(y))$. Let $K_s = \bigcap_{j\geq 0} \ker E_{s,j}$, let $J_s : C_s/K_s \to C_s$ be the isometric orthogonal representative of the state-indexed all-future coimage, and let $\bar{A}_s : C_s/K_s \to C_{\sigma(s)}/K_{\sigma(s)}$ be the induced typed successor (with $\bar{A}_s\pi_s = \pi_{\sigma(s)}A_s$). Then the following hold.

1. **Exponential approximation in operator norm and kernel identification.** For all $n \in \mathbb{N}$, $\|F_{s,n}\| \leq 1$. The series limit $F_{s,\infty} = I - \sum_{j=0}^{\infty} 2^{-(j+1)}\widetilde{E}_{s,j}^*\widetilde{E}_{s,j}$ satisfies $\|F_{s,\infty}\| \leq 1$ and $\|F_{s,n} - F_{s,\infty}\| \leq 2^{-n}$, and its fixed-point subspace is the all-future invisible kernel $F_{s,\infty}x = x \iff \forall j \geq 0,\ E_{s,j}x = 0$.
2. **The $n2^{-n}\|x\|$ truncation bound and strong recovery of the current coimage.** For any $x \in C_s$ and prefix length $n \in \mathbb{N}$, the difference between the approximation $\hat{x}_{s,n} := \widehat{\mathcal{P}}_{s,n}(E_{s,0}x, \dots, E_{s,n-1}x)$ based on the first n actual readings and the ergodic average for the full-history limit satisfies

$$\left\|\hat{x}_{s,n} - \mathcal{A}_n(F_{s,\infty}, x - F_{s,\infty}x)\right\| \leq n\,2^{-n}\|x\|, \tag{2.15}$$

   The full-history average $\mathcal{A}_n(F_{s,\infty}, x - F_{s,\infty}x)$ converges strongly to $J_s(\pi_s x)$, at the rate $O((n+1)^{-1})$ under the range source condition of Theorem 2.7(1). Together these give $\lim_{n\to\infty} \hat{x}_{s,n} = J_s(\pi_s x)$.
3. **Typed coimage successor from the shifted prefix.** Drop the reading at step 0 and feed the next n readings $(E_{s,1}x, \dots, E_{s,n}x)$ into the recovery operator $\widehat{\mathcal{P}}_{\sigma(s),n}$ for the next state. The sequence converges strongly in $C_{\sigma(s)}$ to the typed successor representative $J_{\sigma(s)}(\bar{A}_s\pi_s x)$. Its current evaluation converges strongly to the actual successor reading $o_{\sigma(s)}(A_s x)$.

Proof. For each j, $\|\widetilde{E}_{s,j}\| \leq \|E_{s,j}\|/(1+\|E_{s,j}\|) < 1$, and the proof of Theorem 2.7(2) gives $\|I - \widetilde{E}_{s,j}^*\widetilde{E}_{s,j}\| \leq 1$. Since $\sum_{j=0}^{n-1} 2^{-(j+1)} = 1 - 2^{-n}$, express $F_{s,n}$ as a convex combination of operators in the unit ball:

$$F_{s,n} = 2^{-n}I + \sum_{j=0}^{n-1} 2^{-(j+1)}(I - \widetilde{E}_{s,j}^*\widetilde{E}_{s,j}),$$

Hence $\|F_{s,n}\| \leq 2^{-n} + (1-2^{-n}) = 1$. The series tail norm is bounded by $\sum_{j=n}^{\infty} 2^{-(j+1)} = 2^{-n}$, so $F_{s,\infty}$ is well defined and $\|F_{s,n} - F_{s,\infty}\| \leq 2^{-n}$. If $F_{s,\infty}x = x$, then $0 = \mathrm{Re}\langle x, x - F_{s,\infty}x\rangle = \sum_{j=0}^{\infty} 2^{-(j+1)}\|\widetilde{E}_{s,j}x\|^2$. All weights $2^{-(j+1)} > 0$ and scaling factors $(1+\|E_{s,j}\|)^{-1} > 0$ are positive, which forces every $E_{s,j}x = 0$. The converse follows directly, proving part 1.

For part 2, with actual readings $y_j = E_{s,j}x$ we have $d_{s,n}(y) = \sum_{j=0}^{n-1} 2^{-(j+1)}\widetilde{E}_{s,j}^*\widetilde{E}_{s,j}x = x - F_{s,n}x$. Apply (2.11) to $F_{s,n}$ and $F_{s,\infty}$. The first term x cancels, leaving the difference of two

Birkhoff averages of nonexpansive operators. For operators satisfying $\|A\| \le 1, \|B\| \le 1$, induction gives $\|A^j x - B^j x\| \le \|A(A^{j-1}x - B^{j-1}x)\| + \|(A-B)B^{j-1}x\| \le \|A^{j-1}x - B^{j-1}x\| + \|A-B\|\|x\|$, hence $\|A^j x - B^j x\| \le j\|A-B\|\|x\| \le n\|A-B\|\|x\|$ $(\forall j \le n)$. Average over $j = 0, \dots, n$ and substitute $\|F_{s,n} - F_{s,\infty}\| \le 2^{-n}$ to get (2.15). Since $n2^{-n}\|x\| \to 0$ and $\mathcal{A}_n(F_{s,\infty}, x - F_{s,\infty}x) \to P_{\ker(I-F_{s,\infty})^\perp}x = P_{K_s^\perp}x = J_s(\pi_s x)$, the triangle inequality gives strong convergence.

For part 3, observe that $E_{s,j+1}x = E_{\sigma(s),j}(A_s x)$. Feeding the shifted readings into $\widehat{\mathcal{P}}_{\sigma(s),n}$ is the current-state recovery of part 2 for target $A_s x$ at state $\sigma(s)$, with limit $J_{\sigma(s)}(\pi_{\sigma(s)}A_s x) = J_{\sigma(s)}(\bar{A}_s \pi_s x)$. Continuity of $o_{\sigma(s)}$ and the equality $o_{\sigma(s)}J_{\sigma(s)}\pi_{\sigma(s)} = o_{\sigma(s)}$ give the last statement. ■

2.3 Terminating partition refinement, incremental equivalence, and coordinates found at a collision

A finite state system can have several actions and readouts into different types. Its all-future quotient can be computed in finite time from the current structure. Adding a readout then requires only incremental refinement of the existing quotient.

Let $X = \mathrm{Fin}(m)$ be the finite source state set and $\Sigma = \mathrm{Fin}(\ell)$ the operation alphabet. Each symbol $a \in \Sigma$ induces a transition $\alpha_a : X \to X$. A finite word $w = a_1 \cdots a_k \in \Sigma^*$ acts by left-to-right composition, with $\alpha_\varepsilon = \mathrm{id}_X$ and $\alpha_{aw}(x) := \alpha_w(\alpha_a(x))$. Heterogeneous readouts are indexed by $r \in \mathcal{J}_{\mathrm{read}} = \mathrm{Fin}(\rho)$. Each readout $o_r : X \to Y_r$ takes values in its own type Y_r with decidable equality. States $x, y \in X$ are all-future indistinguishable, written $x \equiv_\infty y$, if for every finite operation word $w \in \Sigma^*$ and readout index $r \in \mathcal{J}_{\mathrm{read}}$ we have $o_r(\alpha_w x) = o_r(\alpha_w y)$.

Define the set of initially indistinguishable pairs and the one-step refinement over the whole alphabet by

$$P_0 = \{(x,y) \in X \times X : \forall r \in \mathcal{J}_{\mathrm{read}},\ o_r(x) = o_r(y)\}, \tag{2.16}$$

$$\mathcal{F}(P) = \{(x,y) \in P : \forall a \in \Sigma,\ (\alpha_a x, \alpha_a y) \in P\}. \tag{2.17}$$

Define a terminating refiner with natural-number fuel $f \in \mathbb{N}$. It returns P if $f = 0$ or $\mathcal{F}(P) = P$. Otherwise it continues on $\mathcal{F}(P)$ with fuel $f - 1$, written $\mathrm{settle}(f, P)$. Denote the resulting kernel by $K_* := \mathrm{settle}(|P_0|, P_0)$.

Theorem 2.9 (Terminating minimal all-future model and incremental equivalence for heterogeneous readouts). In the finite-state, multi-operation system above the following construction uses the current readouts and one-step effects.

1. **Termination with endogenous fuel and exactness of the all-future kernel.** K_* is a fixed point of $\mathcal{F}$ $(\mathcal{F}(K_*) = K_*)$, and every $x, y \in X$ satisfies

$$(x,y) \in K_* \iff x \equiv_\infty y. \tag{2.18}$$

   Hence K_* is an equivalence relation. For each $x \in X$ choose the least source address $\mathrm{code}(x) := \min\{y \in X : (x,y) \in K_*\}$ in its class. Define the least-address quotient $M_* := \{x \in X : \mathrm{code}(x) = x\}$, the surjective encoding $\mathrm{encode} : X \to M_*$, the induced action $\bar{\alpha}_a(u) := \mathrm{encode}(\alpha_a u)$, and the induced readout $\bar{o}_r(u) := o_r(u)$. For every finite word $w \in \Sigma^*$, readout index $r \in \mathcal{J}_{\mathrm{read}}$, and state $x \in X$, we have $\bar{\alpha}_w(\mathrm{encode}(x)) = \mathrm{encode}(\alpha_w x)$ and $\bar{o}_r(\bar{\alpha}_w(\mathrm{encode}(x))) = o_r(\alpha_w x)$.

2. **Universal minimality and unique factorization.** Take any representation type $\widetilde{M}$, map $\varphi : X \to \widetilde{M}$, action $\tilde{\alpha}_a : \widetilde{M} \to \widetilde{M}$, and readout $\tilde{o}_r : \widetilde{M} \to Y_r$ satisfying the commutation conditions $\tilde{\alpha}_a \circ \varphi = \varphi \circ \alpha_a$ $(\forall a \in \Sigma)$ and $\tilde{o}_r \circ \varphi = o_r$ $(\forall r \in \mathcal{I}_{\mathrm{read}})$. Then encode factors uniquely through the actual image $\varphi(X)$ of φ. The unique map $g : \varphi(X) \to M_*$ satisfies $g(\varphi(x)) = \mathrm{encode}(x)$ and preserves the induced action and readouts. Each future evaluation $x \mapsto o_r(\alpha_w x)$ also has a unique factorization on M_*.
3. **Incremental refinement equals full recomputation.** Add any readout $o_{\mathrm{new}} : X \to Y_{\mathrm{new}}$ with decidable equality and let $K_*(\mathcal{I}_{\mathrm{read}} \cup \{\mathrm{new}\})$ be the newly computed kernel starting from scratch. Alternatively, start from K_*, keep pairs with equal new readings $P_0' = \{(x, y) \in K_* : o_{\mathrm{new}}(x) = o_{\mathrm{new}}(y)\}$, and refine over the same alphabet with fuel $|P_0'|$, obtaining $K'_{\mathrm{inc}} := \mathrm{settle}(|P_0'|, P_0')$. The two sets satisfy

$$K'_{\mathrm{inc}} = K_*(\mathcal{I}_{\mathrm{read}} \cup \{\mathrm{new}\}). \tag{2.19}$$

Proof. By definition $\mathcal{F}(P) \subseteq P$. Whenever $\mathcal{F}(P) \neq P$, $\mathcal{F}(P)$ is a proper subset of P, and the finite cardinality decreases by at least $|\mathcal{F}(P)| \leq |P| - 1$. Induction on fuel f shows that if $|P| \leq f$, then $\mathrm{settle}(f, P)$ reaches a fixed point of $\mathcal{F}(Q) = Q$ by the time the fuel is exhausted. If fuel runs out, the set itself is empty. Taking $f = |P_0|$ establishes $\mathcal{F}(K_*) = K_*$. If $(x, y) \in K_*$, condition $K_* \subseteq P_0$ gives equality for all readouts under the empty word. By $\mathcal{F}(K_*) = K_*$, every first symbol a gives $(\alpha_a x, \alpha_a y) \in K_*$. Induction on word length using left-to-right composition $\alpha_{aw}(x) = \alpha_w(\alpha_a(x))$ gives $x \equiv_\infty y$. Conversely, if $x \equiv_\infty y$, the empty word gives $(x, y) \in P_0$. For any pair set P containing all all-future equal pairs, $x \equiv_\infty y \implies \forall a,\, \alpha_a x \equiv_\infty \alpha_a y$ implies $(x, y) \in \mathcal{F}(P)$. No iteration drops such a pair, so $(x, y) \in K_*$ and (2.18) follows. Each equivalence class is finite and nonempty, so its least member $\mathrm{code}(x)$ determines it uniquely and $\mathrm{code}(\mathrm{code}(x)) = \mathrm{code}(x)$ holds. Invariance and readout agreement under $(x, \mathrm{code}(x)) \in K_*$ give the commutation identities for every word and readout.

For part 2, one-step action and readout commutation imply $\tilde{\alpha}_w(\varphi(x)) = \varphi(\alpha_w x)$ by induction on word length. If $\varphi(x) = \varphi(y)$, for every w, r we have $o_r(\alpha_w x) = \tilde{o}_r(\tilde{\alpha}_w(\varphi(x))) = \tilde{o}_r(\tilde{\alpha}_w(\varphi(y))) = o_r(\alpha_w y)$ and hence $x \equiv_\infty y$. From (2.18) obtain $\mathrm{encode}(x) = \mathrm{encode}(y)$, making $g(\varphi(x)) := \mathrm{encode}(x)$ well defined. Every point of the image has a source representative, so g is unique. Computing with representatives shows that it commutes with the induced action and readouts. Since encode is surjective, the factorization of each future readout through encode is also unique.

For part 3, K'_{inc} is a fixed point of $\mathcal{F}$ contained in $P_0' \subseteq K_* \cap \{(x, y) : o_{\mathrm{new}}(x) = o_{\mathrm{new}}(y)\}$. Thus $(x, y) \in K'_{\mathrm{inc}}$ gives equal old and new readings at the empty word. Induction using the fixed point gives equality at every future word of the expanded system. Conversely, if (x, y) has equal expanded readings at every future word, it has equal old readings as well, so (2.18) places it in K_*. At the empty word it satisfies $o_{\mathrm{new}}(x) = o_{\mathrm{new}}(y)$, so it lies in P_0'. Expanded all-future equivalence is closed under every α_a, so each round of settle keeps the pair, giving $(x, y) \in K'_{\mathrm{inc}}$. Apply (2.18) to the expanded system to prove (2.19). ■

For very large or evolving state spaces, enumerating the full transition table can be impractical. The action defect in Theorem 2.2 and incremental refinement in Theorem 2.9 give on-demand coordinate refinement after a collision and continuation from a complete residual.

Theorem 2.10 (Locating a missing source coordinate by a future collision and continuing byte for byte from the full residual). Suppose the state space is in bijection with the product $S \cong (\prod_{c\in\mathcal{B}} V_c) \times R_{\mathrm{res}}$ of a finite addressable source-coordinate table $\mathcal{B}$, whose coordinates have decidable equality, and a shared packed residual R_{res}. Fix two completed branches b_1, b_2 from a common source root, with the same packed residual component $r \in R_{\mathrm{res}}$. Let $\Pi \subseteq \mathcal{B}$ be the selected coordinate subtable and $V_\Pi : S \to \prod_{c\in\Pi} V_c$ its projection view.

1. **A collision exposes a missing source coordinate.** Suppose the branches have the same old view $V_\Pi(b_1) = V_\Pi(b_2)$ before execution but different actual successor views $V_\Pi(b_1^+) \neq V_\Pi(b_2^+)$ after one deterministic step. No function of V_Π predicts the successor view. A finite basis scan of the pre-execution complement $\mathcal{B} \setminus \Pi$ finds a missing coordinate c_* satisfying $c_*(b_1) \neq c_*(b_2)$, without a table of future transitions. The extended plan $\Pi' = \Pi \cup \{c_*\}$ preserves every old distinction ($V_{\Pi'}(x) = V_{\Pi'}(y) \implies V_\Pi(x) = V_\Pi(y)$) and separates (b_1, b_2).
2. **Lossless execution with the full packed residual.** Encode the source-derived coordinate plan Π' jointly with the complementary coordinates and shared packed residual R_{res}. Decoding and re-encoding form a bijection. One step and any finite autonomous trajectory then agree with full-state execution in each independent readout, the six whole-ledger writebacks, and the canonical successor.

Proof. If F satisfied $F(V_\Pi(b_i)) = V_\Pi(b_i^+)$, equality $V_\Pi(b_1) = V_\Pi(b_2)$ would force equal successor views. Source evolution is a deterministic map on S. Since b_1, b_2 share the packed residual $r \in R_{\mathrm{res}}$, distinct successors imply distinct $\prod_{c\in\mathcal{B}} V_c$ coordinates. The coordinates in Π agree, so the finite complement $\mathcal{B} \setminus \Pi$ contains a coordinate c_* satisfying $c_*(b_1) \neq c_*(b_2)$. A basis-order scan finds the first one. Projecting onto the Π subtuple preserves every old distinction, and the new coordinate c_* gives $V_{\Pi'}(b_1) \neq V_{\Pi'}(b_2)$. The executable representation includes the Π' view, $\mathcal{B} \setminus \Pi'$ complement, and R_{res} residual, so decoding and re-encoding are inverses. Induction on the number of steps gives equality on arbitrary finite autonomous trajectories. ■

In the MaleCNS simulation, snapshots at visits 55 and 56 agree byte for byte in the remaining 51 fields when the controller gate field and visit counter are omitted. One step under the fixed source law separates 26 non-gate fields. Scanning the pre-execution records locates two missing coordinates, gate field 35 and the visit counter. The full record of visit 56 compresses losslessly from 27,538,174 bytes to 17,170,139 bytes, a reduction of 37.65%. Resuming $56 \to 57 \to 58$ from this packet reproduces all 58 originals and events in each transition, as well as the corresponding microtrajectories, byte for byte.

2.4 A finite three-clock carrier, eight-byte replay keys, and a cofinal all-future fiber for closed syntax

In an execution environment with admission guards, an operation word can advance the state or return a deterministic error when the current occurrence or physical width fails a guard. Distinctions at the error boundary require including those outcomes in the state model. Consider a guarded parser with a private state and three native guards: the visit clock v, the neural tick clock t_{NN}, and the physical integration clock t_{phys}. The byte width w of the physical counter internally

bounds both success and error trajectories. Successful states admit a fixed eight-byte key and can be replayed from an anchor.

Theorem 2.11 (The finite three-clock carrier, universal minimality of $\mathbb{N} \oplus \mathrm{String}$, and eight-byte success-key replay). Fix a source boundary, installed materials, and an anchor snapshot s_{anc}. For example, take a sealed anchor with starting visit count $v_0 = 4$, neural tick $t_{\mathrm{NN},0} = 400$, and physical integration count $t_{\mathrm{phys},0} = 4000$, with a physical counter of w bytes and in particular $w = 8$. Consider the four-letter alphabet Σ_4 consisting of the autonomous step and three directed operations, and the 69 heterogeneous state readouts exposed by the system. Readout 64 returns the successful visit count or the error string.

1. **A common finite carrier for successes and errors.** Any successful mixed word agrees with the pure autonomous prefix of the same length. One successful step increments the visit count by one ($v' = v+1$). An ordinary step adds 100 neural ticks and 1000 physical integration steps. The two gate operations at visits $v = 55, 58$ leave both clocks unchanged. The physical integration count stays within the unsigned w-byte bound $t_{\mathrm{phys}} < 256^w$. An execution error first occurs at an illegal effect after a bounded successful prefix. Its string is absorbing under all subsequent letters. The resulting finite addresses and types are

$$\mathrm{Address} = \mathrm{Fin}(B_{\mathrm{pre}}) \oplus (\mathrm{Fin}(B_{\mathrm{pre}}) \times \mathrm{Fin}(4)), \qquad B_{\mathrm{pre}} = \lfloor 256^w/1000 \rfloor + 3, \tag{2.20}$$

   Executing bounded prefixes and first-step effects from the anchor surjects onto all reachable successes and errors. The construction requires no supplied state or error table.
2. **Minimal all-future completeness of $\mathrm{Key} = \mathbb{N} \oplus \mathrm{String}$.** On the reachable carrier define the encodings $\mathrm{key}(\mathrm{ok}\, s) = \mathrm{inl}(s.\mathrm{visit})$ and $\mathrm{key}(\mathrm{err}\, \eta) = \mathrm{inr}(\eta)$. Then key is injective. The on-demand decoder $\mathrm{decode}(\mathrm{inl}\, v)$ replays $v - v_0$ autonomous steps from the anchor to reconstruct the full snapshot and the physical, learning, and event trajectories. For every pair of reachable states x, y,

$$\mathrm{key}(x) = \mathrm{key}(y) \iff \forall u \in \Sigma_4^*,\ \forall r \in \mathrm{Fin}(69),\ o_r(\alpha_u x) = o_r(\alpha_u y). \tag{2.21}$$

   Any representation preserving the 69 original readouts must preserve every distinction made by key.
3. **Eight-byte success keys and the six whole-ledger fields.** At the sealed anchor with $w = 8$, a successful visit address $v < 2^{56}$ fits in the eight-byte little-endian integer $256v$, with least significant byte tag 0. An error key uses tag 1 followed by the variable-length canonical UTF-8 error string. Recovery consults the cache of errors that actually occurred. The parser rejects out-of-range and noncanonical encodings and unregistered tags. Serialized strings agree exactly when every one of the 69 readouts agrees after every word over the four-letter alphabet. For a generated successful record, replay from the anchor recovers all readouts and the six whole-ledger destinations.

Proof. Under the source admission law the mixed step returns success only if its selected action agrees with the native action at the current visit count v. It then calls the same deterministic updater and increments v by 1. Induction on word length shows that any successful word of length

k generates the same snapshot as k pure autonomous steps, with visit count $v_0 + k$. The physical integration count $< 256^w$ bounds the number of steps that advance this clock by $\lfloor 256^w/1000 \rfloor$. Since the visit count is strictly increasing, each of the two operations with zero physical-clock advance can occur at most once. The total number of steps is strictly below $B_{\mathrm{pre}} = \lfloor 256^w/1000 \rfloor + 3$. Once a step returns $\mathrm{err}\,\eta$, monadic bind passes through $\mathrm{err}\,\eta$ for every later letter. Each error state therefore comes from a successful prefix of length $< B_{\mathrm{pre}}$ and one of the four first-error letters, proving the surjectivity in (2.20). Two successful states with visit count v arise from the same number $v - v_0$ of deterministic autonomous steps and thus have equal snapshots. This makes key injective with replay as a left inverse, proving (2.21) from left to right. Conversely, if all future outputs agree, or even if the 69 readouts agree on the empty word, readout 64 gives the successful visit count or the exact error string and forces $\mathrm{key}(x) = \mathrm{key}(y)$. A representation preserving these readouts must therefore separate different key values. Finally, $v < 2^{56}$ ensures $256v < 2^{64}$ fits in eight bytes with low byte `0`, disjoint from error packets beginning with `1`. The canonical round trip transports (2.21) to the bytes. Successful replay checks the original visit address. An error key looks up a bounded-prefix address at which that string actually occurred. For a registered success, replay agrees on the full snapshot and the execution receipt, hence on the original readouts and ledger. ■

The previous encoding replays a state from its visit address. In arithmetic queries admitted in growing packets, the same all-future relation reads executed requests while preserving each execution record by position. Write $\mathcal{W}_j$ for the formal word module of admitted packet j. The source run generates paired lifts and language transports $S_j : \mathcal{W}_j \to \mathcal{W}_{j+1}$. They define a shift S on the finitely supported direct sum $\mathcal{W} = \bigoplus_j \mathcal{W}_j$. In packet j, the current environment ρ_j and actual increment η_j give the prior value and effect $D_j(u) = (\mathrm{ev}_{\rho_j}\, u_j, \mathrm{eff}_{\rho_j,\eta_j}\, u_j)$ of a word. The effect is evaluation in the updated environment minus evaluation in the old one. The sequence of registered admissions advances and retains every intervening step. Source-Native Process Algebra, Theorems 9.9–9.12, gives compatibility of the paired lifts, language transports, and readouts [16].

Proposition 2.12 (Cofinal observation, the next request, and the all-future zero fiber of closed-syntax intervals). Form the all-future quotient by the common kernel of all $D_j S^n$ in the run above. Read the type and value of the full state together. The execution prefixes come from one run. Each station's joint observation records the prior value and effect, recovering the complete formal word before reading the next frame, including its old and newborn records. At an already-paid computation step the macro-increment of the original environment is zero. At a birth it is the actual source effect. Subtracting the residual r obtained by actual inversion from state x changes the linear window readout o_{win} by $-o_{\mathrm{win}}(r)$. Together with the corrected next readout this determines the new joint fiber.

Fix the arithmetic requests actually generated in that source run and their execution traces. Let the receipt word of the closed request e_q at position q be $\kappa_q(\delta_{e_q} - \delta_{\mathrm{const}(v_q)})$, where v_q is the actual evaluation and $\kappa_q \in \mathbb{Z}$ the registered coefficient. Transport the receipt sum of any consecutive interval of actual requests to one successor packet. The sum lies in the all-future zero fiber. Its absolute address in the next packet retains the request numbers, coefficients, chosen decoders, and complete execution segments.

Proof. The dependent pair $(\mathcal{X}, m)$ retains both carrier and value. The erasure identity of a

full occurrence aligns the environment and expression. The syntax budget and causal depth then distinguish frames of execution. The prefix observations restrict compatibly at each station, so two completion words have the same image if and only if all stationwise joint readouts agree. The source run keeps the environment address at a paid computation step and applies the generated effect at a birth. Subtracting the inverse residual gives $o_{\mathrm{win}}(x-r)-o_{\mathrm{win}}(x)=-o_{\mathrm{win}}(r)$ by linearity. Theorem 2.2(3) then reads the one-step joint fiber.

In an arbitrary decoding environment ρ, recursion on closed syntax gives $\mathrm{ev}_{\rho}(e_{q})=v_{q}$. Therefore each receipt word evaluates to zero in every environment, and the same holds for their finite sum. Paired lifts and language transports preserve this relation at every step. For all ρ,η, a semantically zero word u satisfies $\mathrm{ev}_{\rho}(u)=0$ and $\mathrm{ev}_{\rho+\eta}(u)-\mathrm{ev}_{\rho}(u)=0$. The lift evaluates the two branches, and the transport changes their types. The receiver square of the actual source action gives zero in every environment of the next packet. Induction on future word length gives zero on this receipt sum for every $D_{j}S^{n}$. The positions in the paid trace embed in order into the next packet. Their absolute addresses can be read back even when their linear sum is zero.

For example, the receipt $\delta_{2+3}-\delta_{5}$ for the closed term $2{+}3$ evaluates to zero in every environment. The open term $x-5$ evaluates to zero at $x=5$ and to one at $x=6$. The all-future zero fiber of the closed receipt gives semantic equivalence, while each execution record retains its own address. ■

3 The sharp $a+1$-frame observation window, rational decoding, and single-prime saturation

All-future evaluation may involve infinitely many time steps. An instrument must collect each window in finitely many steps. For the spatial, mass, and clock observer in Example 1.2(a), what is the observability index of the observable quotient [22; 45]? Can a nonlinear map use fewer frames? The shortest window has exactly $a+1$ frames. The upper bound comes with closed-form rational decoding. The lower bound uses Corollary 2.5 in Section 2.1 to rule out every update map on the full carrier that reads fewer than $a+1$ frames, without assumptions of linearity, continuity, or measurability. An explicit two-frame collision illustrates the lower bound. We then give a complementary characterization of the shortest window on an infinite free source, where a single prime gives the saturation length $\mathrm{cut}(p)+1$.

3.1 What the original observer reads in space, mass, and clock

Definition 3.1 (The full spatial, mass, and clock carrier and its original action). Define the complex Hilbert space

$$G=\ell^{2}(\mathbb{N};\mathbb{C})\oplus_{2}\mathbb{C}\oplus_{2}\mathbb{C},\qquad v=(h,m,c),\tag{3.1}$$

with spatial coefficients, mass, and clock as its three components and squared norm $\|v\|^2 = \sum_{k=0}^{\infty} |h_k|^2 + |m|^2 + |c|^2$. The canonical embedding of a finitely supported source word w is

$$\iota(w) = \left(w, \sum_k w_k, \sum_k (k+1) w_k \right). \tag{3.2}$$

In the completed space G, m, c are independent coordinates. Equation (3.2) specifies how they are generated on the finitely supported source subspace, where both sums are finite. For general ℓ^2 vectors the two values are supplied as independent coordinates. Define the original one-step time action A and the material copy action B_a at integer scale $a \geq 1$ by

$$A(h, m, c) = (Sh, m, c + m), \qquad B_a(h, m, c) = (U_a h, m, ac), \tag{3.3}$$

where the right shift satisfies $Se_k = e_{k+1}$ and the dilation satisfies $U_a e_k = e_{a(k+1)-1}$.

Fix the original population bound $N \in \mathbb{N}$, a nonunit copy scale $a \in \{2, \dots, N+1\}$, and $s \geq 0$ additional population steps. Let $n = N+s+1$ be the number of columns in one frame, so $n \geq a \geq 2$. The first n source basis vectors are $u_k := \iota(e_k) = (e_k, 1, k+1)$ for $0 \leq k < n$. In each frame the original observer reads all n probe vectors $g_j := B_a u_j$.

$$g_j = (e_{a(j+1)-1},\ 1,\ a(j+1)), \qquad 0 \leq j < n. \tag{3.4}$$

The single-frame observation $o(v) = \sum_{k=0}^{n-1} x_k u_k$ is the unique element of $\operatorname{span}\{u_0, \dots, u_{n-1}\} \subset G$ whose coefficients $x = (x_0, \dots, x_{n-1}) \in \mathbb{C}^n$ solve

$$\sum_{k=0}^{n-1} \Gamma_{jk} x_k = r_j(v), \qquad \Gamma_{jk} = \langle g_j, g_k \rangle = \delta_{jk} + 1 + a^2 (j+1)(k+1), \qquad r_j(v) = \langle g_j, v \rangle \tag{3.5}$$

uniquely. Since $B_a o(v) = \sum_{k=0}^{n-1} x_k g_k$, these equations match every probe inner product $\langle g_j, B_a o(v) \rangle = \langle g_j, v \rangle$. The physical Gram matrix Γ is positive definite. Its quadratic form is $\langle x, \Gamma x \rangle = \sum_{j<n} |x_j|^2 + \left|\sum_{j<n} x_j\right|^2 + a^2 \left|\sum_{j<n} (j+1) x_j\right|^2$. Invertible linear maps thus relate the single-frame observation $o(v)$, the coefficient vector x, and the response vector $r(v)$.

Example 3.2 (The full response when spatial samples vanish). Take $a = n = 2$ and the unit source word at position 2, $v = \iota(e_2) = (e_2, 1, 3)$. The probes read spatial indices 1 and 3, yielding $(h_1, h_3) = (0, 0)$. Equation (3.5) gives the physical Gram matrix, full response, and representation coefficients:

$$\Gamma = \begin{pmatrix} 6 & 9 \\ 9 & 18 \end{pmatrix}, \qquad r(v) = \begin{pmatrix} 7 \\ 13 \end{pmatrix}, \qquad x = \Gamma^{-1} r(v) = \begin{pmatrix} 1/3 \\ 5/9 \end{pmatrix}. \tag{3.6}$$

Two further steps under the original action give $r(Av) = (9, 18)$ and $r(A^2 v) = (11, 21)$. Treating the spatial samples as the entire initial readout would report a zero vector at time 0. The independent mass and clock also provide the reference values used to recover the subsequent spatial components.

3.2 Explicit decoding from $a+1$ frames recovers mass, clock, and space

Write the response in column j of frame t as $r_j(t) = \langle g_j, A^t v\rangle$. With spatial probe index $i_j = a(j+1)-1$, expanding $A^t(h,m,c) = (S^t h, m, c+tm)$ gives

$$r_j(t) = \mathbf{1}_{t\le i_j} h_{i_j - t} + m + a(j+1)(c+tm). \tag{3.7}$$

For $t > i_j$, the shift moves the signal past i_j, so the spatial term is exactly zero.

Theorem 3.3 (Explicit linear decoding from all columns in $a+1$ frames). The responses in all n columns of the consecutive prefix $t = 0, \dots, a$, comprising $a+1$ frames, determine the independent mass m, the independent clock c, and the first na spatial coordinates $h_0, \dots, h_{na-1}$ by closed-form linear formulas.

$$m = \frac{r_1(a) - r_0(0) - r_0(a)}{a^2 - 1}, \qquad c = \frac{r_0(a) - m}{a} - am; \tag{3.8}$$

For every $0 \le k < na$, take block $j = \lfloor k/a \rfloor \in \{0, \dots, n-1\}$ and frame $t = a(j+1) - 1 - k \in \{0, \dots, a-1\}$. Then

$$h_k = r_j(t) - m - a(j+1)(c+tm). \tag{3.9}$$

Let $T(P_a v)$ denote the linear reconstruction obtained by setting the spatial coordinates of index $\ge na$ to zero, where $P_a v = (o(v), \dots, o(A^a v))$. Then $TP_a v$ is the canonical orthogonal representative in $K_\infty^\perp$ of the all-future quotient in Theorem 2.4, and the full all-future invisible kernel is

$$K_\infty = \{(h,0,0) \in G : \forall k < na,\ h_k = 0\}. \tag{3.10}$$

In particular, $P_a v = P_a u$ if and only if the observations of the two states agree at every future time $t \ge 0$. Theorem 3.4 proves that this is the shortest window for the observable quotient.

Proof. At frame $t = a$, the first column has index $i_0 = a - 1 < a$, so the spatial term of $r_0(a)$ vanishes and $r_0(a) = m + a(c + am)$. The second column at $t = a$ reads spatial index $i_1 - a = (2a-1) - a = a - 1$, the same source index as the first column at $t = 0$, $i_0 - 0 = a - 1$. Substitute the three responses from (3.7) and subtract:

$$r_1(a) - r_0(0) - r_0(a) = (h_{a-1} + m + 2a(c+am)) - (h_{a-1} + m + ac) - (m + a(c+am)) = (a^2-1)m.$$

Since $a \ge 2$, $a^2 - 1 \ge 3 \ne 0$. Dividing by $a^2 - 1$ yields the first formula in (3.8), and substitution into $r_0(a)$ gives the second. Next partition $\{0, \dots, na-1\}$ into n disjoint blocks $\{aj, \dots, a(j+1)-1\}$ of length a. Each index k in a block has a unique frame $t = i_j - k \in \{0, \dots, a-1\}$. At that frame $t \le i_j$, and rearranging (3.7) yields (3.9).

If $v \in K_\infty$, all its responses in the first $a+1$ frames vanish. Equations (3.8)–(3.9) then give $m = c = 0$ and zero in the first na spatial coordinates. Conversely, every vector of the form (3.10) retains spatial support in $\{na, na+1, \dots\}$ under every right shift. The largest spatial probe index is $na-1$, and mass and clock stay zero. Thus $r_j(t) = 0$ for all $t \ge 0$, proving (3.10). As TP_a preserves the first na spatial entries and (m,c) and zeros the rest, it is the orthogonal projection onto $K_\infty^\perp$. ■

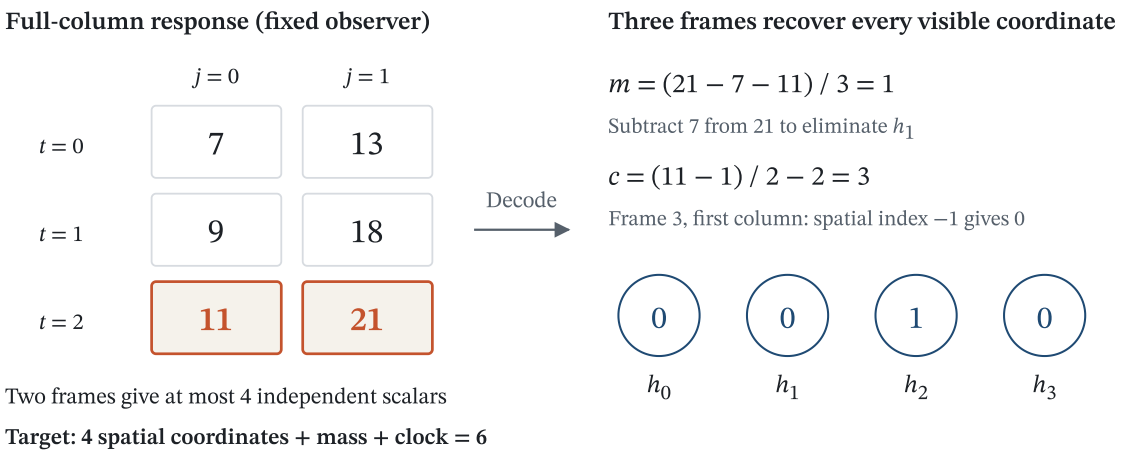


Figure 2. Three frames recover four spatial coordinates, mass, and clock. For $a = n = 2$, the first two frames provide at most four independent complex scalars. The second column of the third frame, shown in vermilion, and the first column of the first frame share the term h_1. Subtraction gives one equation in mass and clock. The first column of the third frame has no spatial term. The two equations give $m = 1, c = 3$, after which each of $h_0, \dots, h_3$ is recovered.

3.3 Every update map on the full carrier needs the last frame

The $a+1$-frame linear decoder gives an upper bound. Could a highly nonlinear neural network or a piecewise lookup map determine the next frame from a frames? Corollary 2.5 in Section 2.1 and the dimension of $K_\infty^\perp$ impose the same lower bound on every such map.

Theorem 3.4 (Shortest-window lower bound for arbitrary maps on the full carrier with the fixed observer). On the entire carrier G of Definition 3.1, let the prefix order be $\ell \in \mathbb{N}$. If there is a map, without any assumption of linearity, continuity, smoothness, boundedness, or measurability, satisfying

$$F : G^{\ell+1} \longrightarrow G^{\ell+1}, \qquad \forall v \in G, \quad F(P_\ell v) = P_\ell(Av), \tag{3.11}$$

then $\ell \geq a$. Together with Theorem 3.3, this gives exactly $a+1$ frames as the shortest prefix admitting an autonomous update.

Proof. Suppose there is a map satisfying (3.11). Taking $v = 0$ yields $F(0) = 0$. Since F is single-valued, $A(\ker P_\ell) \subseteq \ker P_\ell$. By maximal invariance in Theorem 2.4(1) and Corollary 2.5, $\ker P_\ell = K_\infty$. Theorem 3.3 identifies the finite-dimensional observable subspace $V = K_\infty^\perp \cong \mathbb{C}^{na+2}$, spanned by the first na spatial coordinates and the independent mass and clock. Since $\ker(P_\ell|_V) = V \cap \ker P_\ell = V \cap K_\infty = \{0\}$, the complex-linear restriction $P_\ell|_V : V \to (\operatorname{ran} o)^{\ell+1}$ is injective. The single-frame range $\operatorname{ran} o$ has complex dimension n, so the codomain has complex dimension $(\ell+1)n$. Injectivity therefore implies

$$na + 2 \leq (\ell+1)n. \tag{3.12}$$

If $\ell < a$, then $\ell+1 \leq a$ and the right-hand side is at most an, giving the contradiction $na+2 \leq an$. Hence any F satisfying (3.11) needs at least $\ell+1 \geq a+1$ frames. The reconstruction of Theorem 3.3 attains the bound via $F(y) := P_a(ATy)$. ■

An explicit collision with two zero frames ($a = n = 2,\ \ell = 1$). Besides the dimension argument, a concrete hidden state yields two zero frames followed by a nonzero third frame. With $a = n = 2$ and spatial probe indices $i_0 = 1, i_1 = 3$, compare $v_* = ((-3, -1, -5, -1, 0, 0, ...),\ 1,\ 0) \in G$ with $0 \in G$. Equation (3.7) gives the following responses.

- At $t = 0$, $r_0(0) = h_1 + 1 + 0 = -1 + 1 = 0$ and $r_1(0) = h_3 + 1 + 0 = -1 + 1 = 0$.
- At $t = 1$, $r_0(1) = h_0 + 1 + 2 = -3 + 3 = 0$ and $r_1(1) = h_2 + 1 + 4 = -5 + 5 = 0$. The first two frames therefore agree, $P_1(v_*) = P_1(0) = (0, 0)$.
- At $t = 2$, which is frame 1 for Av_*, $r_0(2) = 0 + 1 + 4 = 5$ and $r_1(2) = h_1 + 1 + 8 = -1 + 9 = 8$. Thus $P_1(Av_*) \neq P_1(A0)$.

Negative spatial impulses cancel the mass and clock contribution in the first two frames. At $t = 2$ the first probe's source index goes before the origin, so its spatial term vanishes and the response $(5, 8) \neq (0, 0)$ is exposed. No single-valued F, however nonlinear, can take the same two zero frames to both a zero and a nonzero output. Theorem 3.4 fixes the all-column observer, a consecutive prefix starting at frame 0, and the full carrier G. A different instrument that directly reads (m, c), or a subspace known to satisfy $m = c = 0$, changes the shortest-window problem. For the stated observer on G, frame $a + 1$ is necessary.

3.4 Two moments suffice for rational encoding and reconstruction

Direct inversion of the dense $n \times n$ matrix Γ produces complicated fractions in exact rational computation. Its low-rank form $\Gamma = I + \mathbf{1}\mathbf{1}^\top + a^2\mathbf{j}\mathbf{j}^\top$, where $\mathbf{j}_k = k + 1$, reduces encoding and recovery of all participants' posterior weights to two sums with integer weights and one positive integer denominator.

For a rational response vector $b = r(v) \in \mathbb{Q}^n$, let the fixed integer power sums be $S_1 = \sum_{j=0}^{n-1}(j + 1)$ and $S_2 = \sum_{j=0}^{n-1}(j + 1)^2$. Let the data moments be $B_0 = \sum_{j=0}^{n-1} b_j$ and $B_1 = \sum_{j=0}^{n-1}(j + 1)b_j$. The denominator and moment solution are

$$\Delta = (n + 1)(1 + a^2 S_2) - a^2 S_1^2 > 0, \tag{3.13}$$

$$M_0 = \frac{(1 + a^2 S_2)B_0 - a^2 S_1 B_1}{\Delta}, \qquad M_1 = \frac{(n + 1)B_1 - S_1 B_0}{\Delta}, \qquad x_j = b_j - M_0 - a^2(j + 1)M_1. \tag{3.14}$$

By Cauchy–Schwarz, $S_1^2 \leq nS_2$. Hence $\Delta = (n + 1) + a^2[(n + 1)S_2 - S_1^2] \geq n + 1 + a^2 S_2 > 0$. Summing the last formula of (3.14) unweighted and with weights $(j + 1)$ gives $\sum_{j<n} x_j = M_0$ and $\sum_{j<n}(j + 1)x_j = M_1$, respectively. Substitution into (3.5) proves $\Gamma x = b$.

Theorem 3.5 (Rational window transmission and recovery of every posterior weight). Fix the population bound N and a nonunit copy scale $a \in \{2, \dots, N + 1\}$. The sender applies (3.14) to each response frame in the consecutive $a+1$-frame window of the original rational state and transmits an $(a + 1) \times n$ rational coefficient matrix. From that matrix, the population parameters, and the copy scale a, the receiver recovers the responses $r_j(t)$ by left multiplication by Γ and then recovers all na visible spatial coordinates and the independent mass and clock by (3.8)–(3.9). In particular, let $v_i := A\iota(e_i) = (e_{i+1}, 1, i + 2)$ be participant i's source point after one

step, and suppose the sender transmits the original population's conditional-mean vector

$$v_y = \sum_{i=0}^{N} p_N(i \mid y)(e_{i+1}, 1, i+2), \tag{3.15}$$

at the maximal copy scale $a = N+1$. The receiver reads every rational posterior weight $p_N(i \mid y)$, $i \in \{0, \dots, N\}$, directly from the reconstructed spatial coordinates. The same protocol recovers all-zero weights from the zero object for an absent readout.

Proof. Formula (3.14) and left multiplication by Γ are inverse maps over the rationals. Theorem 3.3 then recovers the first na spatial coordinates and (m, c). In (3.15), each participant $i \in \{0, \dots, N\}$ occupies a distinct spatial basis vector e_{i+1}, so their contributions cannot overlap. At the maximal copy scale $a = N+1$, the recoverable range has $na \geq (N+1)^2 \geq N+2$ coordinates, since a nonunit scale requires $N \geq 1$. All indices $1, \dots, N+1$ lie in the recovered window, and coordinate $i+1$ equals $p_N(i \mid y)$. By linearity, zero input gives all-zero weights. ■

3.5 The shortest $\mathrm{cut}(p)+1$-frame single-prime saturation window and closure of function slices

Theorem 3.4 uses the dimension of a finite-dimensional orthogonal complement. On an infinite free $\mathbb{Z}$-module, finite saturation in time can give a shortest finite window. In the discrete shift system on an infinite free $\mathbb{Z}$-module with prime-indexed counting channels [16], each prime p defines a step readout by the inequality $2(s+2) \geq p$. Its finite window is determined by the first step at which the readout has saturated to mass.

Consider the free $\mathbb{Z}$-module $C_{\mathrm{src}} = \bigoplus_{s \in \mathbb{N}} \mathbb{Z}$ with state basis $\{\delta_s\}_{s \in \mathbb{N}}$. The native one-step shift is $A_{\mathrm{nat}}(\delta_s) = \delta_{s+1}$, and the integer mass functional is $m_{\mathbb{Z}}(\delta_s) = 1$ $(\forall s \in \mathbb{N})$. For each prime p, set $\mathrm{cut}(p) := \lfloor \max(0, p-3)/2 \rfloor \in \mathbb{N}$. For example, $p = 2, 3$ give $\mathrm{cut}(p) = 0$, $p = 5$ gives 1, and $p = 7$ gives 2. On basis vectors the integer readout $e_p : C_{\mathrm{src}} \to \mathbb{Z}$ is $e_p(\delta_s) := \mathbf{1}_{2(s+2) \geq p} = \mathbf{1}_{s \geq \mathrm{cut}(p)}$. Its value at step t is therefore $e_p(A_{\mathrm{nat}}^t \delta_s) = \mathbf{1}_{s+t \geq \mathrm{cut}(p)}$.

Theorem 3.6 (The shortest $\mathrm{cut}(p)+1$-frame single-prime saturation window and explicit shorter-window counterexamples). On the infinite free carrier $C_{\mathrm{src}} = \bigoplus_{s \in \mathbb{N}} \mathbb{Z}$, for every prime p the following hold.

1. **Monic saturation recurrence and the all-future kernel.** For $t \geq \mathrm{cut}(p)$, the stage readout $e_p \circ A_{\mathrm{nat}}^t$ has saturated to the mass functional $m_{\mathbb{Z}}$ on every basis vector. It therefore satisfies the finite recurrence $e_p \circ A_{\mathrm{nat}}^{\mathrm{cut}(p)+1} = e_p \circ A_{\mathrm{nat}}^{\mathrm{cut}(p)}$, whose final coefficient is 1. This induces an autonomous linear update on the prefix window $P_{p,\mathrm{cut}(p)} = (e_p, e_p A_{\mathrm{nat}}, \dots, e_p A_{\mathrm{nat}}^{\mathrm{cut}(p)}) : C_{\mathrm{src}} \to \mathbb{Z}^{\mathrm{cut}(p)+1}$. Its kernel is the all-future evaluation kernel:

$$\ker P_{p,\mathrm{cut}(p)} = \bigcap_{t=0}^{\infty} \ker(e_p \circ A_{\mathrm{nat}}^t), \tag{3.16}$$

   and the actual window image gives an injective realization of the single-prime all-future quotient that intertwines the actions.
2. **An explicit escaping source word for every shorter window.** For every strictly smaller prefix bound $b < \mathrm{cut}(p)$, which requires $\mathrm{cut}(p) \geq 1$, or equivalently an odd prime $p \geq 5$, the

basis vector $w_{p,b} := \delta_{\mathrm{cut}(p)-b-1} \in C_{\mathrm{src}}$ has readout zero at all stages $0, \dots, b$ and readout 1 at step $b+1$.

$$P_{p,b}(w_{p,b}) = (0, \dots, 0), \qquad (e_p \circ A_{\mathrm{nat}}^{b+1})(w_{p,b}) = 1 \neq 0, \tag{3.17}$$

Hence $\ker P_{p,b} \neq \bigcap_{t\geq 0} \ker(e_p \circ A_{\mathrm{nat}}^t)$. By Corollary 2.5, a prefix shorter than $\mathrm{cut}(p)+1$ frames cannot support an autonomous update for this prime channel on all of C_{src}.

Proof. If $t \geq \mathrm{cut}(p)$, then $s + t \geq t \geq \mathrm{cut}(p)$ for every $s \in \mathbb{N}$, so $e_p(A_{\mathrm{nat}}^t \delta_s) = 1 = m_{\mathbb{Z}}(\delta_s)$. By $\mathbb{Z}$-linear extension, $e_p \circ A_{\mathrm{nat}}^t = m_{\mathbb{Z}}$. Taking $t = \mathrm{cut}(p)$ and $\mathrm{cut}(p)+1$ yields $e_p A_{\mathrm{nat}}^{\mathrm{cut}(p)+1} = e_p A_{\mathrm{nat}}^{\mathrm{cut}(p)}$. Thus after one step the window vector $(y_0, \dots, y_{\mathrm{cut}(p)})$ becomes $(y_1, \dots, y_{\mathrm{cut}(p)}, y_{\mathrm{cut}(p)})$. If its first $\mathrm{cut}(p)+1$ readouts vanish, induction makes all later readouts vanish, proving (3.16) and the quotient isomorphism. Conversely, for $b < \mathrm{cut}(p)$, the index $s_* = \mathrm{cut}(p) - b - 1 \in \mathbb{N}$ becomes $s_* + j = \mathrm{cut}(p) - b - 1 + j < \mathrm{cut}(p)$ after j steps for $0 \leq j \leq b$, so $e_p(A_{\mathrm{nat}}^j w_{p,b}) = 0$. At $j = b+1$, $s_* + (b+1) = \mathrm{cut}(p)$ and $e_p(A_{\mathrm{nat}}^{b+1} w_{p,b}) = 1 \neq 0$, proving (3.17). ■

Remark (All-future closure of finitely supported slices of the function ring). Let S be the state set, $\mathcal{G}$ the generator set, and B a commutative coefficient ring. On the finitely supported function module $(S \times \mathcal{G}) \to_0 B$, suppose one step moves only the state component $s \mapsto \sigma(s)$ and preserves the generator $g \in \mathcal{G}$ and the coefficient [16]. The current-slice readout in $\mathcal{G} \to_0 B$ induces an all-future quotient that identifies states exactly when all their successor slices agree. The coimage projection recovers the entire current function slice without loss.

4 Updating the shortest window, explicit precision budgets, and three errors on changing spaces

Decoding a shortest window recovers a single state. In a continuously evolving system, the original population bound N grows as new records arrive, measurement introduces a continuous perturbation η, and successive frames may lie in L^2 spaces with different empirical measures. We now propagate the shortest window of Section 3. This gives an update compatible with the original action, an exact expression for how noise changes over one step, a threshold for recovering discrete support, and a three-term orthogonal error decomposition across changing empirical spaces.

4.1 Native update and sampling for an expanded window

Let P^N and T_N be the shortest-window operator and its linear reconstruction at population bound N and maximal copy scale $a = N+1$. The native update applies A from (3.3) to the complete state and embeds the result in the coordinate range of the next population $N+1$. Define the update between the two windows by

$$\mathcal{U}_N(y) := P^{N+1}(AT_N y), \tag{4.1}$$

where P^{N+1} uses the all-column observer for the actual next population.

Theorem 4.1 (The shortest window under the native update). For every observable object with zero tail, that is, $T_N P^N v = v$, the window update commutes with the native action:

$$\mathcal{U}_N(P^N v) = P^{N+1}(Av), \qquad T_{N+1}\mathcal{U}_N(P^N v) = Av. \tag{4.2}$$

For any $v \in G$, write its current invisible tail as $D_N v = v - T_N P^N v$. The full successor decomposes as

$$Av = T_{N+1}\mathcal{U}_N(P^N v) + AD_N v. \tag{4.3}$$

In particular, the objects generated by the original finite population and their conditional means (3.15) have zero tail residual $D_N v = 0$. Equation (4.2) directly recovers their native successors.

Proof. A one-coordinate right shift keeps the current visible block inside the expanded range of the next population. The spatial support of $T_N P^N v$ lies in $\{0, \dots, (N+1)^2 - 1\}$. After applying A, it lies in $\{1, \dots, (N+1)^2\}$, and the next maximal-scale window has $(N+2)^2 > (N+1)^2 + 1$ visible spatial coordinates. Thus $AT_N P^N v$ lies in the fixed subspace of $T_{N+1}P^{N+1}$, and $T_{N+1}P^{N+1}(AT_N P^N v) = AT_N P^N v$. Apply the linear map A to $v = T_N P^N v + D_N v$ to obtain (4.3). When $D_N v = 0$, this gives (4.2). ■

Theorem 4.1 shows that $\mathcal{U}_N$ autonomously predicts the evolution of the already observed component. Recovering additional spatial coordinates from the previously invisible tail $D_N v$ requires the added columns actually sampled at the next population. The energy of the newly visible block plus the energy of the new tail then equals the old tail energy $\|D_N v\|^2$.

4.2 Mass–clock coupling and the $2N + 3$ native noise benchmark

The action $A(h, m, c) = (Sh, m, c + m)$ adds the mass error to the clock error. The change in squared noise is $|\delta m|^2 + 2\,\mathrm{Re}(\overline{\delta c}\,\delta m)$, which can have either sign. For $\delta m = 1, \delta c = -1$, for instance, the combined squared mass and clock noise drops from 2 to 1. The native state gives an exact benchmark with a positive increment.

Theorem 4.2 (Current and next-frame errors for a noisy shortest window). Fix the original observer of Section 3 and a scale $a > 1$. Suppose the received window is $P_a v + \eta$, with decoded noise $T\eta = (u, \delta m, \delta c) \in G$. The reconstruction errors at the current and next frames are

$$\|v - T(P_a v + \eta)\|^2 = \|Dv\|^2 + \|T\eta\|^2, \qquad \|Av - AT(P_a v + \eta)\|^2 = \|Dv\|^2 + \|AT\eta\|^2, \tag{4.4}$$

where the noise energy after the action is

$$\|AT\eta\|^2 = \|u\|^2 + |\delta m|^2 + |\delta c + \delta m|^2. \tag{4.5}$$

Consider the conditional distribution at a readout y of positive weight. Let V be the random source object from the original population, with conditional mean v_y. All original participants have mass 1, so the centered difference $V - v_y$ has mass 0. The action preserves its norm, $\|A(V - v_y)\| = \|V - v_y\|$, giving $\mathrm{Var}(AV \mid y) = \mathrm{Var}(V \mid y)$. The conditional expected errors at the current frame and at the next frame under the same distribution are therefore

$$\mathbb{E}[\|V - T(P_a v_y + \eta)\|^2 \,|\, y] = \mathrm{Var}(V \mid y) + \|Dv_y\|^2 + \|T\eta\|^2, \tag{4.6}$$

$$\mathbb{E}[\|AV - AT(P_a v_y + \eta)\|^2 \,|\, y] = \mathrm{Var}(V \mid y) + \|Dv_y\|^2 + \|AT\eta\|^2. \tag{4.7}$$

In particular, take as the perturbation the noise-free window $\eta = P_a v_{\mathrm{nat}}$ of the native state at population bound N, whose decoded mass is $\delta m = 1$ and clock is $\delta c = N + 1$. The noise energy

and the full conditional expected error both increase over one step by

$$\|AT\eta\|^2 - \|T\eta\|^2 = 2N + 3. \tag{4.8}$$

Proof. The tail residual Dv has spatial support at indices $\geq na$ and has zero mass and clock. The decoded noise $T\eta$ has spatial support at indices $< na$, along with its (m, c) components, so $Dv \perp T\eta$. After applying A, $ADv = (SDv, 0, 0)$ is supported at indices $\geq na + 1$ and satisfies $\|ADv\| = \|Dv\|$. The spatial support of $AT\eta = (Su, \delta m, \delta c + \delta m)$ extends at most to na, so orthogonality persists. Pythagoras gives (4.4)–(4.5). Apply Lemma 2.6 to split off the conditional variance from the squared bias of the mean. Each $v_i = (e_{i+1}, 1, i + 2)$ has mass 1, so $v_i - v_y$ has mass 0. Thus $A(v_i - v_y) = (S(v_i^{\text{space}} - v_y^{\text{space}}), 0, v_i^{\text{clock}} - v_y^{\text{clock}})$ has the same norm as $v_i - v_y$, giving $\mathrm{Var}(AV \mid y) = \mathrm{Var}(V \mid y)$ and (4.6)–(4.7). Finally, the native-state perturbation decodes to $\delta m = 1, \delta c = N + 1$. Substitution into (4.5) and subtraction in (4.6)–(4.7) yield $|(N + 1) + 1|^2 - |N + 1|^2 = (N + 2)^2 - (N + 1)^2 = 2N + 3$, proving (4.8). ■

4.3 The a^4 cotest Gram matrix, explicit gains, and discrete support recovery

Recovering a discrete list of participants from noisy continuous measurements calls for explicit bounds available before measurement. Write the decoding formulas (3.8)–(3.9) of Theorem 3.3 as inner-product functionals of a single-frame perturbation $\eta_t \in \operatorname{ran} o \subset G$. The induced error in the probe response is $r_j(B_a\eta_t) = \langle g_j, B_a\eta_t\rangle = \langle B_a^* g_j, \eta_t\rangle$. Thus the decoder pairs the perturbation with the adjoint probe columns, whose cotest Gram matrix is

$$\tilde{g}_j := B_a^* g_j = (e_j,\, 1,\, a^2(j+1)), \qquad \langle \tilde{g}_j, \tilde{g}_k\rangle = \delta_{jk} + 1 + a^4(j+1)(k+1). \tag{4.9}$$

The transmitter's physical responses are governed by the a^2 Gram matrix in (3.5). The receiver's noise gains are governed by the a^4 Gram matrix in (4.9). For any decoder row coefficient array $b = (b_{tj})_{0\leq t\leq a, 0\leq j<n}$, equation (4.9) evaluates the functional energy $\mathcal{E}(b) := \sum_{t=0}^{a} \left\|\sum_{j<n} \overline{b_{tj}}\tilde{g}_j\right\|^2$ by rational arithmetic for rational rows. Let b_k, b_m, b_c be the rows for the first na spatial coordinates, mass, and clock, respectively. Define the explicit gains

$$\kappa_{\text{gain}} := \sum_{k<na} \mathcal{E}(b_k) + \mathcal{E}(b_m) + \mathcal{E}(b_c), \qquad \kappa_{\text{next}} := \sum_{k<na} \mathcal{E}(b_k) + \mathcal{E}(b_m) + \mathcal{E}(b_c + b_m). \tag{4.10}$$

Theorem 4.3 (Explicit precision budget and recovery of discrete conditional support). Every window perturbation $\eta = (\eta_0, \dots, \eta_a)$ satisfies

$$\|T\eta\|^2 \leq \kappa_{\text{gain}} \sum_{t=0}^{a} \|\eta_t\|^2, \qquad \|AT\eta\|^2 \leq \kappa_{\text{next}} \sum_{t=0}^{a} \|\eta_t\|^2. \tag{4.11}$$

For conditional mean recovery with $N + 1$ equally weighted participants, $N \geq 1$, and maximal copy scale $a = N + 1$, let the separating threshold be $\tau_N := \frac{1}{2(N+1)}$. For any readout y of positive weight, suppose the perturbation energy satisfies the prior budget

$$\kappa_{\text{gain}} \sum_{t=0}^{a} \|\eta_{y,t}\|^2 < \tau_N^2, \tag{4.12}$$

Then the spatial threshold set $\{i \le N : \mathrm{Re}(T(P_a v_y + \eta_y))_{i+1}^{\mathrm{space}} > \tau_N\}$ computed from the noisy window recovers the complete discrete conditional support F_y without error. This determines the rational conditional probabilities $p_N(i \mid y)$ for all participants.

Proof. Apply Cauchy–Schwarz on the product space to each decoder output coordinate and sum over coordinates to obtain the first inequality of (4.11). Using the row $b_c + b_m$ for the updated clock gives the second. For the transmitted conditional mean (3.15), $Dv_y = 0$. Under (4.12), $\|T\eta_y\| < \tau_N$, so the absolute error of the real part of each spatial coordinate is less than τ_N. For an absent participant $i \notin F_y$, the true coordinate is 0, so the noisy real part is $< \tau_N$. For a present participant $i \in F_y$, write $m_y \le N + 1$ for the fiber size. The true coordinate is $1/m_y \ge 1/(N+1) = 2\tau_N$, so the noisy real part is $> 2\tau_N - \tau_N = \tau_N$. The discrete gap of width $2\tau_N$ separates the coordinates despite the continuous perturbation. The threshold recovers F_y without loss, and equal prior weights uniquely give the uniform conditional weight $1/m_y$. ■

4.4 Orthogonal time, observation, and decoding errors across changing empirical spaces

Return to the state-dependent space C_s of Theorem 2.8 in Section 2.2. When time evolution changes the underlying empirical reference measure, the adjoining empirical L^2 carriers C_{now} and C_{next}, which are instances of C_s on successive empirical windows, are different Hilbert spaces. The prediction error can then have three distinct sources.

Theorem 4.4 (The orthogonal decomposition of time, observation, and decoding errors). For the finite empirical Hilbert spaces $(C_{\mathrm{now}}, C_{\mathrm{next}})$, let $K : C_{\mathrm{next}} \to C_{\mathrm{now}}$ be the isometric pullback preserving empirical measure, and let $\Lambda = K^* : C_{\mathrm{now}} \to C_{\mathrm{next}}$ be its adjoint conditional time transfer. Let $J : M_{\mathrm{next}} \to C_{\mathrm{next}}$ be the isometric embedding of the next all-future observation quotient as its visible subspace. Define the optimal compressed target $m' := J^*\Lambda u \in M_{\mathrm{next}}$, the time residual operator $R_{\mathrm{time}} := I - K\Lambda$, and the next observation residual operator $R_{\mathrm{obs}} := I - JJ^*$. Then every current task vector $u \in C_{\mathrm{now}}$ splits into three pairwise orthogonal terms:

$$u = KJm' + R_{\mathrm{time}}u + KR_{\mathrm{obs}}\Lambda u, \tag{4.13}$$

and for any candidate d decoded in the next observation quotient M_{next} the total squared error satisfies

$$\boxed{\|u - KJd\|^2 = \|R_{\mathrm{time}}u\|^2 + \|R_{\mathrm{obs}}\Lambda u\|^2 + \|m' - d\|^2.} \tag{4.14}$$

Proof. Since K is isometric, $K^*K = I_{C_{\mathrm{next}}}$. Thus $K\Lambda = KK^*$ is the orthogonal projection in C_{now} onto the closed subspace $\operatorname{ran} K$, yielding $u = K\Lambda u + R_{\mathrm{time}}u$ with $R_{\mathrm{time}}u \perp \operatorname{ran} K$. Similarly, $J^*J = I_{M_{\mathrm{next}}}$ makes JJ^* the orthogonal projection in C_{next} onto $\operatorname{ran} J$. This yields $\Lambda u = Jm' + R_{\mathrm{obs}}\Lambda u$ with $R_{\mathrm{obs}}\Lambda u \perp \operatorname{ran} J$. Applying the isometry K to this second decomposition and substituting into the first gives (4.13). For every $d \in M_{\mathrm{next}}$ we have $u - KJd = KJ(m' - d) + R_{\mathrm{time}}u + KR_{\mathrm{obs}}\Lambda u$. The first and third terms lie in $\operatorname{ran} K$, so both are orthogonal to $R_{\mathrm{time}}u$. Within $\operatorname{ran} K$, the isometry preserves inner products: $\langle KJ(m' - d), KR_{\mathrm{obs}}\Lambda u\rangle = \langle J(m' - d), R_{\mathrm{obs}}\Lambda u\rangle = 0$. Expanding the squared norm of the three pairwise orthogonal terms gives (4.14). ■

Example 4.5 (A time collision imposes a loss even under complete future observation). Take the unit impulse $u = (1, 0)$ on two equally weighted participants, and suppose

one time step merges both participants into the same empirical state. The conditional time mean is $\Lambda u = 1/2$, and the time loss is already $\|R_{\text{time}}u\|^2 = \frac{1}{2}[(1-1/2)^2 + (0-1/2)^2] = 1/4$. By (4.14), even with lossless all-future observation at the next stage ($R_{\text{obs}} = 0$) and the optimal decoder $d = m' = 1/2$, every decoder using the next state incurs at least $1/4$ squared error. Recovering the impulse $(1, 0)$ through (4.13) requires retaining both orthogonal residual terms $R_{\text{time}}u$ and $KR_{\text{obs}}\Lambda u$.

5 Streaming reception, exact merge–update commutativity, and the coarsening risk gap

After decoding one window, a streaming receiver faces a continuing sequence of new records. The population grows, conditional means change with the sample counts, and a later readout may merge fine classes into a coarse table. We construct a complete reception state F that retains counts. Writing weighted averages in terms of counts and sum moments gives sufficient statistics in a commutative monoid [8]. This makes updating and coarsening commute on both the infinite-dimensional spatial, mass, and clock carrier and the discrete posterior. It also preserves the unnormalized counted residual $c_y\varepsilon_y$ across every finite continuation. A single precision budget at initialization then suffices for exact discrete posterior recovery at every finite step.

5.1 Retained state and conservation of counted residuals

Definition 5.1 (The complete reception state). Given a population $I_N = \{0, \dots, N\}$ and its original readout $q : \mathbb{N} \to Y$, write the true count and conditional mean in a fine class $y \in Y$ as

$$c_y = \big|\{i \le N : q(i) = y\}\big|, \qquad v_y = \frac{1}{c_y} \sum_{i \le N:\, q(i)=y} v_i, \qquad v_i = (e_{i+1}, 1, i+2), \tag{5.1}$$

with $c_y = 0, v_y = 0$ for an empty fiber. A complete reception state F stores the set of observed keys, the counts c_y, the conditional weights of all participants, and the raw rational coordinate vectors $r_y \in G$. Such a vector includes the received spatial coefficient array and the independent mass and clock. Write the reception residual in each class as $\varepsilon_y := r_y - v_y$. At initialization, the threshold in Theorem 4.3 recovers the discrete conditional weights, while r_y retains the noisy rational coordinates actually received.

When participant $N+1$, $b = v_{N+1} = (e_{N+2}, 1, N+3)$, arrives with the actual receipt key $y_* = q(N+1)$, the receiver updates that class by

$$c'_{y_*} = c_{y_*} + 1, \qquad r'_{y_*} = r_{y_*} + \frac{b - r_{y_*}}{c_{y_*} + 1}, \tag{5.2}$$

Untouched classes keep their values, with zeros appended for the new spatial coordinates. The update multiplies the conditional weights on the old support by $c_{y_*}/(c_{y_*}+1)$ and assigns weight $1/(c_{y_*}+1)$ to $N+1$. For a newly opened class with $c_{y_*} = 0$, this gives $c'_{y_*} = 1, r'_{y_*} = b$. This is a realization of the classical one-pass mean update [42; 8] on the infinite-dimensional spatial, mass, and clock carrier.

Theorem 5.2 (One initialization gives correct finite continuations and conserves counted residuals). Suppose the key set and conditional weights of the initial reception state equal the actual readout image and conditional distribution for population N, and every subsequent step receives the actual new participant's receipt. For every finite number of steps $k \in \mathbb{N}$, the resulting key set and conditional weights equal the actual readout image and conditional distribution for the enlarged population $N+k$. In the updated class the reception residual contracts and its counted value is conserved:

$$\varepsilon'_{y_*} = \frac{c_{y_*}}{c_{y_*}+1}\varepsilon_{y_*}, \qquad c'_{y_*}\varepsilon'_{y_*} = c_{y_*}\varepsilon_{y_*}, \tag{5.3}$$

Untouched residuals remain fixed. A newly opened class has zero counted residual.

Proof. Before the update, the weight on each point of the old support in the affected class is $1/c_{y_*}$. Scaling by $c_{y_*}/(c_{y_*}+1)$ gives $1/(c_{y_*}+1)$, the weight also assigned to the new point. This is the uniform conditional distribution on the enlarged fiber. Adjoining $q(N+1)$ maintains the readout image. The true conditional mean v_{y_*} follows the same update $v'_{y_*} = v_{y_*} + (b - v_{y_*})/(c_{y_*}+1)$. Subtracting it from (5.2) gives $\varepsilon'_{y_*} = \varepsilon_{y_*} - \varepsilon_{y_*}/(c_{y_*}+1) = \frac{c_{y_*}}{c_{y_*}+1}\varepsilon_{y_*}$. Multiplying by $c_{y_*}+1$ gives $c'_{y_*}\varepsilon'_{y_*} = c_{y_*}\varepsilon_{y_*}$. For a new class, $c_{y_*} = 0$ and $r'_{y_*} = b = v'_{y_*}$, so $\varepsilon'_{y_*} = 0$. Induction on k proves the result. ■

5.2 Exact commutativity of merging and updating along a trajectory

Let a forgetful map $f : Y \to Z$ merge fine readout keys into coarse keys. For each coarse key $z \in Z$, write the merged count and convex weights as

$$C_z = \sum_{f(y)=z} c_y, \qquad w_{z,y} = \begin{cases} c_y/C_z, & f(y) = z \text{ and } C_z > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{5.4}$$

The complete merge $\mathcal{M}_f$ combines the observed key sets and the conditional distributions. It takes the raw coordinates in each coarse class to the weighted average $r_z = \sum_y w_{z,y} r_y$, using the same weights for the spatial array, mass, and clock. Algebraically, $(c_y, r_y) \mapsto (c_y, c_y r_y)$ represents the weighted mean by additive sufficient statistics in the commutative monoid $\mathbb{N} \times G$ [8]. This monoid identity extends to the entire reception state, including its keys, discrete posterior, and all spatial, mass, and clock coordinates.

Theorem 5.3 (Exact merge–update commutativity for the complete reception state). Under the initialization of Theorem 5.2 from the same source, for any fine-to-coarse key map $f : Y \to Z$ and any actual new receipt y_*, the complete reception state F satisfies

$$\boxed{\mathcal{M}_f(\mathrm{next}_{y_*}\mathsf{F}) = \mathrm{next}_{f(y_*)}(\mathcal{M}_f\mathsf{F}).} \tag{5.5}$$

The equality covers the key set, the counts, the conditional weights of all participants, and every rational spatial, mass, and clock coordinate. Along any finite sequence of new receipts, stepwise updates at the fine resolution followed by merging equal merging first and then updating with the coarsened receipts.

Proof. Inserting the new key and adding to its count commute with merging. For raw coordinates, multiply (5.2) by $c'_{y_*} = c_{y_*} + 1$ to obtain the additive update of a single counted moment.

$$c'_{y_*} r'_{y_*} = c_{y_*} r_{y_*} + b. \tag{5.6}$$

Thus updating first and then merging gives the new counted moment of coarse class z $\sum_{f(y)=z} c'_y r'_y$ by adding b to the old value $C_z r_z = \sum_{f(y)=z} c_y r_y$ exactly when $z = f(y_*)$. The other moments $C_z r_z$ stay fixed. This agrees coordinate by coordinate with merging to (C_z, r_z) first and then applying (5.6) at the coarse key $f(y_*)$. The same equality holds for empty coarse classes $C_z = 0$ by the zero convention. In both orders the conditional weights are uniform on the same coarse fiber, so the entire frame is equal. Induction on the trajectory length gives commutativity for every finite sequence. ■

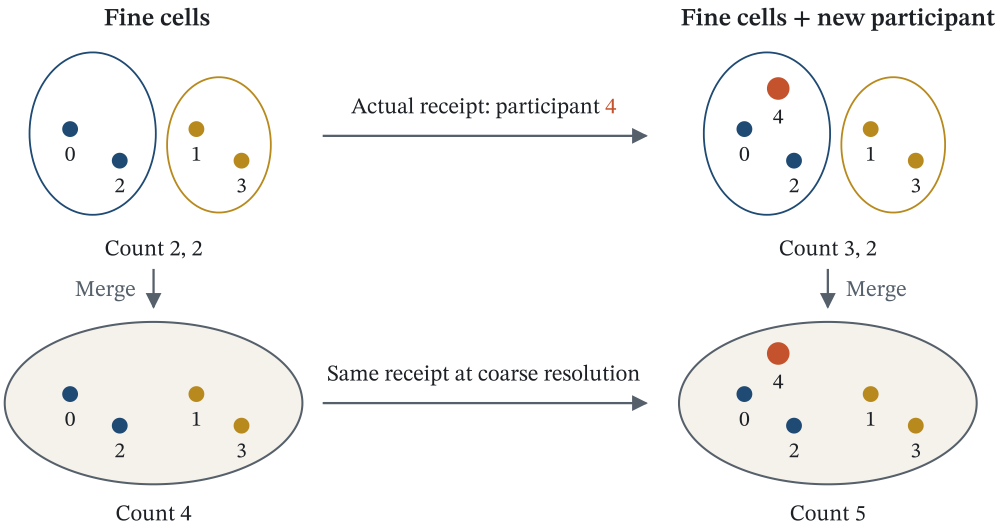


Figure 3. Merging and updating commute for fine and coarse tables. Dark blue and gold ellipses enclose the two fine readout fibers, the warm gray ellipse encloses their merger, and the vermilion point is the new participant 4. The upper path updates the fine classes and then merges them. The lower path merges first and then updates the coarse class. The two paths give the same counts, conditional posterior, and full counted coordinate moment $C'_z r'_z$.

5.3 Mixed vector residuals, variance innovation, and noise contraction

Theorem 5.4 (Coarsened recovery error for the original field and its one-step recurrence). Let v_z be the true conditional mean of coarse class z, and let $\varepsilon_y = r_y - v_y$ be the fine residuals. The coarse residual is the weighted vector sum $\varepsilon_z := r_z - v_z = \sum_y w_{z,y} \varepsilon_y$. The recovery error of the coarse reception state on the entire population I_N decomposes as

$$E_N := \frac{1}{N+1} \sum_{i=0}^{N} \|v_i - r_{f(q(i))}\|^2 = \underbrace{\frac{1}{N+1} \sum_{i=0}^{N} \|v_i - v_{f(q(i))}\|^2}_{\text{intrinsic coarse conditional variance}} + \sum_z \frac{C_z}{N+1} \left\| \sum_y w_{z,y} \varepsilon_y \right\|^2. \tag{5.7}$$

Suppose the new participant $b = v_{N+1}$ falls in coarse class $z = f(y_*)$, with current count C_z, true mean v_z, and mixed residual ε_z. Define the variance innovation $J_N := \frac{C_z}{C_z+1}\|b - v_z\|^2$. The total error at the next frame satisfies

$$\boxed{(N+2)E_{N+1} = (N+1)E_N + J_N - \frac{C_z}{C_z+1}\|\varepsilon_z\|^2.} \tag{5.8}$$

Proof. By the tower property, $v_z = \sum_y w_{z,y} v_y$. Subtracting this from $r_z = \sum_y w_{z,y} r_y$ gives $\varepsilon_z = \sum_y w_{z,y}\varepsilon_y$. Separating conditional variance from mean bias on each coarse fiber by Lemma 2.6 gives (5.7). By Theorems 5.2–5.3, the decomposition holds at every finite successor. At the next frame, the unnormalized variance and residual sums of untouched classes remain fixed. In the affected class z, the centered sum-of-squares update gives the exact increment $J_N = \frac{C_z}{C_z+1}\|b - v_z\|^2$ in unnormalized conditional variance, with both sides equal to 0 when $C_z = 0$. The contraction $\varepsilon'_z = \frac{C_z}{C_z+1}\varepsilon_z$ from Theorem 5.2 changes its unnormalized residual energy from $C_z\|\varepsilon_z\|^2$ to $(C_z+1)\left\|\frac{C_z}{C_z+1}\varepsilon_z\right\|^2 = \frac{C_z^2}{C_z+1}\|\varepsilon_z\|^2$. The difference is $-\frac{C_z}{C_z+1}\|\varepsilon_z\|^2$. Combining the contributions yields (5.8). ■

Equations (5.7) and (5.8) describe the effects of merging residuals and adding a new sample, respectively. In (5.7), first take the vector sum of the residuals and then square its norm. Two equal-count fine classes with opposite residuals u and $-u$ merge to residual zero. If the residuals are u and u, the coarse residual is u. The sum of the fine squared errors is the same in the two cases, yet the actual coarse errors differ. In (5.8), one receipt changes the error in two opposing ways. The new sample contributes the variance innovation $+J_N$, while the larger count decreases the old reception noise by $-\frac{C_z}{C_z+1}\|\varepsilon_z\|^2$.

5.4 Rational counted tables recover support after any finite number of updates

An executable receiver can avoid continually renormalizing a floating-point or fractional weight table by storing a four-component rational counted-moment entry for every observed key:

$$\mathrm{Entry}_y := (c_y,\ H_y,\ M_y,\ T_y) = (c_y,\ c_y r_y^{\mathrm{space}},\ c_y r_y^{\mathrm{mass}},\ c_y r_y^{\mathrm{clock}}). \tag{5.9}$$

A new receipt requires additions only: $c_y \leftarrow c_y + 1$, $H_y(N+2) \leftarrow H_y(N+2) + 1$, $M_y \leftarrow M_y + 1$, and $T_y \leftarrow T_y + (N+3)$. Coarsening by f adds the tuples with the same coarse key componentwise. Conservation of the counted residual means the initial sampling precision suffices for any finite k, with no tightening as k grows.

Theorem 5.5 (Full posterior recovery from the rational counted table after any finite number of steps). Suppose the initial table comes from the maximal-scale window in Theorem 3.5. Each initially nonempty fine class satisfies the precision budget (4.12), and later frames update by addition using only their actual receipts. Then for any finite $k \geq 0$, in the fine table on population $N + k$ or any table merged under a coarsening f, read the half-unit threshold set

$$\widehat{F}_y = \{i \leq N + k : \mathrm{Re}\, H_y(i+1) > 1/2\}, \tag{5.10}$$

and assign uniform weights on $\widehat{F}_y$. This recovers the entire fine or coarse conditional distribution for that frame without error. Feeding in the next actual receipt gives the same next quotient model as the original autonomous update. The counted moments also realize the original-field error identities (5.7)–(5.8).

Proof. By Theorem 5.2, the sample count may grow while the unnormalized counted error stays fixed at its initial value. The true initial counted spatial moment $c_y v_y^{\mathrm{space}}$ is either 0 or 1 at participant coordinate $i+1$. By the initial budget (4.12), each initially nonempty fine class satisfies $\|\varepsilon_y\| < \tau_N = \frac{1}{2(N+1)}$, so

$$\|c_y \varepsilon_y\| < \frac{c_y}{2(N+1)} \le \frac{1}{2}. \tag{5.11}$$

Theorem 5.2 gives the same counted residual $c_y\varepsilon_y$ after any number of subsequent updates. Newly opened classes have counted residual 0. Thus for all $k \ge 0$, the real part of an absent participant's coordinate, with true value 0, remains $< 1/2$. For a present participant, with true value 1, it remains $> 1/2$. The threshold (5.10) is exact at every finite successor.

For an arbitrary coarsening f, use the common initial budget (4.12) when summing fine residuals. Adding a bound of $1/2$ for each fine class would lose the needed precision. For every initially nonempty coarse class z the count $C_z = \sum_{f(y)=z} c_y \le N+1$, so the triangle inequality gives

$$\left\| \sum_{f(y)=z} c_y \varepsilon_y \right\| \le \sum_{f(y)=z} c_y \|\varepsilon_y\| < \frac{\sum_{f(y)=z} c_y}{2(N+1)} = \frac{C_z}{2(N+1)} \le \frac{1}{2}. \tag{5.12}$$

By the commutativity of Theorem 5.3, this coarse counted residual also remains fixed throughout subsequent updates. The coarse table therefore recovers all coarse supports and posteriors at threshold $1/2$ and gives the native next model directly. ■

5.5 Excess coarsening risk at the next state and its execution record

Equation (5.7) uses the raw noisy coordinates. To isolate the information lost in coarsening at the next frame, use the exact posterior means recovered by Theorem 5.5. Define the source-object coarsening gap on the uniform population I_N by

$$\mathrm{Gap}_G(q,f) := \frac{1}{N+1} \sum_{i=0}^{N} \|v_{q(i)} - v_{f(q(i))}\|^2. \tag{5.13}$$

Theorem 5.6 (Excess risk after updating the coarse table). Under the initialization budget of Theorem 5.5, take any finite number of updates and any coarsening f. Use the posteriors recovered from the fine and coarse successor tables to define two conditional-mean decoders, both with the original weights and target v_i on the next population I_{N+k+1}. The optimal squared risk for the coarse table equals the fine-table risk plus $\mathrm{Gap}_G(q,f)$ on that population. If the fine readout distinguishes every participant, the fine risk is zero and the total coarse risk equals the coarsening gap. The identity coarsening $f = \mathrm{id}$ has zero gap.

Proof. Theorem 5.5 recovers the true posteriors from both successor tables. Their decoders therefore give $v_{q(i)}$ and $v_{f(q(i))}$, respectively. Split $v_i - v_{f(q(i))}$ into $v_i - v_{q(i)}$ and $v_{q(i)} - v_{f(q(i))}$. The first vector sums to zero within each fine fiber and the second is constant there. The cross term

vanishes. Squaring and summing gives excess risk $\mathrm{Gap}_G(q, f)$. If the fine readout is injective, then $v_{q(i)} = v_i$, so the fine risk is zero. For the pointwise clock readout coarsened by parity, participants $0, 2$ are both even. The coarse mean cannot equal both of their different spatial and clock vectors. Hence both the total risk and the conditional entropy $H(q \mid fq)$ are strictly positive.

In the fixed clock-to-parity case, the actual counted table gives this total-risk identity. The receipt of the same event also supplies the original whole ledger and the canonical successor. ■

6 Persistent Recovery Gains, Birth Parity, Singular Gram Matrices, and Completely Monotone Forgetting

To compare the recovery performance of two observation channels, fix the population, the target task, and the historical weights. With the population, the full-carrier task, and the original weights fixed, we characterize when coarsening causes a strictly positive loss and give an example in which the excess risk on a scalar projection is zero. For a family of copy channels coupled through the scan index modulo 2, we prove two laws. Retaining a fixed pair of parity channels with their original provenance yields a strict recovery gain for the same task at every finite generated successor. Replacing the previous channel by the newest one reverses the recovery advantage with the parity of the birth depth, and a channel born at odd depth worsens recovery of the old task. We then track kernel fibers of potentially singular Hermitian Gram matrices across cofinal stages and give a formalized Gamma–Laplace derivation of shifted power-law forgetting.

6.1 Left Inverses for Action Words, Risk Decomposition for a Fixed Task, and Zero Scalar Excess Risk

Compose the native time step A and the material copy actions B_a at various scales. Any resulting admissible finite word w acts on the full carrier G by an affine clock update and an isometric spatial shift:

$$F_w(h, m, c) = (U_w h,\, m,\, \alpha_w c + \beta_w m), \qquad \alpha_w \in \mathbb{N}_{>0}, \quad \beta_w \in \mathbb{N}, \tag{6.1}$$

where $U_w e_j = e_{\alpha_w(j+1)+\beta_w-1}$ is an injective isometric spatial embedding. The native time step has $(\alpha, \beta) = (1, 1)$, and copying by B_a has $(a, 0)$. The map F_w has the explicit continuous linear left inverse

$$L_w(h, m, c) := (U_w^* h,\, m,\, (c - \beta_w m)/\alpha_w), \qquad L_w F_w = I_G. \tag{6.2}$$

The left inverse ensures that the task after the action, $Z_i := F_w v_i$, preserves every distinction among the original participants.

On the same population I_N, let $q : \mathbb{N} \to Y$ be the fine readout and $f : Y \to Z$ the coarsening map. Write $\mu_y = \mathbb{E}[Z \mid q = y]$ and $\nu_z = \mathbb{E}[Z \mid fq = z]$ for the fine and coarse conditional means. Define the fine risk, coarse risk, and recovery gap for the fixed task by

$$\mathrm{Risk}_{\mathrm{fine}} = \frac{1}{N+1} \sum_{i=0}^{N} \|Z_i - \mu_{q(i)}\|^2, \qquad \mathrm{Risk}_{\mathrm{coarse}} = \frac{1}{N+1} \sum_{i=0}^{N} \|Z_i - \nu_{f(q(i))}\|^2,$$

$$\mathrm{gGap} = \frac{1}{N+1}\sum_{i=0}^{N} \|\mu_{q(i)} - \nu_{f(q(i))}\|^2. \tag{6.3}$$

Theorem 6.1 (Orthogonal risk decomposition and the exact criterion for strict loss at fixed task and historical weights). The fine and coarse conditional means attain the global minimum squared risk for their respective observations and satisfy

$$\boxed{\mathrm{Risk}_{\mathrm{coarse}} = \mathrm{Risk}_{\mathrm{fine}} + \mathrm{gGap}.} \tag{6.4}$$

For every full G-valued task $Z_i = F_w v_i$ generated by an admissible action word, the following four conditions are equivalent. (i) gGap $= 0$. (ii) The merging gap for the original task is $\mathrm{Gap}_G(q, f) = 0$. (iii) The coarsening map f leaves all nonempty fine fibers distinct. (iv) The conditional information loss is $H(q \mid fq) = 0$.

Proof. Split $Z_i - \nu_{f(q(i))}$ into $(Z_i - \mu_{q(i)}) + (\mu_{q(i)} - \nu_{f(q(i))})$. On each fine fiber $q(i) = y$, the second term is constant and the first sums to zero. The cross term therefore vanishes, giving (6.4). Since the linear map F_w commutes with finite conditional expectation, $\mu_y = F_w v_y$ and $\nu_z = F_w v_z$. If gGap $= 0$, every participant has positive weight $1/(N+1) > 0$, so for each i we have $F_w(v_{q(i)} - v_{f(q(i))}) = 0$. Applying the left inverse L_w from (6.2) gives $v_{q(i)} = v_{f(q(i))}$, hence $\mathrm{Gap}_G(q, f) = 0$. Suppose two disjoint nonempty fine fibers F_{y_1}, F_{y_2} merge into a coarse fiber F_z. Each participant has a distinct orthogonal spatial basis vector e_{i+1}, so the spatial coordinates of v_{y_1} on the support of F_{y_2} are all 0, while v_z has coordinate $1/|F_z| > 0$ there. This gives $v_{y_1} \neq v_z$, a contradiction. Thus $\mathrm{Gap}_G = 0$ holds exactly when each nonempty fine cell stays distinct, equivalently when the conditional fine distribution within each nonempty coarse cell is a point mass ($H(q \mid fq) = 0$). ■

Theorem 6.1 uses a full G-valued task with a left inverse. Coarsening can reduce spatial resolution while leaving the risk of a single scalar projection, such as the clock coordinate, unchanged.

Example 6.2 (The clock risk stays fixed while the full risk rises by $1/5$). Take five equally weighted participants $i = 0, \ldots, 4$. The fine observation separates the parity classes $\{0, 2, 4\}$ and $\{1, 3\}$, and coarsening merges the two cells. The task is copying at scale 2, $Z_i = B_2 v_i = (e_{2(i+2)-1}, 1, 2(i+2))$. The five clocks are $(4, 6, 8, 10, 12)$. The conditional clock mean is $(4+8+12)/3 = 8$ in the even cell, $(6+10)/2 = 8$ in the odd cell, and 8 in the coarse cell. Thus the squared clock risk on both sides is $(16 + 4 + 0 + 4 + 16)/5 = 8$. The conditional clock mean 8 and the mean squared clock risk 8 are distinct statistics that coincide numerically in this symmetric arithmetic progression. The equality of the fine means gives zero excess clock risk clockGap $= 0$, and the original spread around the mean is still 8. In the orthogonal spatial component, the fine risk is $(5-2)/5 = 3/5$ and the coarse risk is $(5-1)/5 = 4/5$. The mass is always 1, giving risk 0

for either readout. The complete decomposition is

$$\begin{aligned} R_{\mathrm{clock,fine}} = R_{\mathrm{clock,coarse}} &= 8, \\ \mathrm{Risk}_{\mathrm{fine}} &= \frac{3}{5} + 8 = \frac{43}{5}, \\ \mathrm{Risk}_{\mathrm{coarse}} &= \frac{4}{5} + 8 = \frac{44}{5}, \\ \mathrm{gGap} &= \frac{1}{5} > 0, \qquad \mathrm{clockGap} = 0. \end{aligned} \tag{6.5}$$

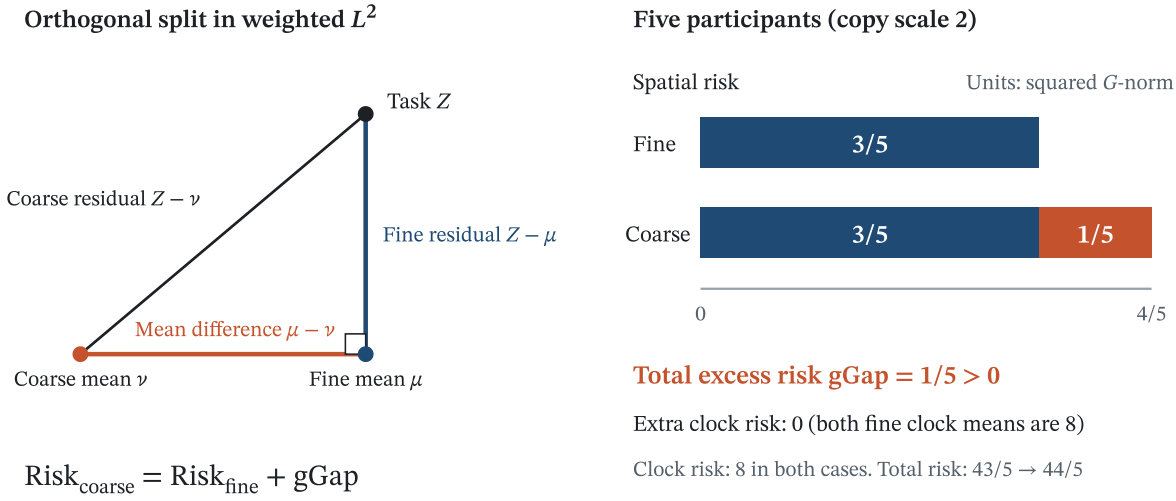


Figure 4. Coarsening risk decomposition for the same task. The red base on the left is the difference $\mu - \nu$ between the fine and coarse conditional means. It is orthogonal to the fine residual $Z - \mu$. The right panel decomposes the spatial risk $3/5 + 1/5 = 4/5$ in the example with five participants. Both readouts have clock risk 8 and zero mass risk. The full risk rises from $43/5$ to $44/5$.

6.2 Persistent Strict Recovery Gains for the Same Task in the Modulo 2 Copy Family

Example 6.2 fixes the task and varies the observation map. We now let the material inventory grow and consider the copy channel family generated by the source run and coupled through the scan index modulo 2. Each copy letter referring to a scan index $a \in \mathbb{N}_{>0}$ acts at two levels. On the full carrier G, the copy operator B_a of Definition 3.1 expands space and the clock ($\alpha_w = a$). At the binary readout level, the readout is the scan index modulo 2. A time step adds 1 modulo 2, and a copy letter referring to scan index a multiplies the bit by a mod 2. This coupling gives exact quantitative comparisons for retaining channels with their original provenance and for replacing them at each generation.

Definition 6.3 (Complete future binary response function). For a current action word w in a fixed admissible inventory and a participant i, define its complete future response $\mathcal{R}_w(i)$ as the function that takes any admissible finite continuation t and returns the binary readout obtained by following w with t. This includes the empty word $t = \varepsilon$. Every later letter acts by a deterministic single-valued map modulo 2, so induction on the word t gives

$$\mathcal{R}_w(i) = \mathcal{R}_w(j) \iff q_w(i) = q_w(j). \tag{6.6}$$

Reading the entire response tree $\mathcal{R}_w$, with its infinitely many future branches, gives the same participant fibers and conditional distributions as reading the single current bit q_w.

The generation depth d is the state index s_d in the original trajectory. The material source of a copy letter born at depth d is the successor state s_{d+1}. The scan index of state s_j is $j+1$. Thus the letters generated at depths 2 and 3 have the following indices.

- Old letter. Generation depth 2 $\longrightarrow$ material source s_3 $\longrightarrow$ scan index 4 (even, binary multiplier 0, acting on G by B_4).
- New letter. Generation depth 3 $\longrightarrow$ material source s_4 $\longrightarrow$ scan index 5 (odd, binary multiplier 1, acting on G by B_5).

As the inventory grows, the inclusion maps transport the letter positions and retain their original material sources.

Proposition 6.4 (The old channel cannot recover the new channel's complete future response, with a lower bound of 3/4). Carry the old letter from depth 2 into depth 3 (state s_3, with 4 equally weighted participants $0, 1, 2, 3$) and compare it with the new letter.

1. No decoder g, whether given a single bit or the entire infinite future response tree, can satisfy $\forall i \in \{0,1,2,3\}$, $g(\mathcal{R}_{\text{old}}(i)) = \mathcal{R}_{\text{fresh}}(i)$.
2. For every decoder of the full G-valued task generated by the new letter from the old channel's binary code, the weighted squared risk satisfies Risk $\geq 3/4$.

Proof. The scan index of the old letter is $4 \equiv 0 \pmod 2$, so multiplication sets every participant's current bit to zero. By (6.6), $\mathcal{R}_{\text{old}}(0) = \mathcal{R}_{\text{old}}(1)$. The common response is 0 at the empty word and 1 after one native time step. The new letter has scan index $5 \equiv 1 \pmod 2$ and retains the original parity distinction between participants $0, 1$. Hence $\mathcal{R}_{\text{fresh}}(0) \neq \mathcal{R}_{\text{fresh}}(1)$, and a single-valued g cannot map their common input to both outputs. For 4 equally weighted participants, the minimum variance of the four orthogonal unit spatial vectors in the single fiber is $(4-1)/4 = 3/4$. Adding the nonnegative mass and clock variances gives full-carrier risk $\geq 3/4$. ■

Theorem 6.5 (Persistent strict recovery gains for the same task at every finite generated successor). Start at the original state s_3. For any finite number of steps $k \in \mathbb{N}$, the kth generated successor is $s_{3+k} = \sigma^k(s_3)$ (population upper index $N_k = 3 + k$, uniform historical weight $p_k(i) = \frac{1}{N_k+1}$). Carry the old and new copy letters with material sources s_3, s_4 (scan indices $4, 5$) along the inventory inclusions to s_{3+k}, and denote the resulting words by $w_{\text{old},k}$ and $w_{\text{fresh},k}$. Fix the same full task generated by the new material, $Z_{k,i} := F_{w_{\text{fresh},k}} v_i$, at that step. Then, for every $k \in \mathbb{N}$, the conditionally optimal recovery risks for the two observations, or for their complete future response functions, satisfy

$$\boxed{\text{Risk}_{\text{old},k} = \text{Risk}_{\text{fresh},k} + \text{gGap}_k, \qquad \text{gGap}_k > 0, \qquad H(q_{\text{fresh},k} \mid q_{\text{old},k}) > 0,} \tag{6.7}$$

and the complete conditional distributions for both observations can be recovered at every step by Section 5.4. At the same occurrence, the original run recovers the new channel's posterior from the counted table and produces the whole-ledger writeback and the actual successor s_{4+k}.

Proof. At each step k, the inventory inclusion preserves the scan index of the referenced material. The word $w_{\mathrm{old},k}$ always multiplies by $4 \equiv 0 \pmod 2$, so every old bit is 0. The word $w_{\mathrm{fresh},k}$ always multiplies by $5 \equiv 1 \pmod 2$, so it continues to distinguish participants 0 and 1, both present with weight $\frac{1}{4+k} > 0$. Take $q_{\mathrm{fresh},k}$ as the fine observation and $f \equiv 0$ as the constant coarsening. Then $f \circ q_{\mathrm{fresh},k} = q_{\mathrm{old},k}$. The two nonempty fibers of $q_{\mathrm{fresh},k}$ merge at every step k under f, and $F_{w_{\mathrm{fresh},k}}$ has the full left inverse (6.2). Theorem 6.1 gives the positive gap and positive conditional entropy in (6.7) for every $k \in \mathbb{N}$. Equation (6.6) preserves the fibers and risks when the readouts are replaced by their complete future response functions. Applying the same counted construction to the two sets of fibers from (6.6) recovers both posteriors. With the existing parity channels retained, a new participant is recorded according to its key, and the successor $s_{3+k} = \sigma^k(s_3)$ of the original occurrence is independent of the choice of decoder. ■

6.3 Alternating Recovery by Birth Parity and Innovation under Population Growth

Theorem 6.5 tracks a fixed pair of historical materials, generated at depths 2 and 3, through all later steps. Consider instead replacing channels in the same modulo 2 copy family. At each birth depth $d \geq 1$, compare the newest channel with the one generated one step earlier. The recovery advantage reverses with the parity of the birth depth.

Theorem 6.6 (Alternating recovery by birth parity). In the modulo 2 copy family of Section 6.2, for any $d \geq 1$, consider successor state s_{d+1}, with $d+2$ equally weighted original participants $0, \ldots, d+1$. Compare the retained copy word $w_{\mathrm{old}}(d)$ from the previous depth d, with material source s_{d+1} and scan index $d+2$, with the copy word $w_{\mathrm{new}}(d)$ just born at this step, with material source s_{d+2} and scan index $d+3$.

1. **Strict gain in optimal recovery at even birth depths ($\forall d \geq 2$ even).** Since $d+2 \equiv 0 \pmod 2$ and $d+3 \equiv 1 \pmod 2$, the old channel's current bit is 0 for every participant and its complete future response is identical across participants. The new channel retains native parity and distinguishes participants $0, 1$. No function recovers the new complete future response participant by participant from the old one. For the same new-letter task $Z_{\mathrm{new}}(d) = F_{w_{\mathrm{new}}(d)} v$, every decoder $g : \mathbb{Z}_2 \to G$ of the old binary code satisfies the lower bound

$$\mathrm{Risk}(Z_{\mathrm{new}}(d);\, g \circ q_{\mathrm{old}}) \geq \frac{d+1}{d+2}, \tag{6.8}$$

   and the conditionally optimal risks satisfy $\mathrm{Risk}^*_{\mathrm{old}}(d) = \mathrm{Risk}^*_{\mathrm{new}}(d) + \mathrm{gGap}_d$, with $\mathrm{gGap}_d > 0$ and conditional information loss > 0.
2. **Strict loss in optimal recovery at odd birth depths ($\forall d \geq 1$ odd).** Since $d+2 \equiv 1 \pmod 2$ and $d+3 \equiv 0 \pmod 2$, the retained old channel preserves the original distinction between participants $0, 1$. The new channel erases the current bit of every participant, setting it to 0. No function recovers the old complete future response participant by participant from the new one. For the same old-letter task $Z_{\mathrm{old}}(d) = F_{w_{\mathrm{old}}(d)} v$, every decoder $g : \mathbb{Z}_2 \to G$ of the new binary code satisfies $\mathrm{Risk}(Z_{\mathrm{old}}(d);\, g \circ q_{\mathrm{new}}) \geq \frac{d+1}{d+2}$, and the new channel produces a strict recovery loss:

$$\mathrm{Risk}^*_{\mathrm{new}}(d) = \mathrm{Risk}^*_{\mathrm{old}}(d) + \mathrm{gGap}_d, \qquad \mathrm{gGap}_d > 0. \tag{6.9}$$

Recover the complete future conditional distributions for both readouts using (6.6). In the even case, the source run jointly verifies the new code's posterior, the whole ledger, and the successor at that step. In the odd case, it jointly verifies both posteriors, the whole ledger, and the successor.

Proof. Arithmetic modulo 2 determines both material multipliers. When $d = 2m \geq 2$, $(d+2) \bmod 2 = 0$ gives $q_{\text{old}} \equiv 0$. The multiplier $(d+3) \bmod 2 = 1$ makes q_{new} equal to the native one-step parity readout, which differs between participants $0, 1$. On $d+2$ equally weighted participants, the conditional variance for the single constant code $q_{\text{old}} \equiv 0$ on the $d+2$ orthogonal unit spatial coordinates is $\frac{(d+2)-1}{d+2} = \frac{d+1}{d+2}$. This value is attained in the spatial projection. The mass variance is 0, and the clock variance is positive ($\frac{(d+3)^2((d+2)^2-1)}{12} > 0$). Lemma 2.6 gives the lower bound (6.8) for every decoder on the full carrier G. At $d = 2$, this recovers the bound $3/4$ of Proposition 6.4. For $Z_{\text{new}}(d)$, apply Theorem 6.1 with fine observation q_{new} and coarsening $f \equiv 0$ to obtain $\text{gGap}_d > 0$.

When $d = 2m+1 \geq 1$, $(d+2) \bmod 2 = 1$ and $(d+3) \bmod 2 = 0$ reverse the roles. The old channel distinguishes participants $0, 1$ and the new channel is the constant 0. By (6.6), the new complete future response is the same for every participant, so it cannot recover the old response. The single-fiber code $q_{\text{new}} \equiv 0$ gives, on the spatial component of $Z_{\text{old}}(d)$, the same lower bound $\frac{d+1}{d+2}$. For the same old task $Z_{\text{old}}(d)$, apply Theorem 6.1 with fine observation q_{old} and constant coarsening $f \equiv 0$, which gives q_{new}, to obtain the reversed excess $\text{gGap}_d > 0$ in (6.9). ■

Theorems 6.5 and 6.6 distinguish two effects of population growth. Retaining an existing odd-indexed probe preserves its recovery gain across any finite continuation. Replacing the old channel by the newest one at every birth loses the parity distinction at odd depths.

Proposition 6.7 (Sample innovation across steps and inverse recovery in the action image). Fix the observation rule and action word, and let the population grow between adjacent steps $N \to N+1$. The arrival of participant $N+1$ contributes a sample innovation. For any fixed decoder d, the unnormalized risk satisfies $(N+2)\text{Risk}_{N+1}(d) = (N+1)\text{Risk}_N(d) + \|Z_{\text{born}} - d(q_{\text{born}})\|^2$. For a fresh conditionally optimal decoder at each step, let c be the previous count of the new participant's key and μ its previous mean. The centered sum-of-squares update [42; 8] gives

$$(N+2)\text{Risk}^*_{N+1} = (N+1)\text{Risk}^*_N + \frac{c}{c+1}\|Z_{\text{born}} - \mu\|^2. \tag{6.10}$$

For example, for copy scale 2 with a single coarse cell, the optimal risks for four and five participants are $23/4$ and $44/5$. The unnormalized total increases from 23 to 44, and the new participant's squared innovation is $\|Z_4 - \mu\|^2 = 105/4$, giving $44 = 23 + \frac{4}{5} \cdot \frac{105}{4}$. If a later step requires pulling the conditional distribution back through the range of the action operator F_w, write $P_w := F_w L_w$ for the orthogonal projection. The conditional squared error has three orthogonal terms for the intrinsic spread, the component outside the range, and the error of the proposed preimage:

$$\mathbb{E}[\|V - F_w x\|^2 \,|\, y] = \text{Var}(V \mid y) + \|(I - P_w)v_y\|^2 + \|F_w(L_w v_y - x)\|^2, \tag{6.11}$$

The unique optimal preimage is $x_* = L_w v_y$. Decompose each v_i into the residual outside the range and its projection onto the range. The residual contributes a fixed squared norm per participant,

and the projected part follows (6.10) in each cell. Each birth therefore increases the unnormalized optimal inverse cost by $\|(I-P_w)b\|^2 + \frac{c}{c+1}\|P_w b - P_w v_y\|^2$.

6.4 All-Future Recovery for Multiple Actors and Singular Gram Innovation

An action word can produce effects $+1$ and -1 at consecutive steps whose net effect is zero. The original execution record of the paired tree retains both effects at their respective positions. Conditional recovery has a related ambiguity. Several requests can produce the same measurement column, making the normal equations singular while the observable projection and innovation remain unique. We retain the intermediate trajectory of the ordered actions and the coefficient fiber of a given measurement.

Theorem 6.8 (All-future quotient for ordered action words, singular Gram innovation, and birth pairing). Fix the paired source and target trees generated by a single source execution and their compatible actions (the pairing construction in Theorem 9.2 of *Source-Native Process Algebra* [16]). Let C be the joint state module of the two trees, and let Σ be the alphabet containing all source, target, and synchronous actions in the trees. Each node position ν has a linear measurement M_ν. The same payload at different positions is counted at each position. Write $\mathcal{M}x = (M_\nu x)_\nu$, with values in a finite orthogonal direct sum, so $\|\mathcal{M}x\|^2 = \sum_\nu \|M_\nu x\|^2$. Here A_a is the native action, and E_a acts isometrically on the node measurement space, actor by actor.

1. **All-word observation and joint recovery.** Combine the coarse reading and all node measurements into o_{joint}. Then $K_{\Sigma^*} = \bigcap_{w\in\Sigma^*} \ker(o_{\mathrm{joint}} A_w)$ is the largest submodule of $\ker o_{\mathrm{joint}}$ invariant under all original letters. The quotient $M = C/K_{\Sigma^*}$ preserves the original readings for all words. Write D_{c} for the coarse all-word readout on M, and D_H for the joint all-future readout obtained by successive restrictions by word length of the same positional measurements. The joint readout (D_{c}, D_H) has a linear two-sided inverse on its image, and the inverse commutes with every letter and action word. On the smallest invariant submodule generated by the same seed, the algebraic all-word readout, coherent cofinal measurements, and coarse readout likewise have a two-sided inverse on their joint image. Write $\bar{A}_w$ for the quotient action and descend the node readout $\mathcal{M}$ to M. For $v \in M$, the same recovered object determines the discrepancy $(E_w \mathcal{M}(\bar{A}_u v))_\nu - \mathcal{M}_\nu(\bar{A}_w \bar{A}_u v)$ between the two measurement orders, which can then be used in the next query.
2. **Recomputing the singular Gram matrix at each stage.** At each cofinal stage m, consider the finite set I_m of requests registered at that stage. For each entry i, take its coefficient κ_i and expression e_i. Evaluate the old value and the effect in the old environment ρ with its actual increment η, and apply the original measurement ℓ to each branch separately. This yields $a_{i,1} = \kappa_i \ell(e_i(\rho), 0)$ and $a_{i,2} = \kappa_i \ell(0, e_i(\rho+\eta) - e_i(\rho))$ in the same complex inner product space. The total feature column is $c_i = a_{i,1} + a_{i,2}$. Define

$$\begin{aligned} (\mathcal{G}_m)_{ij} &= \langle a_{i,1}, a_{j,1}\rangle + \langle a_{i,1}, a_{j,2}\rangle + \langle a_{i,2}, a_{j,1}\rangle + \langle a_{i,2}, a_{j,2}\rangle = \langle c_i, c_j\rangle, \\ r_i(v) &= \langle c_i, v\rangle. \end{aligned} \tag{6.12}$$

For every v, the matrix $\mathcal{G}_m$ is Hermitian positive semidefinite and $r(v) \in \operatorname{ran} \mathcal{G}_m$. Let $u_*(v)$ be the unique solution in $(\ker \mathcal{G}_m)^\perp$. Then

$$\mathcal{G}_m u = r(v) \iff u - u_*(v) \in \ker \mathcal{G}_m. \tag{6.13}$$

Every solution yields the same orthogonal innovation

$$\mathcal{I}_m(v) := v - \sum_{i \in I_m} u_{*,i}(v) c_i. \tag{6.14}$$

The successor packet retains the written request addresses and signed coefficients. The next stage recomputes $\mathcal{G}_{m+1}$ and $\mathcal{I}_{m+1}$ from its registered requests and new environment.

3. **Observability of both branches of a birth request.** The source run selects the decoding environment $\mathcal{E}_{\mathrm{dec}} = \mathcal{E}_0 + \delta_{\mathrm{birth}}$. For a registered entry with coefficient κ, let the old value and increment be $b = e(\mathcal{E}_0)$ and $d = e(\mathcal{E}_{\mathrm{dec}}) - b$, respectively. The original execution writes $\kappa(b+d)$. Replace e by its difference with the constant $e(\mathcal{E}_{\mathrm{dec}})$ to obtain the paired readout $(-\kappa d, \kappa d)$. The additive decoder sends this pair to zero. The two-branch model still reads the nonzero increment when present. For a fixed written value s, the full fiber of initial pairs yielding that value is $\{(s - t, t) : t \text{ ranges over the value group}\}$, where $t = \kappa d$ retains the effect of this execution.

Proof. For part 1, if $x \in K_{\Sigma^*}$, any letter a followed by a word w is still an admissible word, so $A_a x \in K_{\Sigma^*}$. Every jointly invariant submodule invisible at the empty word is contained in this kernel, proving maximality. If $D_{\mathrm{c}} v = D_H v = 0$, the coarse readings vanish for every word. Restricting the cofinal readout by word length and node shows that all node measurements also vanish for every word. Hence $v = 0$ in M. The joint readout is therefore invertible on its image. Compatibility with each letter makes this inverse commute with the actions. For the seed orbit, applying the zero-kernel characterizations of the three readouts in turn again gives a trivial joint kernel. The full proof appears in Theorem 9.3 and Supplement S3.2 of *Source-Native Process Algebra*. The *Faithful Representations* paper provides the source code entry for the linear two-sided inverse on the image [16; 15].

For part 2, the synthesis operator $Su = \sum_i u_i c_i$ satisfies $\mathcal{G}_m = S^* S$ and $r(v) = S^* v$. In the finite-dimensional coefficient space, $\ker(S^* S) = \ker S$, so $\operatorname{ran}(S^* S) = (\ker S)^\perp = \operatorname{ran} S^*$. This proves consistency and (6.13). For every solution u of the normal equations, $v - Su$ is orthogonal to all c_i. A kernel offset z satisfies $Sz = 0$, so (6.14) is independent of the coefficient representative. The readout at this stage obtains the two columns from $e_i(\rho + \eta) = e_i(\rho) + (e_i(\rho + \eta) - e_i(\rho))$, preserving the cross inner products and the original written value. If the old event segment starts at address b_0 in the next packet, its jth entry reads the original expression and coefficient at position $b_0 + j$. These determine both columns in the new environment, after which the matrix is recomputed. For $c_1 = c_2 = c \neq 0$, the normal equations fix only $u_1 + u_2$, the kernel contains $(1, -1)$, and the innovation is still $v - \langle c, v \rangle c / \|c\|^2$.

For part 3, the sum of the paired old value and effect equals the value written in that execution. Subtracting the constant $b + d$ changes the old value to $b - (b + d) = -d$ and preserves the effect d. Multiplication by the coefficient gives the stated pair. The addition equation $u + t = s$

uniquely determines the first coordinate as $s - t$, giving the entire fiber. For example, with $e = x, \kappa = 1, \mathcal{E}_0(x) = 1, \delta_{\text{birth}}(x) = 1$, the normalized word $x - 2$ has the pair $(-1, 1)$. The total reading is zero and the effect branch is nonzero. ■

Ordered trajectories and subsequent queries. For a word $w = a_1 \cdots a_k$ and its trajectory $x_s = A_{a_s} x_{s-1}$, set $\delta_s = E_{a_s} \mathcal{M}(x_{s-1}) - \mathcal{M}(x_s)$. Insert the successive intermediate terms $E_{a_{s+1} \cdots a_k} \mathcal{M}(x_s)$ and cancel them to obtain $E_w \mathcal{M}(x_0) - \mathcal{M}(A_w x_0) = \sum_{s=1}^{k} E_{a_{s+1} \cdots a_k} \delta_s$. The complete trajectory stores (a_s, x_{s-1}, δ_s) at each position. Embed the jth position of the recorded sequence into the corresponding free word module over the integers. The native successor acts on these basis elements by shifting. Fix a query. At all subsequent steps n and all finite truncations b, compare two words using the node readouts O_b. Then $Q(u) = Q(v)$ if and only if $O_b S^n u = O_b S^n v$ for all n, b. Together with the retained representative in the fiber, $Q(u)$ recovers the original word and carries it into the next query. The source program retains the readout of the current computation, the updated descendant environment, and the resulting material. The difference of the two actual evaluations is a separate request that writes the ordered effect and inverse residual to the same record. For the environment generation and writing equations, see Theorems 9.5–9.9 and 10.1–10.2 of *Source-Native Process Algebra* [16].

6.5 Gamma Mixtures of Heterogeneous Exponential Relaxation and Completely Monotone Shifted Power-Law Forgetting

Multichannel receivers and models of neural memory provide examples of decay on different scales [14; 44]. If the retention in a single channel is e^{-xt}, a Gamma distribution of decay rates $x > 0$ gives a shifted power law with a long tail (Lomax or Pareto II). For this transform and complete monotonicity on the positive half-line, see [4; 43; 29]. The following formal proof connects integration against the probability measure, time derivatives of every order, and convergence of sampling at every time point.

Theorem 6.9 (Gamma mixture identity for a shifted power law, effective decay rate, and complete monotonicity in time). Let the Gamma distribution have shape parameter $\alpha > 0$ and rate parameter $r > 0$, and let time satisfy $t \geq 0$.

1. **Laplace identity for the Gamma density and probability measure.** Let $\gamma_{\alpha,r}(x) = \frac{r^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-rx} \mathbf{1}_{(0,\infty)}(x)$ be the normalized Gamma density on the positive half-line, with probability measure $\text{Gamma}(\alpha, r)$. The integral of the microscopic exponential retention kernel $x \mapsto e^{-xt}$ against this measure, or equivalently, for any random decay rate $X \sim \text{Gamma}(\alpha, r)$, its expectation $\mathbb{E}[e^{-Xt}]$, is the shifted power-law retention curve $\Psi_{\alpha,r}(t) := (1 + t/r)^{-\alpha}$:

$$\int_{(0,\infty)} e^{-xt} \, \mathrm{dGamma}_{\alpha,r}(x) = \frac{r^\alpha}{\Gamma(\alpha)} \int_0^\infty x^{\alpha-1} e^{-(r+t)x} \, \mathrm{d}x = r^\alpha (r+t)^{-\alpha} = \left(1 + \frac{t}{r}\right)^{-\alpha}. \quad (6.15)$$

2. **Complete monotonicity on nonnegative time and the half-rate point.** Write $\alpha^{\overline{n}} := \prod_{k=0}^{n-1} (\alpha + k)$ for the rising factorial (Pochhammer polynomial), with $\alpha^{\overline{0}} = 1$. Pull the power $f_\alpha(x) = x^{-\alpha}$ on $x > 0$ back along the affine time change $x = 1 + t/r$. For all orders $n \in \mathbb{N}$ and

all $t \geq 0$, its derivatives have the closed form and strict alternating signs

$$\frac{\mathrm{d}^n}{\mathrm{d}t^n}\left(1+\frac{t}{r}\right)^{-\alpha} = (-1)^n\, r^{-n}\, \alpha^{\overline{n}} \left(1+\frac{t}{r}\right)^{-\alpha-n}, \qquad (-1)^n \Psi^{(n)}_{\alpha,r}(t) > 0; \tag{6.16}$$

The logarithmic retention $\log \Psi_{\alpha,r}(t) = -\alpha \log(1+t/r)$ induces the instantaneous effective decay rate $\lambda_{\alpha,r}(t) := -\frac{\mathrm{d}}{\mathrm{d}t} \log \Psi_{\alpha,r}(t) = \frac{\alpha/r}{1+t/r}$. This rate is positive and strictly decreasing for $t \geq 0$. Its initial value is $\lambda_{\alpha,r}(0) = \alpha/r$, and at the characteristic time $t = r$ it is exactly half that value, $\lambda_{\alpha,r}(r) = \frac{\alpha}{2r}$.

3. **Transferring uniform error bounds for finite exponential sums and sample-path convergence.** Suppose a finite multichannel exponential sum $M_K(t) = \sum_{k=1}^K w_k e^{-x_k t}$ satisfies, on a time set $\mathcal{T} \subseteq [0,\infty)$, the bound $\sup_{t\in\mathcal{T}} \left|M_K(t) - \int_0^\infty e^{-xt} \gamma_{\alpha,r}(x)\,\mathrm{d}x\right| \leq \delta$. Then $\sup_{t\in\mathcal{T}} \left|M_K(t) - (1+t/r)^{-\alpha}\right| \leq \delta$. In particular, the empirical Laplace averages of pairwise independent, identically distributed samples converge almost surely to $(1+t/r)^{-\alpha}$ on any finite observation grid. With a countable dense time grid and a uniform Lipschitz bound, convergence holds at every nonnegative time on the same set of sample paths of probability one (Appendix B.7).

Proof. Since $\alpha > 0, r > 0, t \geq 0$, the combined rate $b := r+t > 0$ and the exponent $\alpha - 1 > -1$. On the positive half-line $(0,\infty)$, substitute $u = bx$ ($\mathrm{d}x = b^{-1}\mathrm{d}u$). The definition of the Gamma integral gives

$$\int_0^\infty x^{\alpha-1} e^{-(r+t)x}\,\mathrm{d}x = (r+t)^{-\alpha} \int_0^\infty u^{\alpha-1} e^{-u}\,\mathrm{d}u = (r+t)^{-\alpha}\Gamma(\alpha).$$

Since $\Gamma(\alpha) > 0$, multiplication by the normalizing factor $r^\alpha/\Gamma(\alpha)$ cancels $\Gamma(\alpha)$. The identity $r^\alpha (r+t)^{-\alpha} = (1+t/r)^{-\alpha}$ for positive real powers proves the density integral equality. Passing to the absolutely continuous measure (the point $\{0\}$ has measure zero) and applying the integration formula for the pushforward law of the random variable gives equality of $\mathbb{E}[e^{-Xt}]$ and the density integral for every parameter, proving (6.15).

For part 2, differentiate first on $x > 0$ to obtain $f_\alpha^{(n)}(x) = (-\alpha)(-\alpha-1)\cdots(-\alpha-n+1)x^{-\alpha-n} = (-1)^n \alpha^{\overline{n}} x^{-\alpha-n}$. Apply the chain rule for translation and multiplication inductively to the affine composition $t \mapsto f_\alpha(1+r^{-1}t)$, using $\frac{\mathrm{d}^n}{\mathrm{d}t^n} g(a+ct) = c^n g^{(n)}(a+ct)$ with $a = 1, c = r^{-1}$. This gives the first identity in (6.16). For every $t \geq 0$, we have $1 + t/r \geq 1 > 0$, $r^{-n} > 0$, $\alpha^{\overline{n}} > 0$, and $(-1)^{2n} = 1$, so the product is positive, proving complete monotonicity in time. Taking the negative derivative of $-\alpha \log(1+t/r)$ gives $\lambda_{\alpha,r}(t) = \frac{\alpha r^{-1}}{1+r^{-1}t}$. Substituting $t = 0$ and $t = r$ verifies the initial rate and half-rate point. The denominator increases with t, so $\lambda_{\alpha,r}(t)$ is nonincreasing. For part 3, combine (6.15), the strong law of large numbers, and the $\epsilon/3$ triangle inequality on a dense grid (Appendix B.7). ■

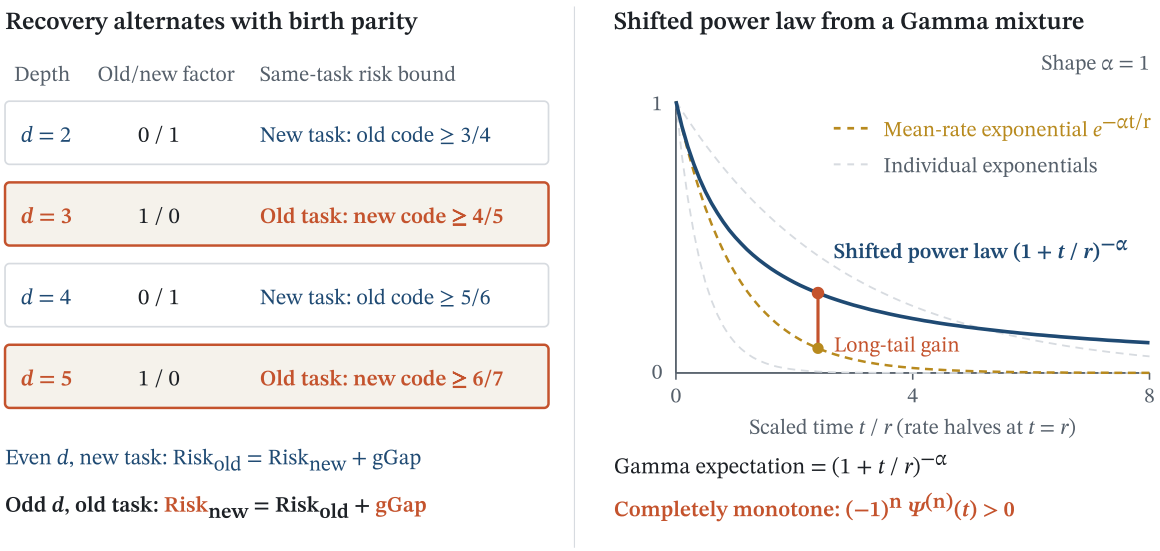


Figure 5. Risk reversal in parity channels and retention under Gamma mixing. The left panel gives the old and new parity multipliers at depths $d = 2, 3, 4, 5$. The new code improves the new task at even depths and worsens the old task at odd depths. The worse code always incurs risk at least $(d+1)/(d+2)$ (Theorem 6.6). The right panel uses $\alpha = 1$. Gamma mixing of heterogeneous exponential components e^{-xt} gives the dark blue shifted power law $(1+t/r)^{-\alpha}$. The dashed gold curve is the single exponential with the same mean rate, $e^{-\alpha t/r}$. The instantaneous decay rate halves at $t = r$ (Theorem 6.9).

7 Dissipative Gate-Level Non-Resetting Measurement Circuits: Blind Recovery of 1024 Commands, On-Time Microexecution, and Dual-Rail Capacitor Feedback

This section implements observation and feedback in a circuit model. A series RLC circuit generates the physical signals. The ADC encodes three sets of reference and operating voltages in a wire packet, and the gate circuit recovers the drive from the packet and produces feedback. Bennett analyzed the cost of erasing old records when memory is reused and discussed simultaneous erasure and measurement [3, pp. 927–929]. We show that a dissipative dual-rail RC gate network can start the next frame directly from the previous capacitor voltages. We quantify its timing guarantees and bound the error of recovering elapsed time from a single packet of finite width.

7.1 Blind Recovery of 1024 Commands, Quotient Isomorphism, and a 10-Bit Lower Bound

Consider a dissipative, multichannel series RLC test fixture with clock, noise, and metering constraints. Its controlled input is the binary drive vector $d = (d_0, \dots, d_9) \in \mathcal{D} = \{0,1\}^{10}$ of 10 independent Boolean channels, giving $|\mathcal{D}| = 2^{10} = 1024$ command configurations. Fix a positive sampling clock period $\tau > 0$. Each physical run evolves according to the driven Kirchhoff circuit. At the tick $t_{\text{sample}} = N_{\text{sample}}\tau$ specified by the synchronous sampling rule, it packages the clock counter and $60 = 10 \times 2 \times 3$ offset-encoded signed ADC words into the digital sample packet $\mathcal{S}(d)$. On each channel $c \in \{0, \dots, 9\}$, the series resistor and inductor branches each provide a zero reference frame (zero), a full-scale span frame (span), and an operating frame (op), for a total of 6 signed words. Each signed ADC word uses 23-, 24-, or 25-bit offset encoding. The validity predicate

characterizes lossless packing and parsing in both directions. The wire format of the whole packet has cardinality $2^{b_{\text{clock}}+60b_{\text{ADC}}}$.

Theorem 7.1 (Blind recovery of 1024 commands, unique isomorphism of the drive indistinguishability quotient, and a 10-bit word-length lower bound). For every valid clocked, noisy, dissipative ADC test fixture source, the following hold.

1. **Local self-calibrating blind decoding and single-channel isolation.** From the 60 ADC words and sampling-tick count of $\mathcal{S}(d)$, the local blind decoder $\text{decode}_{\text{ADC}}$ recovers the drive without access to its original input. For all 1024 drives, it satisfies the left-inverse identity $\text{decode}_{\text{ADC}}(\mathcal{S}(d)) = d$. In particular, a drive exciting just one channel decodes to true on that channel and false on the other nine.
2. **Unique commuting isomorphism of the drive indistinguishability quotient.** The equivalence kernel of the digital sampling map $\mathcal{S} : \mathcal{D} \to \operatorname{ran}\mathcal{S}$ equals the indistinguishability relation $\sim_{\text{read}}$ of the downstream Boolean drive readouts. The induced quotient isomorphism $\mathcal{D}/\!\sim_{\text{read}} \simeq \operatorname{ran}\mathcal{S}$ is a bijection, and each downstream drive readout factors uniquely through $\operatorname{ran}\mathcal{S}$. For any other complete representation $\tilde{\mathcal{S}} : \mathcal{D} \to \operatorname{ran}\tilde{\mathcal{S}}$ of the same hardware source preserving the same readout kernel, the commuting identity $\Phi(\mathcal{S}(d)) = \tilde{\mathcal{S}}(d)$ for each drive determines a unique bijection $\Phi : \operatorname{ran}\mathcal{S} \xrightarrow{\sim} \operatorname{ran}\tilde{\mathcal{S}}$.
3. **Sampling image of size 1024 and a 10-bit lower bound.** The ambient type of clock counts is the infinite set $\mathbb{N}$. The source-generated sampling image has cardinality $|\operatorname{ran}\mathcal{S}| = 1024$. By injectivity, every finite-carrier representation $\rho : \mathcal{D} \to \Omega_{\text{rep}}$ and b-bit word encoding $\rho_b : \mathcal{D} \to \text{Fin}(2^b)$ preserving the drive readout kernel satisfy

$$|\Omega_{\text{rep}}| \geq 1024, \qquad b \geq 10. \tag{7.1}$$

Proof. For part 1, on each channel $c \in \{0, \ldots, 9\}$, the blind decoder uses the three-frame self-calibration ratio $(V_{c,\text{op}} - V_{c,0})/(V_{c,\text{span}} - V_{c,0})$ to cancel channel gain and zero-offset drift. At sampling phase $N_{\text{sample}}\tau$, it demodulates the normalized amplitude estimate $\hat{u}_c$. Under the bounded thermal noise and quantization-step budget of a valid fixture, the combined error between the driven steady-state response and the Boolean bit $d_c \in \{0, 1\}$ satisfies $|\hat{u}_c - d_c| < 1/8 < 1/2$. The midpoint test $\hat{u}_c > 1/2$ therefore recovers d_c without error on each of the ten channels, giving $\text{decode}_{\text{ADC}}(\mathcal{S}(d)) = d$. Thus $\mathcal{S}$ is injective. The ten Boolean readouts read each bit of d, so two drives are indistinguishable if and only if $d_1 = d_2$, equivalently $\mathcal{S}(d_1) = \mathcal{S}(d_2)$. The natural bijection between the quotient and the sampling image gives $\mathcal{D}/\!\sim_{\text{read}} \simeq \operatorname{ran}\mathcal{S}$ and $|\operatorname{ran}\mathcal{S}| = |\mathcal{D}| = 2^{10} = 1024$. If $\tilde{\mathcal{S}}$ preserves the same readout kernel, it is also injective, and $\mathcal{S}(d) \mapsto \tilde{\mathcal{S}}(d)$ is the unique bijection making the triangle commute for each drive. Part 3 follows by injectivity and the finite cardinality comparison $2^{10} = 1024 \leq 2^b$. ■

7.2 128-Bit Gate-Level Energy Arithmetic and On-Time Microexecution for Every Raw Packet

For each channel, the receiver applies a division-free integer RL energy threshold test to the six signed ADC words $(r_0, r_{\mathrm{span}}, r_{\mathrm{op}}, \ell_0, \ell_{\mathrm{span}}, \ell_{\mathrm{op}})$ in the sample packet:

$$r_{\mathrm{span}} - r_0 > 0, \qquad \ell_{\mathrm{span}} - \ell_0 > 0,$$
$$(r_{\mathrm{span}} - r_0)^2(\ell_{\mathrm{span}} - \ell_0)^2 < 4\Big[(r_{\mathrm{op}} - r_0)^2(\ell_{\mathrm{span}} - \ell_0)^2 + (\ell_{\mathrm{op}} - \ell_0)^2(r_{\mathrm{span}} - r_0)^2\Big],$$

which tests whether the sum of the normalized squared resistive and inductive terms exceeds $1/4$. The supported 25-bit signed samples give differences bounded by $|\Delta| \le 3.2 \times 10^7$, so the right side can reach approximately 8.39×10^{30}, between 2^{102} and 2^{103}. For instance, in the finest code set both zero references to $-16{,}000{,}000$ and both span and operating words to 16,000,000. The right side then exceeds the range of 64-bit arithmetic. The following 128-bit gate program covers every valid sample word and returns a timely acceptance or rejection for every raw packet.

Theorem 7.2 (128-bit gate-level energy arithmetic and on-time micro-execution for every raw wire packet). Fix any supported ADC resolution code, counter width, and upper-bound parameter. Install the gate program compiled for that code and width.

1. **Overflow-free 128-bit gate arithmetic for six words per channel.** The AND-inverter graph (AIG) circuits for 128-bit addition, subtraction, multiplication, and signed and unsigned comparisons are correct bit for bit on all 128-bit operands. After embedding the six sample words in 128-bit two's-complement form, the receiver performs subtraction, signed squaring with modular ring semantics, quartic products, summation, scaling by 4, and comparisons of both sides. This six-stage pipeline agrees field by field with the unbounded integer RL energy inequality above. The five stages starting at squaring are free of unsigned overflow.
2. **On-time completion and no early accepted output for every raw packet.** The clocked microprogram starts with an empty frontier and writes one declaration at a time into a fixed buffer, with fan-in confined to the written prefix. For every raw wire packet p, it reaches a terminal state by the deadline

$$L_{\mathrm{prog}} = |G_{\mathrm{pkt}}| + |G_{\mathrm{cal}}| + |G_{\mathrm{sub}}| + 3|G_{\mathrm{mul}}| + |G_{\mathrm{add}}| + \max(|G_{\mathrm{slt}}|, |G_{\mathrm{ult}}|) + 9$$

in ticks, as determined by the gate declaration array sizes of the installed stages. The output is an accepted command or a null rejection result, in agreement with the high-level energy test. An accepted packet cannot expose a completed output at any tick $t < L_{\mathrm{prog}}$. A rejected packet may be explicitly rejected at the packet-header or calibration stage and then remain idle. The entire microcontroller has a finite state space and may reenter the same receiver program for any subsequent packet.

Proof. For part 1, two's-complement subtraction represents the signed integer difference $\Delta \in [-3.2 \times 10^7, 3.2 \times 10^7]$. Squaring uses the ring homomorphism modulo 2^{128}, giving $(a \bmod 2^{128})^2 \equiv a^2 \pmod{2^{128}}$. Since $0 \le \Delta^2 \le 1.024 \times 10^{15} < 2^{128}$, the result is an unsigned nonnegative integer. The subsequent quartic products, sum, and scaling by 4 attain a maximum $\le 8.39 \times 10^{30} < 2^{128}$. Together with the two signed positive-span comparisons and the final unsigned comparison, this

gives bitwise agreement with the integer test. For part 2, L_{prog} sums the gate declaration counts for packet checking, calibration, subtraction, three multiplications (squaring, cross multiplication, and scaling by 4), addition, and parallel signed and unsigned comparisons. It adds 7 stage-commit ticks and 2 final Boolean conjunction control ticks, for a total of 9 control ticks. If the header or calibration check fails, the program enters an absorbing rejection state. Otherwise all graph declarations and commits run in sequence within L_{prog}. Every packet therefore has a terminal result by the deadline. For induction, define the expected-result projection $\mathcal{R}_{\text{micro}}(s)$, which does not occupy any storage in the hardware. At all nine stage boundaries it satisfies the one-step invariant $\mathcal{R}_{\text{micro}}(\text{step}(s)) = \mathcal{R}_{\text{micro}}(s)$. For each accepted packet, initially $\mathcal{R}_{\text{micro}}(s_0) = \text{some}\, d$. The invariant rules out entry into an early terminal rejection state, since that would give $\text{some}\, d = \text{none}$. At all steps without early rejection the rank of the remaining work drops by exactly 1. Thus for all $t \le L_{\text{prog}}$ we have $\text{remaining}(s_t) = L_{\text{prog}} - t$. At $t < L_{\text{prog}}$ the remaining rank is > 0 and no output appears. At $t = L_{\text{prog}}$ the rank reaches zero and the program emits drive d. ■

7.3 Shared AIG Receiver, Non-Resetting Dual-Rail Capacitor Handoff, and Non-Zeno Progress

Compile packet-header admission and the ten channel energy tests into a shared AND-inverter graph (AIG), with one validity bit and ten channel decisions. Each registered input line and each capacitive dual-rail gate branch $p \in \{+, -\}$ have physical load capacitance $C_{v,p} = C_{\text{int}} + N_{\text{load}}(v, p)C_{\text{pin}} > 0$, where N_{load} counts downstream loads. Constant branches provide fixed voltages. The receiver feeds its gate-level decision back to the controlled RLC circuit.

Theorem 7.3 (Commutation of installed micro-execution and delayed physical evolution at every finite step). Take the source RLC hardware, boot initial state, and valid microprogram installation $\mathcal{J}$ generated from them, with processing ticks $L_{\mathcal{J}} > 0$. One step on the installed current state x equals the delayed physical successor after erasing the compile-time tick bound witness. At switching, the next initial state directly inherits the continuous endpoint of the old run at sampling plus processing delay:

$$\text{erase}(\text{step}_{\mathcal{J}}\, x) = \text{physicalNext}_{L_{\mathcal{J}}}(\text{erase}\, x), \qquad (V, I)_{\text{next}}(0) = (V, I)_{\text{old}}(t_{\text{sample}} + L_{\mathcal{J}}\tau). \tag{7.2}$$

This semiconjugacy holds after every finite number of steps $k \in \mathbb{N}$, tracking the unique current frontier of the same physical history.

Proof. The installed state x pairs the underlying continuous physical trajectory state with a proof field for the compile-time microclock deadline. At tick $L_{\mathcal{J}}$, the packet generated by the circuit yields the current command, whose channels are complemented for feedback. A step of the installed transition takes the old circuit's continuous port state at $t_{\text{sample}} + L_{\mathcal{J}}\tau$, namely (V, I), as the initial state of the next driven interval. Erasing the proof field therefore gives pointwise equality with the underlying delayed physical transition. Induction on k proves commutation at every step. ■

Theorem 7.4 (Stable classification, capacitor handoff across frames, and non-Zeno progress for the shared AIG). Fix the hardware, gate technology parameters, and downstream load graph. The initial voltages on registered inputs and gate branches lie within their valid supply rail intervals.

1. **Stable classification of every raw packet.** For any raw packet of the specified wire format, each bit generates a continuous line drive. At the prescribed sampling tick, the shared AIG gives 11 stable outputs. Its classification agrees with the micro-execution of Theorem 7.2. The result may be "no decision" while some voltages are still unclassified. After settling, a validity bit of zero indicates rejection, and a validity bit of one makes the ten channel decisions available.
2. **Handoff for every finite number of frames of an actual run.** Start from a snapshot generated by the hardware. Each frame reads its own packet and uses the decision and feedback to generate the next frame. The joint state J_k records the controlled RLC state, clock capacity, voltages on registered lines, and voltages on capacitive branches. Let $\mathcal{C}_{\text{store}}$ be the corresponding storage locations. For every k, the frontier at frame k consists of J_k alone, and stable reception recovers the drive d_k of that frame's snapshot. Let $t_{\text{end}}(J_k)$ be the receiver time from the old ADC snapshot to the next snapshot. It equals the processing wait t_{read} after the old snapshot plus the time from switching to sampling in the next circuit interval. Every storage capacitor hands off its voltage as follows, with no dedicated reset-to-zero phase:

$$\boxed{\forall (v,p) \in \mathcal{C}_{\text{store}}, \qquad \mathcal{V}_{J_{k+1}}(v,p;\,0) = \mathcal{V}_{J_k}(v,p;\,t_{\text{end}}(J_k)).} \tag{7.3}$$

   The RLC circuit hands off at the earlier feedback switch, with the continuous initial state in (7.2). The old packet drives the lines until the next ADC snapshot. The receiver then switches to the new packet from the inherited voltages.
3. **Non-Zeno progress.** If Δ_j is the physical interval between successive switches of the controlled circuit, then $\Delta_j \geq \tau > 0$ for each frame. Hence for every $T_{\max} \in \mathbb{R}$ there is a finite number of frames k such that

$$\text{elapsed}(k) = \sum_{j=0}^{k-1} \Delta_j > T_{\max}. \tag{7.4}$$

Proof. The line drive for each raw bit relaxes continuously from its given old voltage toward the target rail. A capacitive cell contracts in the voltage band set by its effective inputs with time constant $R_{\text{gate}}C_{v,p}$. Its allocated settling time is $1152R_{\text{gate}}C_{v,p}$. Accumulate the relevant settling times by topological depth in the acyclic gate graph, then choose a capture tick with the clock. This gives all 11 logical outputs. The packet-header, span, and energy tests yield the validity bit and the ten decision bits, which give the terminal output of the microprogram. The argument holds packet by packet, including rejected inputs.

On the hardware's own snapshots, the receiver recovers the current drive exactly. Feedback from that drive generates a new RLC interval with the old circuit's switching endpoint as its initial state. During the wait, the receiver continues evolving along the old waveform. At the next ADC snapshot, the endpoints of all its storage capacitors supply the initial voltages of the next frame. The rail

intervals remain invariant, so the next frame again meets the classification and handoff conditions. Induction over frames gives (7.3) and both handoffs for every finite continuation. Finally, each interval of the circuit contains at least one sampling clock period, and the processing delay is nonnegative. Thus elapsed$(k) \geq k\tau$, and choosing $k > T_{\max}/\tau$ gives (7.4). ■

The new packet overwrites the old node voltages through resistors, with Joule dissipation during the transition. The separate paper on finite-supply physical computation further analyzes the work–heat identity for capacitors with finite stored energy and signed initial states.

7.4 Drive Parity Collisions and a Time-Distortion Lower Bound for Fixed-Width Packets

The circuit gives current feedback across any finite number of frames. For an observer who retains just the compressed drive signal or one current fixed-width packet and tries to recover the current point in the physical history, the fiber formula in Lemma 2.6 gives a distortion lower bound in squared seconds (s^2).

In the same loaded run, the jth actual state starts at elapsed physical time t_j, with $t_0 = 0, t_{j+1} = t_j + \Delta_j$. The physical duration of each step satisfies

$$\Delta_j \geq \tau > 0, \qquad t_j - t_i \geq (j-i)\tau \quad (0 \leq i \leq j). \tag{7.5}$$

Theorem 7.5 (Time loss in squared seconds under two-fiber drive compression, with the duration-dependent cross term). Among the first $n = N + 1$ equally weighted original occurrences, feedback complements the drive at every step. The drive observation, and likewise the complete future drive sequence, gives two parity fibers F_0, F_1 with cardinalities m_0, m_1, where an empty fiber contributes zero. The conditionally optimal squared error for recovering elapsed physical time t_i is

$$\mathsf{D}_{\mathrm{drive}} = \frac{\tau^2}{3n}\sum_{b=0}^{1} m_b(m_b^2-1) + \mathrm{Excess}, \qquad \mathrm{Excess} := \frac{1}{2n}\sum_{b:\, m_b>0}\frac{1}{m_b}\sum_{i,j\in F_b}\left((t_i-t_j)^2 - \tau^2(i-j)^2\right) \geq 0, \tag{7.6}$$

and, for every time decoder, including one that reads the complete future drive sequence, the conditional entropy $H = H(I \mid \mathrm{drive})$ gives

$$\mathrm{Risk} \geq \mathsf{D}_{\mathrm{drive}} \geq \frac{\tau^2}{3}(e^{2H}-1). \tag{7.7}$$

If the recovery target advances to the next step's elapsed time $t_i + \Delta_i$, let R_{drv} be the residual operator of the projection conditional on the drive. The next time error is $\mathsf{D}_{\mathrm{drive}} + 2\,\mathrm{Re}\langle R_{\mathrm{drv}}t, R_{\mathrm{drv}}\Delta\rangle + \|R_{\mathrm{drv}}\Delta\|^2$.

Proof. Apply the first formula of (2.9) to t_i. Separate the baseline squared spacing $\tau^2(i-j)^2$ and the excess term Excess, nonnegative for each pair by (7.5). Indices in the same parity fiber are spaced by 2, so $i - j = 2(u - v)$. The finite sum identity gives $\sum_{u,v=0}^{m_b-1}(u-v)^2 = \frac{1}{6}m_b^2(m_b^2-1)$. Multiplication by $2^2 = 4$ and division by $2nm_b$ give the coefficient $\frac{4}{12n} = \frac{1}{3n}$, proving (7.6). Substituting $\sum_b \frac{m_b}{n}(m_b^2-1) = (\sum_b \frac{m_b}{n}m_b^2) - 1$ into the last inequality of (2.9), $e^{2H} \leq \sum_b \frac{m_b}{n}m_b^2$,

gives (7.7). For the target $t + \Delta$, expand the squared Hilbert residual norm $\|R_{\mathrm{drv}}(t + \Delta)\|^2 = \|R_{\mathrm{drv}}t + R_{\mathrm{drv}}\Delta\|^2$ to obtain the cross term. ■

Now take the full contents of one current record packet as the input. By Section 7.1, the wire packet has a fixed total bit count

$$B = b_{\mathrm{clock}} + 60b_{\mathrm{ADC}}, \qquad M := 2^B. \tag{7.8}$$

The packet header encodes the bounded within-frame sampling tick exactly. The task t_i in the theorem is the time elapsed since boot.

Theorem 7.6 (Capacity lower bound in squared seconds and eventual failure of fixed-width single-packet decoders). For the first $n = N + 1$ equally weighted original occurrences of the same loaded physical run, every time decoder $d : \mathrm{Packet} \to \mathbb{C}$ taking just one current record packet has squared time-recovery error satisfying

$$\mathrm{Risk}_N(d) \geq \frac{\tau^2}{12}(e^{2H(I|\mathrm{packet})} - 1) \geq \frac{\tau^2}{12}\left[\left(\frac{n}{M}\right)^2 - 1\right]. \tag{7.9}$$

Thus for any prescribed tolerance $\epsilon \in \mathbb{R}$, there is a finite history threshold $N_0 \in \mathbb{N}$ such that for all $N \geq N_0$, even when a different decoder is chosen for each history length N, every single-packet decoder d has error strictly above the squared tolerance:

$$\forall N \geq N_0, \quad \forall d : \mathrm{Packet} \to \mathbb{C}, \qquad \mathrm{Risk}_N(d) > \epsilon^2. \tag{7.10}$$

Proof. There are at most $M = 2^B$ packet codes as the history length n grows. Once $n > M$, some packet fiber contains distinct times from the history. For a nonempty fiber of cardinality m, list its members in increasing order of occurrence, $i_0 < i_1 < \cdots < i_{m-1}$. For any $u > v$, we have $i_u - i_v \geq u - v$, so (7.5) gives $(t_{i_u} - t_{i_v})^2 \geq \tau^2(u - v)^2$. Summing yields $\sum_{i,j\in F_y}(t_i - t_j)^2 \geq \frac{\tau^2}{6}m^2(m^2-1)$. The minimum index spacing for a general packet fiber is 1, compared to 2 for a parity fiber. This changes the denominator from 3 to 12. Substitute into (2.9) to obtain the first inequality. The entropy decomposition gives $H(I \mid \mathrm{packet}) = \log n - H(\mathrm{packet}) \geq \log n - \log M = \log(n/M)$. Substituting it in the exponential term gives the second inequality of (7.9). Since $\tau > 0$ and $M = 2^B$ is fixed, when $n = N + 1 > M\sqrt{1 + 12\max(0, \epsilon^2)/\tau^2}$ the lower bound on the right is independent of d and strictly exceeds ϵ^2. This proves (7.10). ■

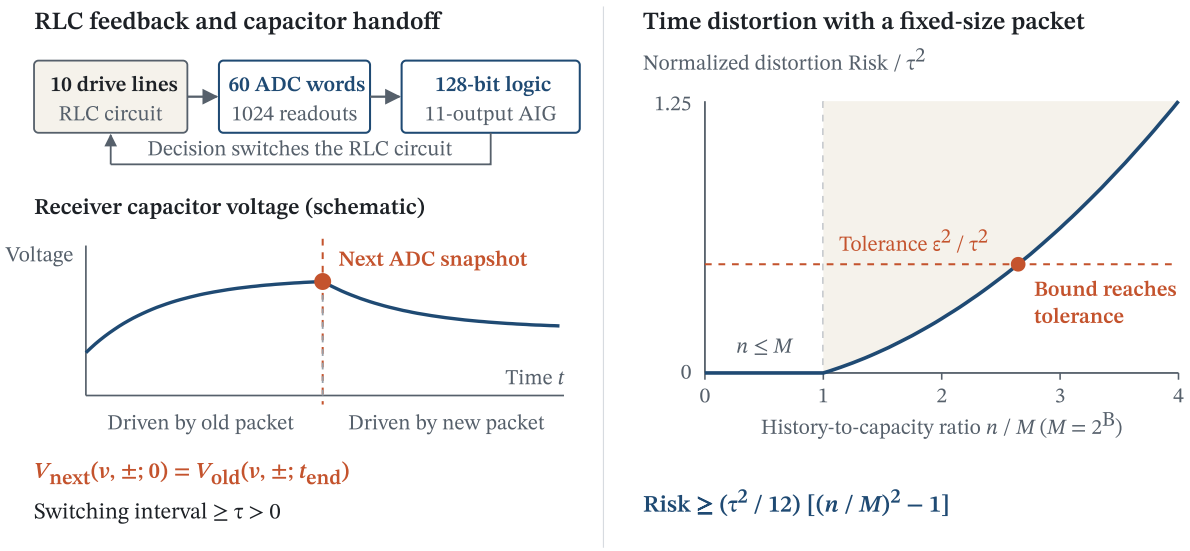


Figure 6. Capacitor handoff between frames and a lower bound on elapsed-time error from one packet. At upper left, the hardware generates a valid packet and the receiver returns feedback. At lower left, a capacitive branch hands off its voltage between adjacent ADC snapshots. The red dot marks when the new packet starts driving the branch. The RLC circuit switches after the feedback decision and then generates the next snapshot. The right panel plots the nonnegative normalized lower bound $\max(0, [(n/M)^2 - 1]/12)$ from (7.9), where $M = 2^B$. The shaded region lies above the lower bound. At the red dot the bound equals the tolerance. To its right, every single-packet decoder strictly exceeds the tolerance.

8 Related work

Observation is a quotient process within the source evolution. Its construction draws on linear systems over commutative rings, automata and coalgebraic minimization, ergodic averages and inverse problems, streaming statistics and the comparison of experiments, completely monotone forgetting models, and the thermodynamics of measurement circuits.

8.1 Observable invariant subspaces, commutative rings, and refinement

For finite-dimensional linear systems over fields, Kalman established the observable decomposition and the theory of minimal realization [23]. Basile and Marro, and later Wonham, developed geometric control methods that characterize unobservable modes by the largest A-invariant subspace in the output kernel [2; 45]. Rouchaleau, Wyman, and Kalman, and then Sontag, developed the module-theoretic realization theory for systems over commutative rings [35; 39].

For discrete state machines, Moore and Nerode characterized minimal quotients through input–output experiments and right-invariant equivalence relations [28; 30]. Hopcroft's minimization algorithm takes $O(n \log n)$ time for deterministic automata over a fixed alphabet. Paige and Tarjan's relational refinement algorithm takes $O(m \log n)$, where m is the number of edges in the relation [19; 31]. Rutten characterized infinite behavioural quotients through final coalgebras [36]. Abadi and Lamport showed that history and prophecy variables can complete refinement mappings under machine closure, internal continuity, and finite invisible nondeterminism. They also gave examples in which history variables alone are insufficient [1].

Four results connect this work to those traditions. First, Theorems 2.2 and 2.4 and Corollary 2.5 characterize the one-step obstruction by $\mathrm{actionDefect}_{A,o} = 0$ on a module over any commutative ring, with no division. The forced value $F(0) = 0$ yields $\ker P_\ell = K_\infty$ as the criterion even for arbitrary nonlinear or nonmeasurable maps of sets. Second, for finite states, multiple operations, and heterogeneous dependently typed readouts, Theorem 2.9 bounds refinement by the number $|P_0|$ of initial state pairs. When a readout is added, incremental refinement reuses the previous quotient and gives exactly the kernel of full recomputation, $K'_{\mathrm{inc}} = K_*(\mathcal{J}_{\mathrm{read}} \cup \{\mathrm{new}\})$. This is a kernel equality and a result about reuse of the previous partition. Hopcroft's and Paige–Tarjan's complexity bounds concern the algorithms just described. Third, for evolving systems too large to enumerate, Theorems 2.10–2.11 construct an escape witness that supplies missing coordinates, show how execution can resume from a complete residual packet, and give an example that replays an 8-byte success key with guards on three clocks. Fourth, the current source action and actual prefix generate finite-prefix recovery (Theorem 2.8) and the continuous request interval for closed syntax (Proposition 2.12). These constructions require no external prophecy variables.

8.2 Mean ergodic theory, Landweber regularization, and dependent-prefix truncation

Strong convergence of the arithmetic means of a nonexpansive operator on a Hilbert space and the $O(1/n)$ rate on vectors of the form $x = z - Az$ are classical mean-ergodic results [41; 34; 24; 9]. For linear inverse problems of the first kind, Landweber's adjoint residual iteration with the self-adjoint nonnegative operator $I - cK^*K$, its convergence rates under source conditions, and its noise-dependent stopping rules are established [26; 12].

Theorem 2.7 treats the general driven nonexpansive step $S_d(u) = Au + d$ and the adjoint step $A_\Theta = I - \Theta^*\Theta$ in one Birkhoff-average argument with zero initial value. This includes non-self-adjoint contractions whose spectrum meets other points of the unit circle and gives a finite-step perturbation budget for operator truncation. Theorem 2.8 treats a system $C_s \to C_{\sigma(s)}$ whose carrier depends on the state and whose future observation operators need no uniform norm bound. Adaptive scaling by $(1 + \|E_{s,j}\|)^{-1}$ and geometric weights $2^{-(j+1)}$ give a recovery operator that reads only the first n actual observations. Its truncation error relative to the full-history average has the explicit exponential bound $n2^{-n}\|x\|$, with strong recovery of the induced successor on the typed coimage. Theorem A.1 uses graph completion and the Fréchet–Riesz theorem on a closed relation quotient to retain a potentially unbounded linear functional. Neither injectivity of the original embedding nor closedness of the relation image is required [9].

8.3 Observability indices, smooth nonlinear observability, and sharp window bounds

For linear multivariable systems the minimum number of steps until the unobservable kernel chain stabilizes is the classical observability index [22; 45]. For smooth nonlinear systems, Hermann and Krener characterized local observability through the rank of the codistribution generated by successive Lie derivatives along smooth vector fields [18]. Takens proved that a generic smooth system on a compact m-dimensional C^2 manifold gives an embedding into a delay space of dimension $2m + 1$ [40].

Section 3 establishes a sharp global threshold for a fixed observer that reads all columns of the infinite-dimensional spatial, mass, and clock carrier G. Theorems 3.3 and 3.5 use a rational two-moment formula with positive denominator $\Delta > 0$ to recover the entire observable quotient in $a+1$ frames. Theorem 3.4 combines Corollary 2.5 with the dimension inequality $na+2 \le (\ell+1)n$ and an explicit two-frame zero-readout collision v_*. On all of G, every autonomous update needs at least $a+1$ frames, even when the map is discontinuous or nonmeasurable. In the discrete setting, Appendix B.4 likewise rules out a nonlinear shortcut for the intrinsic minimum stabilization horizon $h_* \le \deg \chi_A$ on a finite-rank free module over a nonzero commutative ring. Theorem 3.6 uses saturation in a single prime channel to give a shortest window of $\mathrm{cut}(p)+1$ frames and an explicit escaping source word on an infinite-rank free $\mathbb{Z}$-module.

8.4 Merging sufficient statistics, comparing experiments, and singular Gram systems

Classical statistical and computational tools include conditional expectation as an optimal orthogonal projection in L^2 [11, §4.1], one-pass means and blockwise variance merging [42; 8], Blackwell comparisons of experiments on a fixed parameter space [5], the plasticity–stability trade-off in synaptic cascades [14], Kailath's innovations approach for additive white noise [21], normal equations and the Moore–Penrose inverse [32; 33], and extensional incremental computation on streams [7; 6].

Sections 5 and 6 give three results for growing inventories with concurrent provenance.

1. Theorems 5.2–5.6 lift count-and-sum moment merging over commutative monoids to the infinite-dimensional carrier G and to full conditional posteriors. The unnormalized counted residual $c_y\varepsilon_y$ stays constant through every finite continuation. With the initial precision budget, testing whether each counted coordinate is $> 1/2$ recovers the discrete support exactly for both fine and coarse tables. Equation (5.8) includes both the variance innovation $+J_N$ and the contraction of the old noise term $-\frac{C_z}{C_z+1}\|\varepsilon_z\|^2$.
2. Theorems 6.1, 6.5, and 6.6 fix the time, the task, and the original history weights. Example 6.2 has zero excess risk in the scalar clock coordinate and positive excess risk in space. In an explicit family of copy channels modulo 2, preserving the lineage gives a persistent positive gain $\forall k, \mathrm{gGap}_k > 0$. Replacing channels at successive birth depths gives a parity reversal with a lower bound on the worse code's risk ($\ge (d+1)/(d+2)$).
3. Theorem 6.8 uses the finite-dimensional identity $\mathrm{ran}(S^*S) = \mathrm{ran}\, S^*$ [32] even when the empirical Gram matrix $\mathcal{G}_m$ is singular. It connects preservation of kernel fibers across cofinal levels, exact writeback of orthogonal innovations, and the affine pairing $(-\mathrm{effect}, \mathrm{effect})$ at birth. These results accompany the process-core and whole-ledger papers [16; 17].

8.5 Complete monotonicity and mixtures of exponential forgetting

Bernstein and Widder established the Laplace–Stieltjes representation of completely monotone functions on the positive half-line by nonnegative measures [4; 43]. Work on cognitive and neural memory includes empirical power-law forgetting [44], synaptic cascades with multiple timescales [14], and the analytical derivation of shifted power-law retention from Gamma-distributed decay rates [29].

Theorem 6.9 and Appendix B.7 formalize this mechanism in dependent type theory. They connect the normalized Gamma density on the positive half-line and the law $X \sim \mathrm{Gamma}(\alpha, r)$ to the Laplace expectation $(1 + t/r)^{-\alpha}$. Pulling back along $x = 1 + t/r$ gives a rising-Pochhammer formula for each derivative and strict complete monotonicity $(-1)^n \Psi^{(n)}_{\alpha,r}(t) > 0$. At the characteristic time $t = r$, the instantaneous decay rate has halved. Pairwise-independent sampling gives convergence on a countable dense time grid. A common Lipschitz bound then gives pointwise convergence throughout $[0, \infty)$. The sampling argument uses the strong law for pairwise independent, identically distributed variables [13].

8.6 Register reset, hybrid Zeno behavior, and a physical rate–distortion bound

Landauer linked many-to-one logical erasure to heat dissipation [25]. Bennett discussed measurement registers that either reset before a measurement or overwrite the old record during the next measurement [3, pp. 927–929]. Sagawa and Ueda related measurement work, erasure work, and mutual information in thermodynamic bounds [37]. In a hybrid system coupling continuous states and discrete events, a Zeno execution makes infinitely many switches in finite time [20]. The quantum Zeno effect concerns repeated projective measurements [27].

Theorems 7.3–7.4 give a dissipative realization with continuous overwriting. Feedback switches the RLC initial conditions, and the receiver capacitors carry their voltages across adjacent ADC snapshots. A positive dwell time excludes hybrid Zeno executions. Theorems 7.5–7.6 combine Shannon conditional entropy [38] with the minimum separation between historical times sharing a packet to lower-bound squared error for elapsed time. The code is the single-frame wire packet. Its capacity governs the distortion bound, whereas thermodynamic work measures energy.

9 Conclusion

For the fixed observer on the spatial, mass, and clock system, the complete observable quotient has a finite-dimensional realization and the all-future kernel determines its shortest window. Frame $a + 1$ makes mass and clock separately identifiable, yielding an explicit decoder and a sharp lower bound for every update map. For finite systems with heterogeneous readouts, adding a readout admits further refinement starting from the old minimum quotient.

A streaming receiver can keep updating exact posteriors as counts and their coordinate moments grow. The counted error from initialization stays bounded. Which channel gives better recovery depends on the conditional means for the same task. In the explicit parity-coupled family, keeping the available resolution gives a strict benefit across all successors. When channels are replaced at each birth depth, the advantage alternates. The circuit exhibits another form of continuing update. Each new snapshot generated by the hardware inherits the capacitor voltages at the end of the preceding snapshot. The receiver can decode each current packet, while using that packet alone to identify its absolute time in the full history incurs the capacity bound in (7.9).

Appendix A. Graph completion, closed relation quotients, and nonzero Riesz representatives

A linear functional may be unbounded in the observation norm, so the original observation completion need not carry it continuously. Include the functional itself as a graph coordinate, then take the quotient by the closure of the actual relations. The resulting space carries the unique continuous readout and, for a nonzero functional, its nonzero Riesz representative. The construction requires neither injectivity nor a preclosed relation image.

Theorem A.1 (Graph completion and the unique Riesz representative on the closed relation quotient). Let C, Rel be complex vector spaces and H a complex inner product space. Let $\iota : C \to H$, $\ell : C \to \mathbb{C}$, and $r : \mathrm{Rel} \to C$ be complex linear maps with $\ell \circ r = 0$. Equip the graph image $\mathrm{ran}(\iota, \ell) \subseteq H \oplus_2 \mathbb{C}$ with the induced inner product. Write $\mathcal{G} := \widehat{\mathrm{ran}(\iota, \ell)}$ for its completion and $j : C \to \mathcal{G}$ for the dense source map. Define the closed relation quotient

$$Q := \mathcal{G} \big/ \overline{\mathrm{ran}(j \circ r)}, \tag{A.1}$$

and let $\theta : C \to Q$ be the composite into the quotient. Then the following hold.

1. Q is a complete complex Hilbert space, $\theta(C)$ is dense in Q, and the quotient source map annihilates the relations ($\theta \circ r = 0$).
2. There is a unique continuous linear functional $\bar{\ell} : Q \to \mathbb{C}$ satisfying

$$\bar{\ell} \circ \theta = \ell; \tag{A.2}$$

3. If $\ell \neq 0$, then $\bar{\ell} \neq 0$, and there is a unique nonzero Riesz representative $\rho \in Q$ such that

$$\langle \rho, \theta x \rangle = \ell(x) \quad (\forall x \in C); \tag{A.3}$$

   the map $\bar{\ell} \mapsto \rho$ is a conjugate-linear isometry.
4. If H is already complete, then $\mathcal{G}$ embeds isometrically in $H \oplus_2 \mathbb{C}$ and is realized as the norm closure of the original graph image. Each source point reads back as $(\iota x, \ell x)$.

Proof. In the graph norm, every $x \in C$ satisfies $|\ell(x)| \leq \sqrt{\|\iota x\|^2 + |\ell(x)|^2} = \|(\iota x, \ell x)\|$. Projection onto the second coordinate on $\mathrm{ran}(\iota, \ell)$ therefore has norm at most 1. It extends uniquely to a continuous linear functional $\tilde{\ell} : \mathcal{G} \to \mathbb{C}$ with $\tilde{\ell} \circ j = \ell$. The hypothesis $\ell \circ r = 0$ makes $\tilde{\ell}$ vanish on $\mathrm{ran}(j \circ r)$. By continuity, $\tilde{\ell}$ also vanishes on the closed subspace $W := \overline{\mathrm{ran}(j \circ r)}$. The functional thus factors uniquely through $\bar{\ell} : Q = \mathcal{G}/W \to \mathbb{C}$, with $\bar{\ell} \circ \theta = \ell$. The image $j(C)$ is dense in $\mathcal{G}$, and the quotient map is a continuous surjection, so $\theta(C)$ is dense in Q. This also gives uniqueness among continuous linear functionals satisfying (A.2). Since $\bar{\ell} = 0$ implies $\ell = \bar{\ell} \circ \theta = 0$, we have $\ell \neq 0 \implies \bar{\ell} \neq 0$. The conjugate-linear Fréchet–Riesz correspondence on the complete complex Hilbert space Q gives the unique nonzero $\rho \in Q$ satisfying (A.3) [9]. If H is complete, so is $H \oplus_2 \mathbb{C}$, and the isometric inclusion extends uniquely to the closure of the original graph. ■

The co-Poisson–Müntz quotient at a conditional zero. In the quarter-period convergent Mellin test space C_z for $0 < \mathrm{Re}\, z < 1/2$, a compactly supported shell test witnesses that Mellin evaluation $\ell_z : C_z \to \mathbb{C}$ is nonzero for every $\mathrm{Re}\, z > 0$. The co-Poisson–Müntz map $r_z : \mathcal{S}(\mathbb{R}) \to C_z$

satisfies the annihilation condition $\ell_z \circ r_z = 0$ precisely at parameters z with $\zeta(2z) = 0$. The nontrivial zeta zeros on the critical line correspond to $\operatorname{Re} z = 1/4$ in this parameterization. At parameters with $\zeta(2z) \neq 0$, the annihilation hypothesis fails. At a parameter satisfying the zero condition, the original L^2 observation norm need not control ℓ_z. Theorem A.1 still gives a unique continuous linear functional $\bar{\ell}_z$ and a nonzero Riesz representative $\rho_z \in Q_z$ on the closed relation quotient Q_z. Both recover ℓ_z on each test function. The companion paper on the Burnol–Riesz source-operator calculus develops this conditional quotient and its resolvent analysis.

Appendix B. Empirical limits, growing histories, density matrices, and complete monotonicity

B.1 Shift invariance and compact lifting of empirical limits

For the same underlying process, let $q : S \to B$ be a readout with values in an abelian group B. Write F_q for the empirical field, the uniform completion generated by all future readout coordinates, and $S_q : F_q \to F_q$ for its induced shift. Let $\mathcal{C}_q$ be the set of cluster probability measures of the finite-prefix empirical distributions along the filter at infinity. If the coarse readout is $q' = f \circ q$ for a $\mathbb{Z}$-linear homomorphism $f : B \to B'$, applying that homomorphism coordinatewise gives the uniformly continuous map $\Phi : F_q \to F_{q'}$. At every prefix length the empirical distributions satisfy $\Phi_* \mu_{q,n} = \mu_{q',n}$.

Theorem B.1 (Shift invariance, surjective lifting through finite refinement, and the diagonal criterion). For any abelian group B, each empirical cluster point $\mu \in \mathcal{C}_q$ is a probability measure invariant under the shift S_q, so $(S_q)_* \mu = \mu$. If the fine codomain B is finite, with no separate finiteness hypothesis on B', then $\mathcal{C}_q$ is nonempty and the pushforward satisfies $\Phi_* \mathcal{C}_q = \mathcal{C}_{q'}$. If q_1, q_2 take values in finite abelian groups, the cluster set $\mathcal{C}_{(q_1,q_2)}$ of the joint readout (q_1, q_2) projects surjectively onto each marginal cluster set $\mathcal{C}_{q_1}, \mathcal{C}_{q_2}$. For the self-joint readout (q, q) of a finite q, marginal cluster points (μ, ν) admit a common joint lift if and only if $\mu = \nu$.

Proof. For a bounded continuous test function g, the difference between the empirical integrals at step n before and after one shift is bounded by the magnitudes of the first and last terms divided by n. It tends to zero as $n \to \infty$. Taking an ultrafilter limit gives $\int (g \circ S_q)\, \mathrm{d}\mu = \int g\, \mathrm{d}\mu$, establishing S_q-invariance. The uniformly continuous map Φ preserves weak-* ultrafilter limits under pushforward, so $\Phi_* \mathcal{C}_q \subseteq \mathcal{C}_{q'}$. When B is finite, F_q is compact, hence the probability measures on F_q form a weak-* compact space. Given $\nu \in \mathcal{C}_{q'}$, choose an ultrafilter along which the coarse empirical distributions converge to it. The fine empirical distributions converge along the same ultrafilter to some $\mu \in \mathcal{C}_q$, and continuity of pushforward gives $\Phi_* \mu = \nu$. Applying this result to the two coordinate projections of the joint readout gives both marginal surjections. Each empirical distribution for (q, q) is supported on the closed diagonal $\Delta_F = \{(u, u) : u \in F_q\}$. Its weak-* limits remain supported on Δ_F, so their marginals coincide. Conversely, if $\mu = \nu$, lift μ by the first marginal surjection. The second marginal must be the same. ■

B.2 Empirical norm scaling and isometric extension of a growing history

Suppose the old and new history windows contain $1 \le n \le m$ equally weighted participants with their respective normalized empirical inner products $\langle u, w\rangle_n = \frac{1}{n}\sum_{i<n} \overline{u_i} w_i$. Let $E : \mathbb{C}^n \to \mathbb{C}^m$ extend by zero, let $P_{m\to n} : \mathbb{C}^m \to \mathbb{C}^n$ restrict to the first n coordinates, and set $f := n/m \in (0, 1]$.

Proposition B.2 (Energy decomposition and restriction norm for a growing history). Extension by zero and restriction satisfy the scaling and orthogonal decomposition

$$\begin{aligned} &\|Eu\|_m^2 = f\|u\|_n^2, \qquad J := f^{-1/2}E \text{ is isometric and } J^* = f^{1/2}P_{m\to n}, \\ &\|v\|_m^2 = f\|P_{m\to n}v\|_n^2 + \|v - EP_{m\to n}v\|_m^2. \end{aligned} \tag{B.1}$$

The unnormalized restriction operator $P_{m\to n}$ has norm $\|P_{m\to n}\|_{m\to n} = f^{-1/2} = \sqrt{m/n}$. This is strictly greater than 1 when $n < m$ and equals 1 when $n = m$.

Proof. Comparing the weights $1/m$ and $1/n$ gives $\|Eu\|_m^2 = \frac{n}{m}\|u\|_n^2 = f\|u\|_n^2$, so $J = f^{-1/2}E$ preserves the norm. Since $\langle Ju, v\rangle_m = \frac{1}{m} f^{-1/2}\sum_{i<n} \overline{u_i} v_i = \frac{1}{n} f^{1/2}\sum_{i<n} \overline{u_i} v_i = \langle u, f^{1/2}P_{m\to n}v\rangle_n$, its adjoint is $J^* = f^{1/2}P_{m\to n}$. The vectors $EP_{m\to n}v$ and $v - EP_{m\to n}v$ have disjoint coordinate supports. The sum of their squared norms gives the last identity in (B.1). For nonzero $u \in \mathbb{C}^n$, taking $v = Eu$ attains $\|P_{m\to n}v\|_n/\|v\|_m = f^{-1/2}$. ■

B.3 A three-term covariance bound for mixtures of complex two-level density matrices

For a quantum or polarization measurement channel, suppose the finite source weights p_i sum to one ($\sum_i p_i = 1$). Each participant contributes a 2×2 positive semidefinite complex density matrix of trace one, $\rho_i = \left(\begin{smallmatrix} a_i & z_i \\ \overline{z_i} & 1-a_i \end{smallmatrix}\right)$, with $a_i \in [0, 1], z_i \in \mathbb{C}$. Write $\bar{a} = \sum_i p_i a_i$ and $\bar{z} = \sum_i p_i z_i$ for the ensemble-average diagonal population and coherence.

Proposition B.3 (Three-term determinant identity and covariance bound). The determinant of the mixed state $\bar{\rho} = \sum_i p_i \rho_i$ splits into three nonnegative terms: diagonal variance, mean intrinsic mixedness, and coherence variance.

$$\det\Big(\sum_i p_i \rho_i\Big) = \mathrm{Var}_p(a) + \sum_i p_i \det \rho_i + \sum_i p_i |z_i - \bar{z}|^2. \tag{B.2}$$

For any real-valued task u_i, the centred covariance therefore satisfies

$$\mathrm{Cov}_p(u, a)^2 + \mathrm{Var}_p(u)\left(\sum_i p_i \det \rho_i + \sum_i p_i |z_i - \bar{z}|^2\right) \le \mathrm{Var}_p(u) \det\Big(\sum_i p_i \rho_i\Big). \tag{B.3}$$

Proof. Expand $\det \bar{\rho} = \bar{a} - \bar{a}^2 - |\bar{z}|^2$ and $\sum_i p_i \det \rho_i = \bar{a} - \sum_i p_i a_i^2 - \sum_i p_i |z_i|^2$. Subtract these expressions and substitute $\mathrm{Var}_p(a) = \sum_i p_i a_i^2 - \bar{a}^2$ and the complex variance identity $\sum_i p_i |z_i - \bar{z}|^2 = \sum_i p_i |z_i|^2 - |\bar{z}|^2$ to obtain (B.2). Bounding the first term by Cauchy–Schwarz in the form $\mathrm{Cov}_p(u, a)^2 \le \mathrm{Var}_p(u)\,\mathrm{Var}_p(a)$ gives (B.3). ■

B.4 Cayley–Hamilton recurrence, least stable horizon, and two-level optimal recovery

Theorem B.4 (Autonomous windows on finite free modules and the three-term empirical error). Let C be a finite-rank free module over a nonzero commutative ring [35; 39]. Let $A : C \to C$ and $o : C \to B$ be linear and put $d = \deg \chi_A$, where χ_A is the characteristic polynomial of A. For $0 \le j \le d$, let b_j be the negative of the coefficient of X^j in $X\chi_A(X)$. Then $A^{d+1} = \sum_{j=0}^{d} b_j A^j$. For any trajectory $m_{t+1} = Am_t$, the length-$d+1$ observation window $W_t = (om_t, \dots, om_{t+d})$ updates autonomously by

$$W_{t+1} = \Big(om_{t+1}, \dots, om_{t+d}, \sum_{j=0}^{d} b_j om_{t+j}\Big), \tag{B.4}$$

and there is an intrinsic least stable prefix horizon $h_* \le d$, the least natural number ℓ satisfying $A(\ker P_\ell) \subseteq \ker P_\ell$. It satisfies $\ker P_{h_*} = K_\infty$ and induces a linear autonomous successor on $\operatorname{ran} P_{h_*}$. For every shorter horizon $\ell < h_*$, there is a $w \in C$ with $P_\ell w = 0$ but $P_\ell(Aw) \neq 0$. This rules out every prefix update $F : B^{\ell+1} \to B^{\ell+1}$ on the full carrier C, including nonlinear maps, as an instance of Corollary 2.5.

When this finite window is placed in the original empirical field, its positive-weight fiber conditional distributions agree with the full-future conditional distributions participant by participant. Let J_0 be the isometric pullback for the map from participants to the current field, with $U = J_0^*$. Let K be the isometric pullback for the map from the current to the next field, with $\Lambda = K^*$, and write $u_1 = Uu, u_2 = \Lambda u_1$. For any next-field decoder $\hat{d}$, the two orthogonal decompositions give

$$\|u - J_0 K \hat{d}\|^2 = \|u - J_0 u_1\|^2 + \|u_1 - K u_2\|^2 + \|u_2 - \hat{d}\|^2. \tag{B.5}$$

Proof. Multiply the Cayley–Hamilton equation $\chi_A(A) = 0$ on the left by A and rearrange using the monicity of χ_A to obtain $A^{d+1} = \sum_{j=0}^{d} b_j A^j$. Composing with o shows that horizon d already satisfies $A(\ker P_d) \subseteq \ker P_d$. Well-ordering gives the least stable horizon $h_* \le d$, and the maximal-invariance part of Theorem 2.4(1) yields $\ker P_{h_*} = K_\infty$. For $\ell < h_*$, failure of stability gives a witness w with $P_\ell w = 0$ and $P_\ell(Aw) \neq 0$. Corollary 2.5 rules out every shorter-prefix update F. Evaluating at $x = 0$ and $x = w$ gives $0 = F(0) = F(P_\ell w) = P_\ell(Aw) \neq 0$. The finite window and the full-future readout have the same fibers and hence the same conditional distributions. Applying the isometric orthogonal-projection decomposition twice in the empirical Hilbert space, as in Theorem 4.4, gives (B.5). ■

B.5 Canonical recovery on the completed copy graph and rank-one innovation energy

Embed the full empirical history field into the common word space by the isometry of Proposition B.2. Each new record supplies an innovation direction h_j orthogonal to the old visible subspace. Write its rank-one orthogonal projection as $B_j v = \frac{\langle h_j, v\rangle}{\|h_j\|^2} h_j$, taking 0 if $h_j = 0$, and write $r_j v$ for the invisible residual after step j.

Theorem B.5 (Recursive rank-one decoding and conservation of innovation energy). The one-step reconstruction equals the normalized inclusion of the old reconstruction plus the

projection onto the new rank-one direction. Iterating through any finite $k \in \mathbb{N}$ gives the canonical inverse for the full history field at that depth. For every fixed target v in the common word space, the residual and innovations satisfy

$$\|r_k v\|^2 + \sum_{j=0}^{k-1} \|B_j v\|^2 = \|r_0 v\|^2. \tag{B.6}$$

Proof. The visible subspace at step $j+1$ is the orthogonal direct sum of the visible subspace at step j and $\mathbb{C}h_j$. Thus $r_j v = r_{j+1} v + B_j v$ with $r_{j+1} v \perp B_j v$. Pythagoras gives $\|r_j v\|^2 = \|r_{j+1} v\|^2 + \|B_j v\|^2$. Summing over $j = 0, \ldots, k-1$ telescopes to (B.6). ■

B.6 A three-participant pulse clock with minimum squared error $2/3$

Take three equally weighted participants and the true clock task $(0, 1, 2)$. The current field reads (a, b, b) with $a \neq b$, merging participants $1, 2$. The next field reads the constant (b, b, b), merging all three. The current-field conditional mean is $(0, 3/2, 3/2)$. The first-level collision loss is $\frac{1}{3}[0^2 + (1-3/2)^2 + (2-3/2)^2] = 1/6$. In the next field the overall mean is 1, and the second-level coarsening adds $\frac{1}{3}[(0-1)^2 + 2(3/2-1)^2] = 1/2$. By (B.5), every next-field decoder $\hat{d}$ has squared risk

$$\mathrm{Risk}(\hat{d}) = \frac{1}{6} + \frac{1}{2} + |1 - \hat{d}(b)|^2 = \frac{2}{3} + |1 - \hat{d}(b)|^2, \tag{B.7}$$

and the constant decoder $\hat{d}(b) = 1$ attains the global minimum $2/3$. The two-participant pulse in Example 4.5 has minimum $1/4$. An injective clock readout gives zero error.

B.7 Affine derivative formulas and all-time pointwise convergence of empirical mixtures

For Theorem 6.9, define the rising factorial over the reals by $\alpha^{\overline{0}} = 1$ and $\alpha^{\overline{n+1}} = \alpha^{\overline{n}}(\alpha + n)$. The falling factorial satisfies $(-\alpha)_{\underline{n}} = (-1)^n \alpha^{\overline{n}}$. If $g : \mathbb{R} \to \mathbb{R}$ is n times differentiable near the relevant point, its affine pullback along $a + ct$ satisfies $\frac{\mathrm{d}^n}{\mathrm{d}t^n} g(a + ct) = c^n g^{(n)}(a + ct)$. Applying the identity to the positive-base power $f_\alpha(x) = x^{-\alpha}$ with $a = 1, c = r^{-1}$ transports complete monotonicity to the shifted power law $\Psi_{\alpha,r}(t) = (1 + t/r)^{-\alpha}$ on the nonnegative time axis $t \geq 0$.

Let $\{X_i\}_{i \in \mathbb{N}}$ be sampled decay rates on a probability space $(\Omega, \mathcal{F}, \mu)$. Define the Laplace test variables $Y_i(t, \omega) := e^{-X_i(\omega) t}$.

1. **Strong law on a finite time grid.** Suppose that at each time $\tau \in \mathcal{T}_{\mathrm{fin}}$ of a finite grid $\mathcal{T}_{\mathrm{fin}} \subset [0, \infty)$, the variables $\{Y_i(\tau)\}_{i \in \mathbb{N}}$ are integrable, pairwise independent, and identically distributed with $Y_0(\tau)$. Suppose also that their expectation is the normalized Gamma Laplace value $\mathbb{E}_\mu[Y_0(\tau)] = (1 + \tau/r)^{-\alpha}$. The strong law and the fact that the intersection of the resulting events has probability 1 show that for μ-almost every sample path ω, the empirical Laplace averages $\bar{Y}_N(\tau, \omega) := \frac{1}{N} \sum_{i=0}^{N-1} e^{-X_i(\omega)\tau}$ converge to $(1 + \tau/r)^{-\alpha}$ simultaneously at every grid point $\tau \in \mathcal{T}_{\mathrm{fin}}$ [13, Thm. 1]. The same intersection argument applies to countably many observation times.
2. **Extension from a dense countable grid under a common Lipschitz bound.** Suppose the countable set $\{\tau_k\}_{k \in \mathbb{N}}$ is dense in $[0, \infty)$, the mixture sequence converges to $\Psi_{\alpha,r}$ at every grid point, and all $M_N(t)$ and $\Psi_{\alpha,r}(t)$ share a Lipschitz constant $L \geq 0$ on $[0, \infty)$. Given

$t \geq 0$ and $\epsilon > 0$, choose τ_k with $L|t - \tau_k| < \epsilon/3$. A three-term triangle inequality with $\epsilon/3$ for each term yields $\lim_{N\to\infty} M_N(t) = (1 + t/r)^{-\alpha}$ for every $t \geq 0$. The conclusion is pointwise convergence for each t simultaneously.

Appendix C. Continuous spatial gradient flow and the molecular electronic-structure companion

The principle of recovering the complete object extends to three-dimensional continuous fields. For the continuous-field interface of chiral L-alanine across all 32 parameter-band patches, the source calculation determines the field on each interval, gives an injective continuous coordinate transformation without folds, and gives positive finite volume. Fluxes on adjacent interior faces cancel, so the Laplacian integral over the whole physical domain equals its outer-boundary flux.

The spatial gradient flow induced by the same source density satisfies its flow differential equation and the inverse-flow identity $\Phi_{-t}(\Phi_t(x)) = x$ for every initial point and every real parameter $t \in \mathbb{R}$. A basin of positive volume converges to a local critical attractor. Here t parametrizes the spatial topological decomposition. The evolving process has its own physical clock. The same source event gives the complete field certificate together with the original ledger, clock, and canonical successor. The companion paper on molecular electronic structure treats grid-free real-space IQA two-body energy closure for a 48-electron wave function, sub-nanohartree certification of four-centre Coulomb integrals, and variational dynamics with moving Gaussian basis functions.

Code and Data Availability

The pinned H0mework commit 5e115b4637d3 contains the Lean 4 proofs, their immediate consumers, and the original evidence. The claim-by-claim map records original commits, public modules, full declaration types, and receipts. The multi-epoch selection retains the original dependencies. Follow the reproduction guide to restore the original tree and run the build, trust, and type checks. With the pinned dependencies available, run `lake -d Lean build ObservationDynamicsRelease`. The public selection includes the algorithm, controls, and original verification reports for the fly case. Reproducing the original cold runs requires the large private inputs identified in those reports. In the editable package, `verification/recheck.py` independently recomputes the rational and combinatorial examples.

Section	Reproduction entry point
§2 Autonomous updates and minimal realization	`PrefixDeterminacy`
§3 Shortest observation window	`OperatorAcquisitionDecode`
§5 Coarsening and reception	`Table`
§6 Parity recovery and graded innovations	`ColumnAction`
§6.5 Completely monotone forgetting	`P147`
§7 Non-resetting measurement circuit	`Semantics`

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